

MAHIDOL UNIVERSITY

Faculty of Information and Communication Technology



Lecture 2
Network Models and Physical Layer

August 2026

ปัญญาของแผ่นดิน Wisdom of the Land

มหาวิทยาลัยมหิดล
Mahidol University

ห้ามมิให้นักศึกษาทำซ้ำ ดัดแปลง หรือใช้ประโยชน์จากงานอันมีลิขสิทธิ์ปรากฏอยู่ในระบบการศึกษาอิเล็กทรอนิกส์ (E-Learning) ของมหาวิทยาลัย ไม่ว่าจะทั้งหมดหรือบางส่วน โดยไม่ได้รับอนุญาตจากมหาวิทยาลัย นอกเหนือจากการศึกษาส่วนบุคคล ทั้งนี้ การทำซ้ำ ดัดแปลง หรือเผยแพร่ต่อสาธารณชนซึ่งงานอันมีลิขสิทธิ์ **จะมีโทษปรับตั้งแต่ 20,000 บาท ถึง 200,000 บาท และหากเป็นการกระทำเพื่อการค้า จะมีโทษจำคุกตั้งแต่ 6 เดือน ถึง 10 ปี หรือปรับตั้งแต่ 100,000 บาท ถึง 800,000 บาท หรือทั้งจำทั้งปรับ**

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Quote of the Week

THE MIND IS EVERYTHING. WHAT YOU THINK YOU BECOME.

BUDDHA



Today Agenda

- Advantages of Layered Data Communication
- Functions of each layer
- Addressing
  - ✓ Physical layer address
  - ✓ MAC address and IP address

- - Break -

- Network Topology
- Data and Signal
- Equation of Sine wave signal
  - ✓ Phase, Frequency, Amplitude



Layered Network

- In layered architecture of a network model, one whole network process is divided into small tasks.
  - ✓ In layered communication system, one layer of a host deals with the task done by or to be done by its peer layer at the same level on the remote host.
- The task is either initiated by layer at the top most level (send) or the lowest level (receive).
- A set of layers and protocols is known as network architecture.

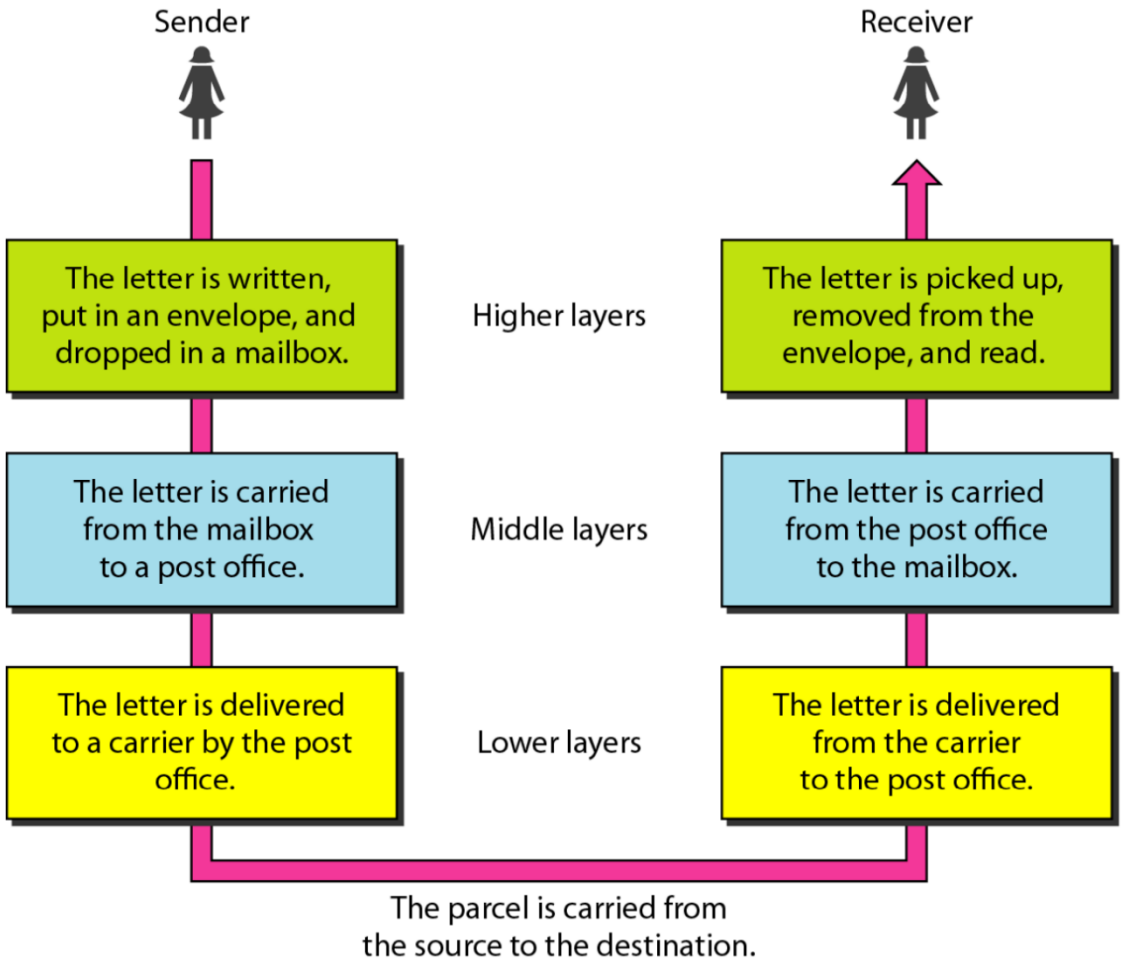
Benefits of Layered Architecture

- **Divide-and-conquer** approach:
  - ✓ **Divide-and-conquer approach** makes a design process in such a way that the unmanageable tasks are divided into **small and manageable tasks**.
  - ✓ In short, we can say that this approach **reduces the complexity of the design.**

- **Modularity**:
  - ✓ **Layered architecture** is more **modular**.
  - ✓ Modularity provides the **independence of layers**, which is easier to understand and implement.
  - ✓ Modularity **allows each module to be easily modified**, analyzed and tested.
  - ✓ One layer **can be changed without affecting other** layers.

Postal Service Layered Architecture

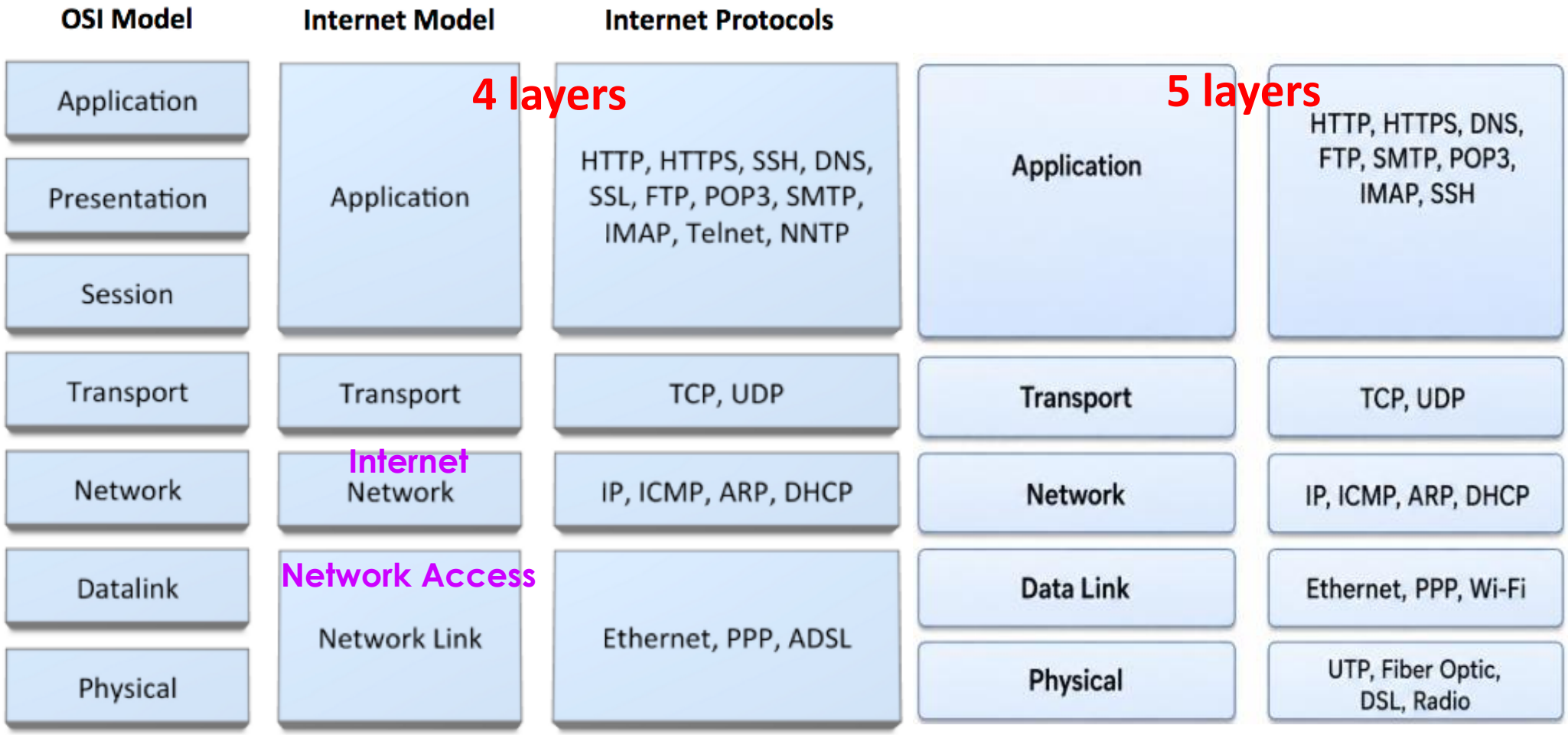
- The process of sending a letter to a friend can be compared to the network model concept as follows:



Network Models

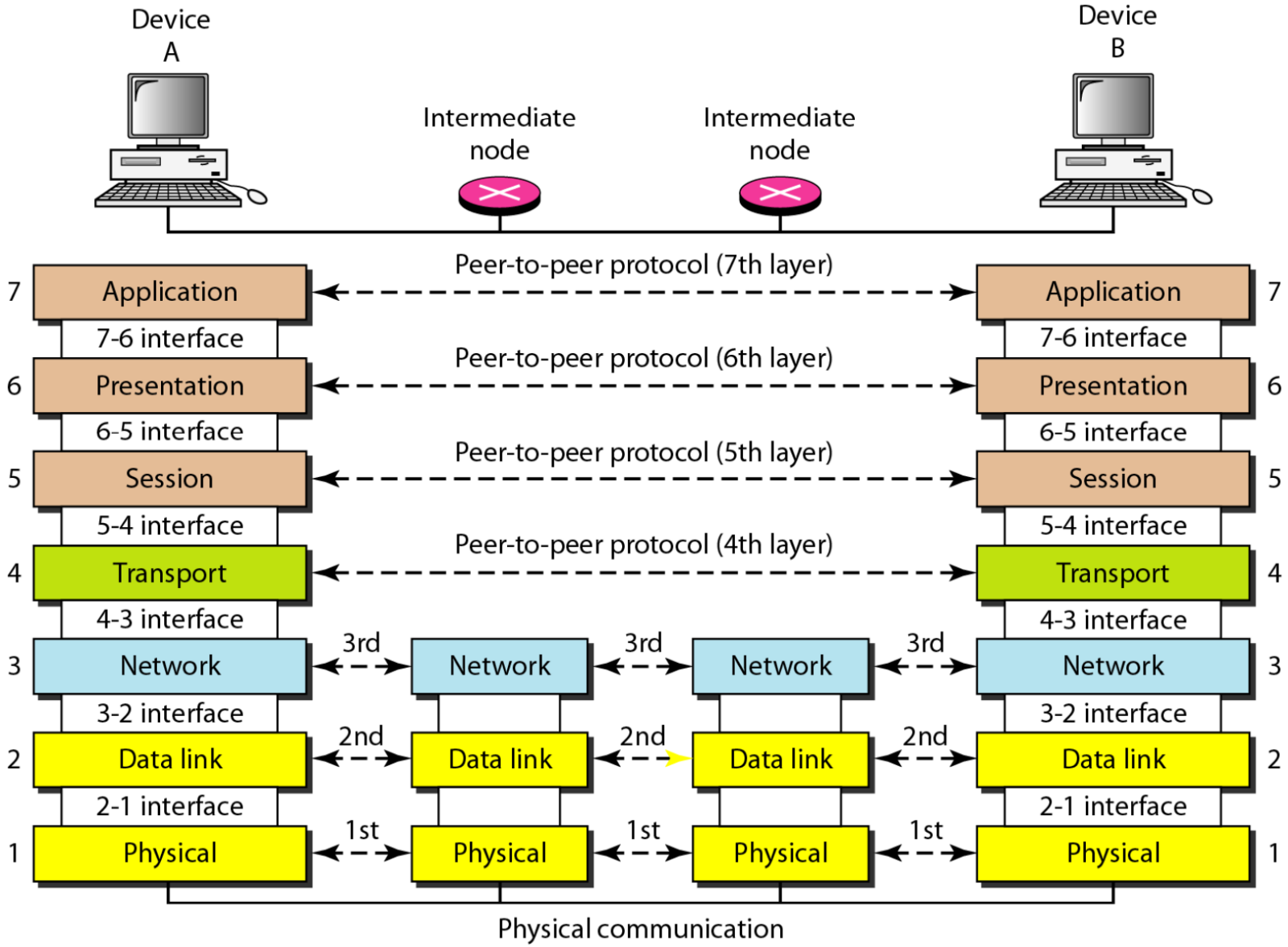
- There are two competing computer network models.
  - **OSI Reference Model**
    - ✓OSI stands for "Open System Interconnection" and the name of this reference model was given by an organization known as "International Organization for Standardization (ISO)".
    - ✓The ISO is responsible for generating and promoting industrial and commercial standards applicable for all the users or universally.
  - **TCP/IP Model**
    - ✓The TCP/IP model consists of four/**five layers**: the application layer, transport layer, network layer, data link layer and physical layer.
    - ✓TCP/IP is the de facto standard for network model of the Internet.

Data Communication Models



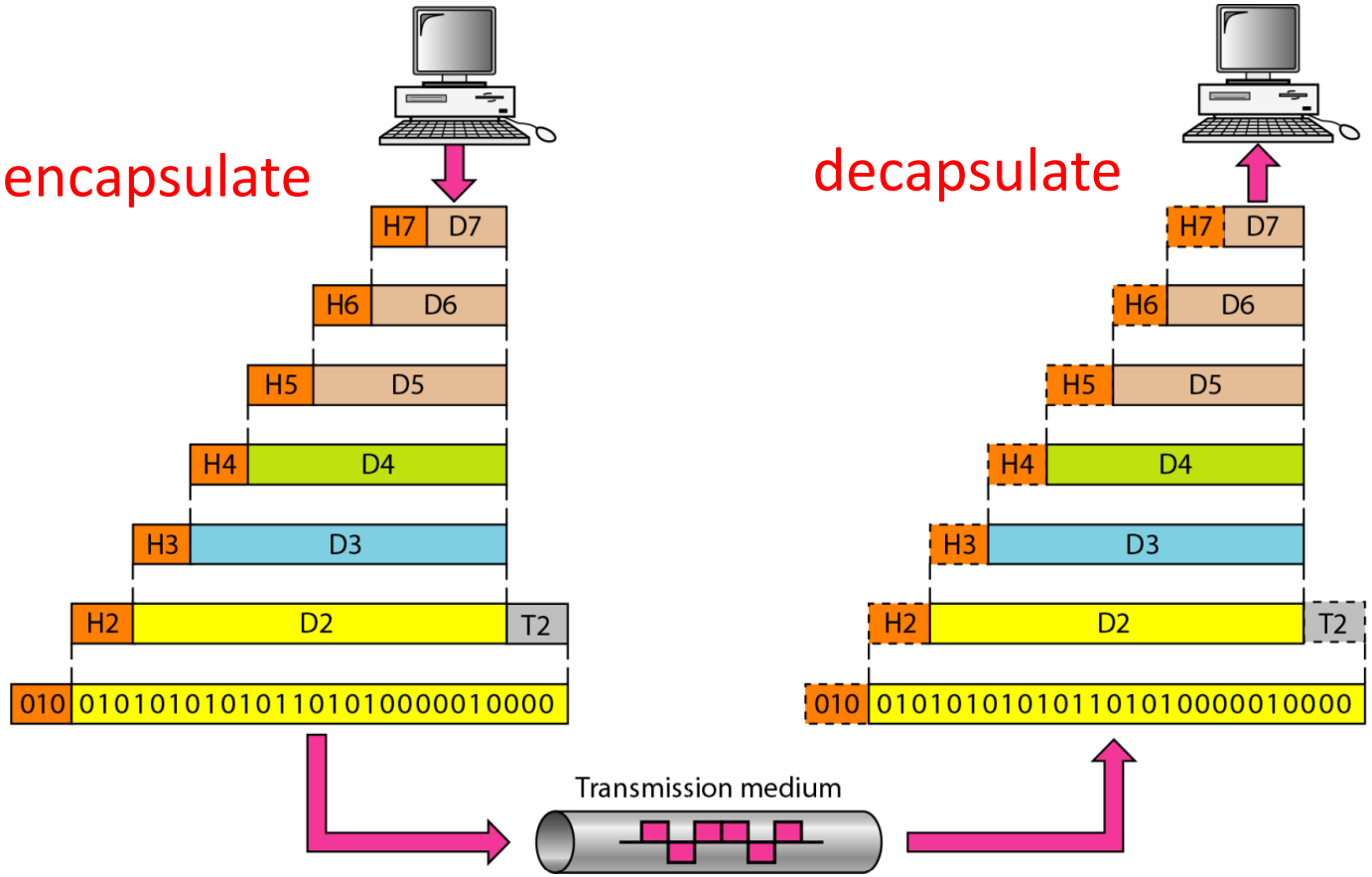
OSI: KnowledgeCatch, https://www.youtube.com/watch?v=8YkL_qc6ozc
Tech in a Teacup, https://www.youtube.com/watch?v=sxjsfsStTf0
OSI & TCP/IP: Cybrixen, https://www.youtube.com/watch?v=s7ui2cdyCtQ

Layered Communication (OSI Model)



Layered Data Flow and Data Encapsulation

- The data in the upper layer is encapsulated and will not be looked at or processed by the lower layer. This gives independency to each layer.
- Each layer puts in its own header (extra information) for same-layer communication between the sender and receiver.



Layer Header

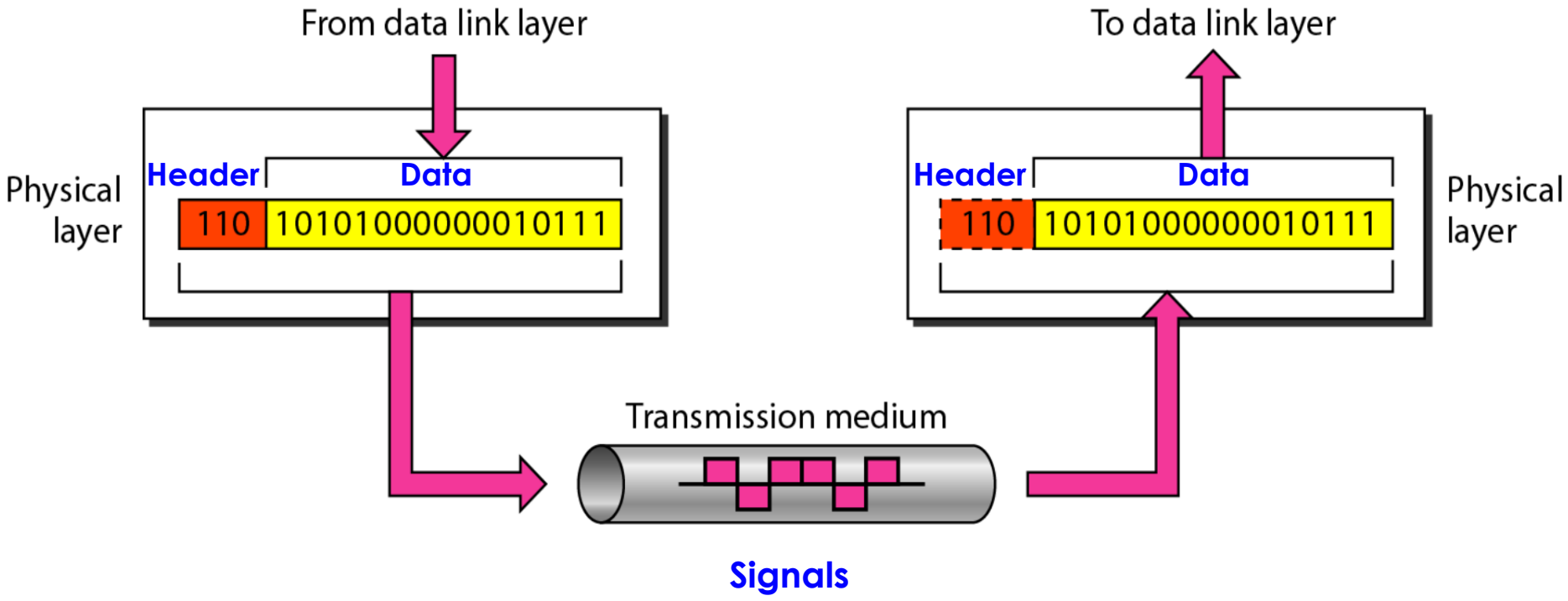
- Layer header contains necessary information to support the responsibility of the layer. Think of it as an envelope or extra label you put on your parcel box in order to instruct the postal service what to do with your parcel at each layer.
- Layer header is added by each layer at the sender side and it is removed by the same layers at the receiver side.





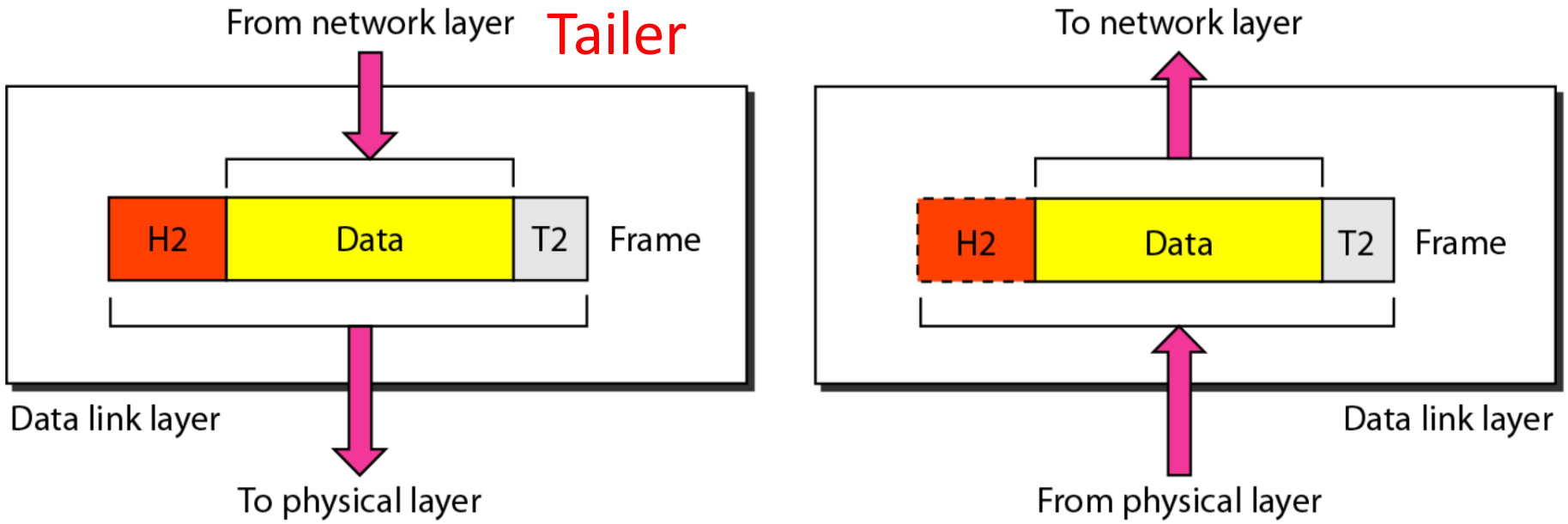
The Physical Layer

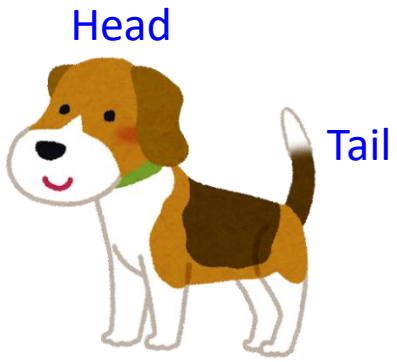
- The physical layer is responsible for movements of individual **bits** from one hop (node) to the next.



The Data Link Layer

- The data link layer is responsible for moving **frames** from one hop (node) to the next hop.

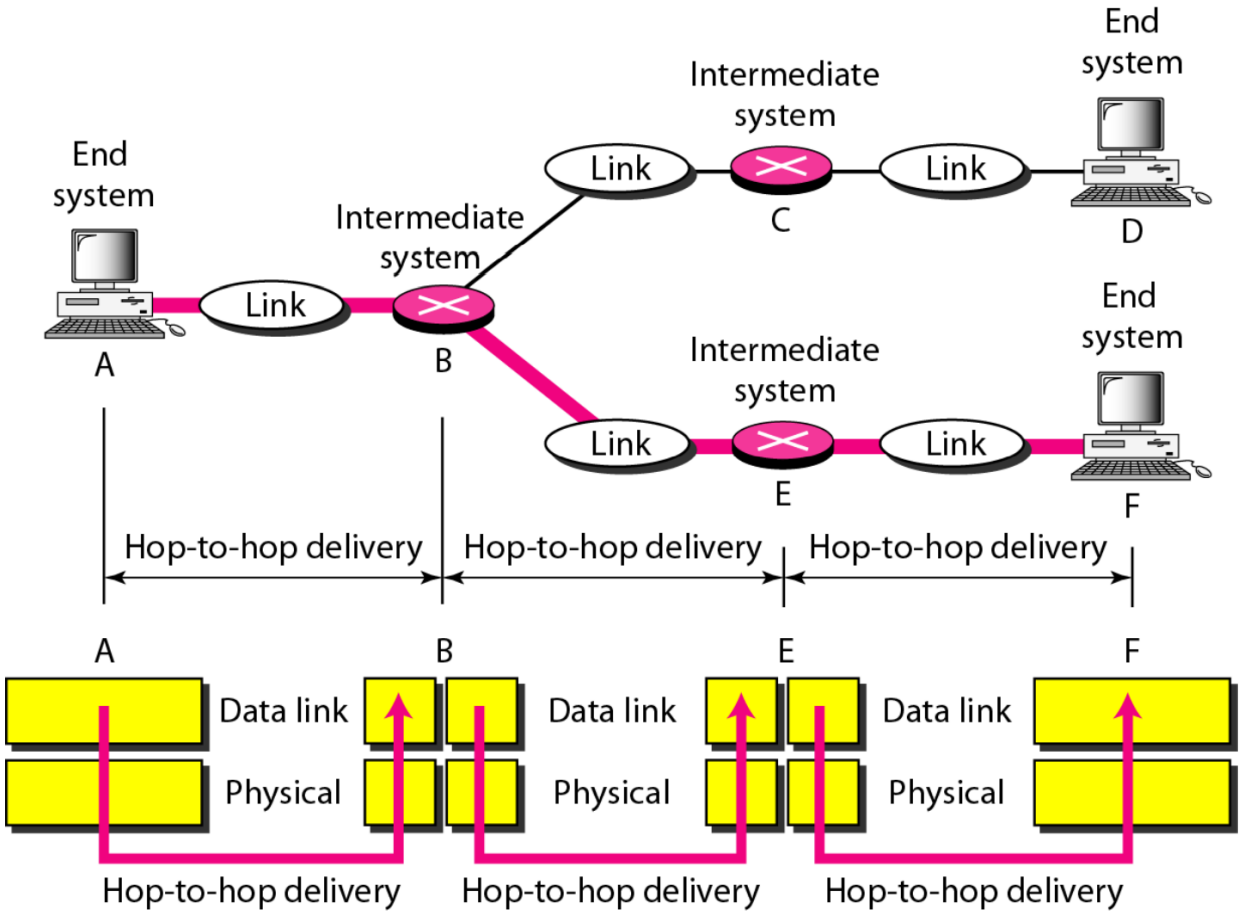




The Data Link Layer

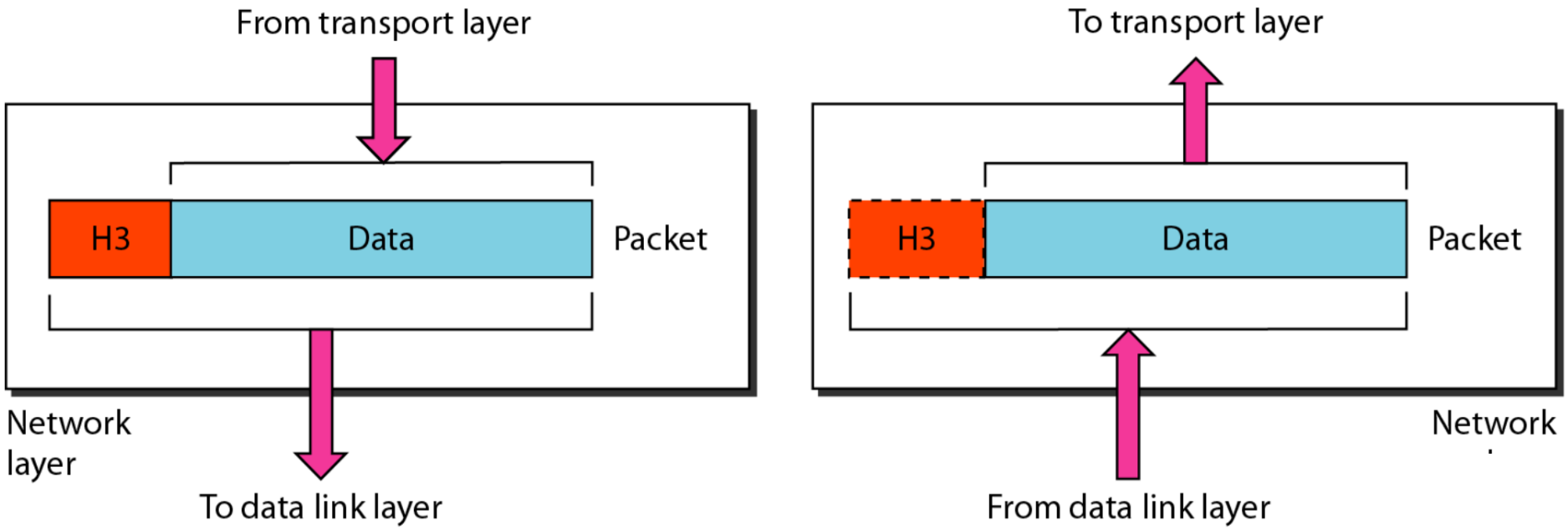
- In order to deliver the data frame from the sender to the recipient, intermediate hops must be able to forward the data frame to the right path.
  → **Forward a frame within the same network**
- A host is identified by a MAC address.

MAC Address == Physical Address



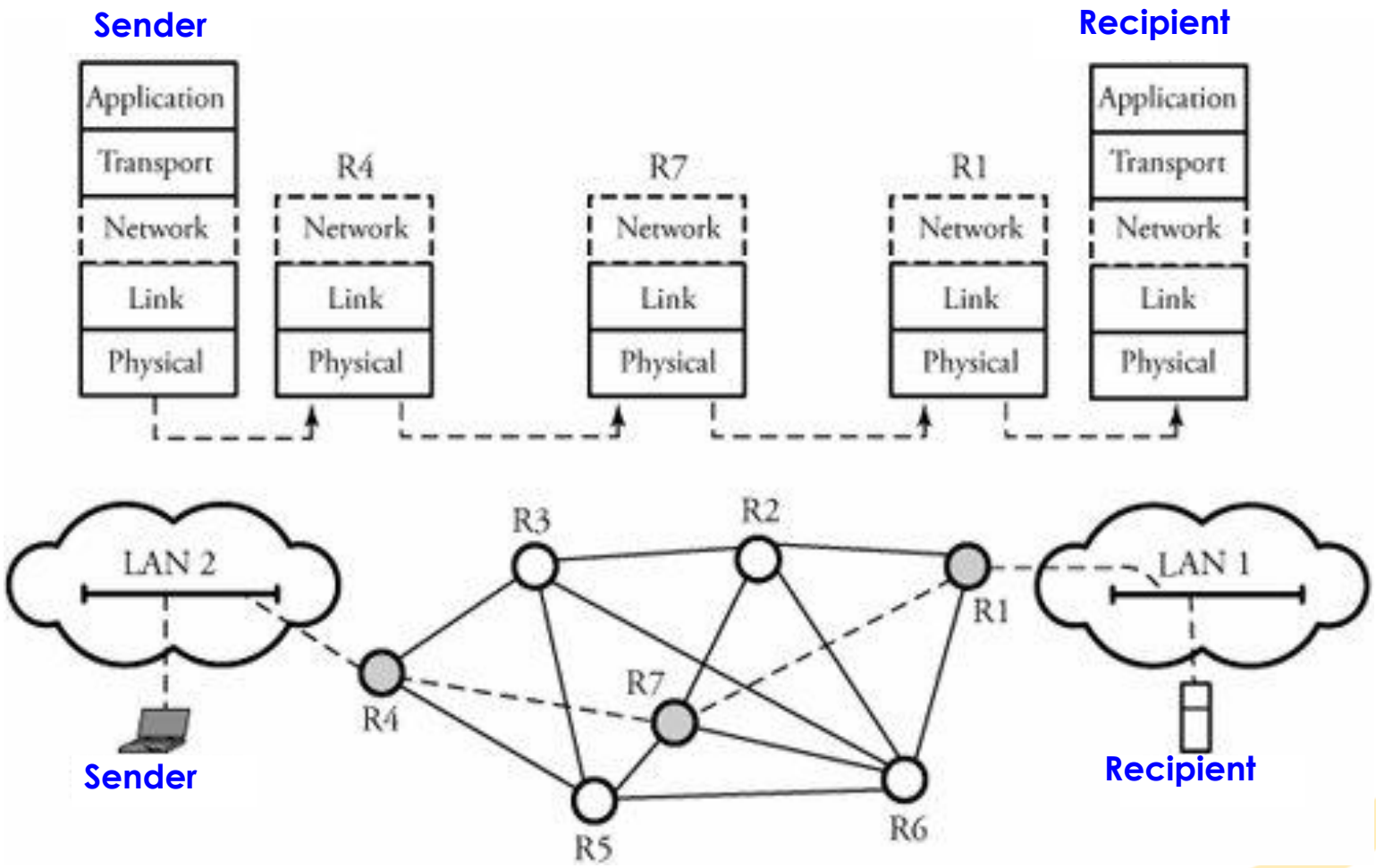
The Network Layer

- A host is identified by an IP address.
- The network layer is responsible for the delivery of individual **packets** from the source IP address (sender) to the destination IP address (recipient) **across the network.**



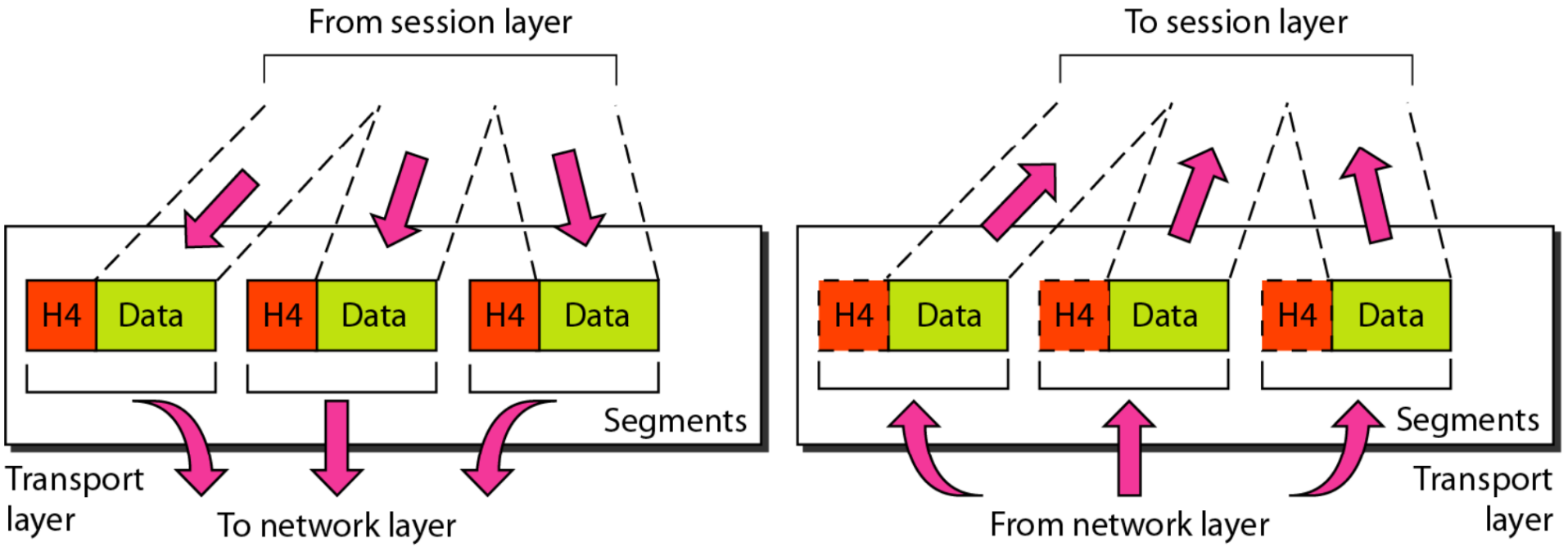
The Network Layer

- Routers have to work together to find **the best route/path** at any time.



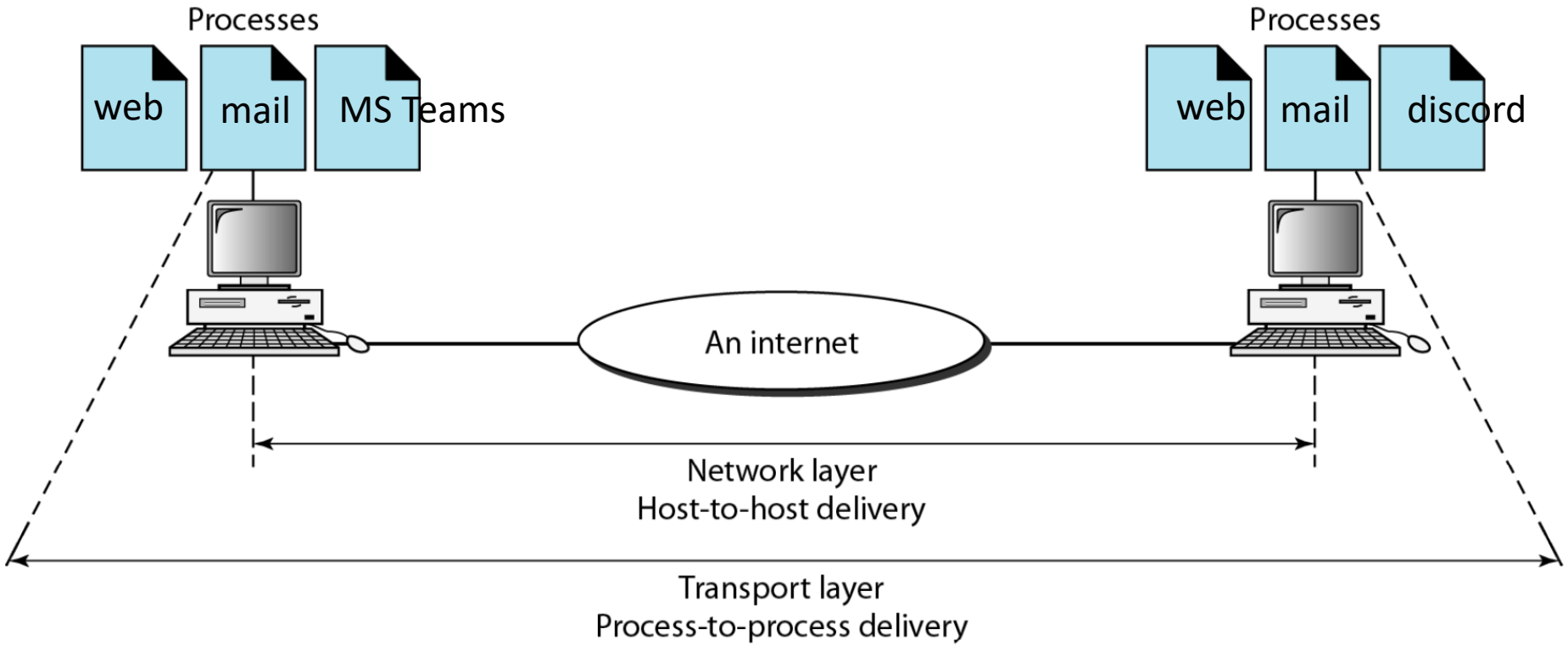
The Transport Layer

- The transport layer is responsible for the delivery of a **segment** from one **process** to another.



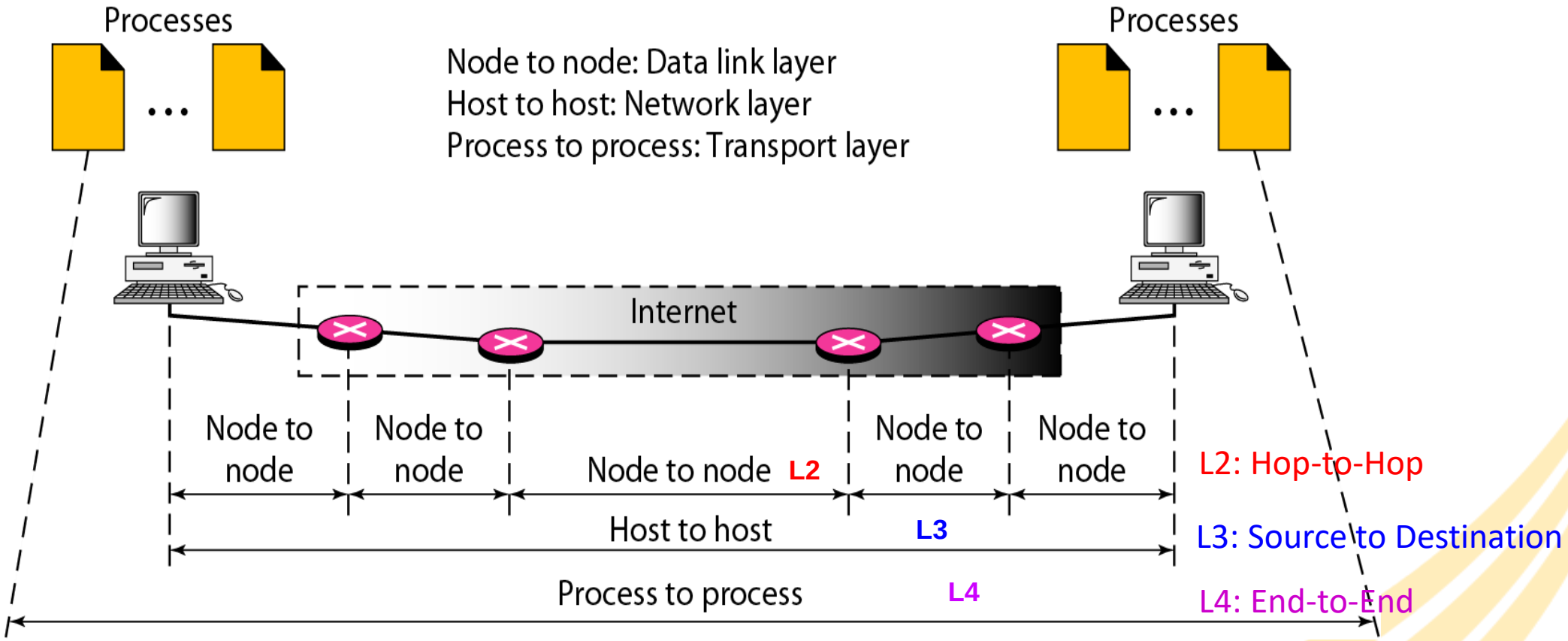
The Transport Layer

- The transport layer provides **multiplexing** service for different processes.
- Each application is specified by an application or process ID (commonly called port number).



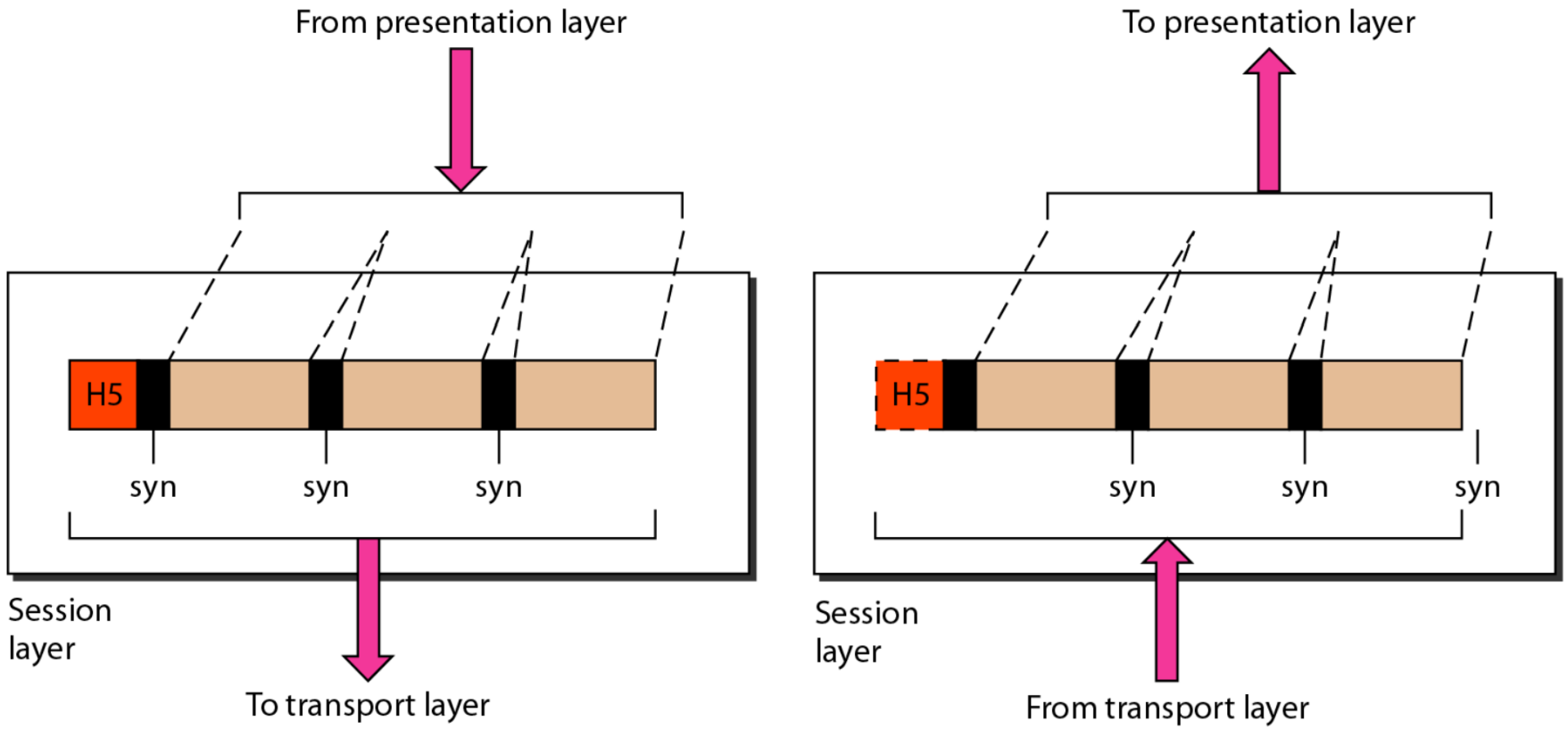
Process-to-Process Communication

Transport layer handles communication between a process (application) in a sending host and a process in a receiving host.



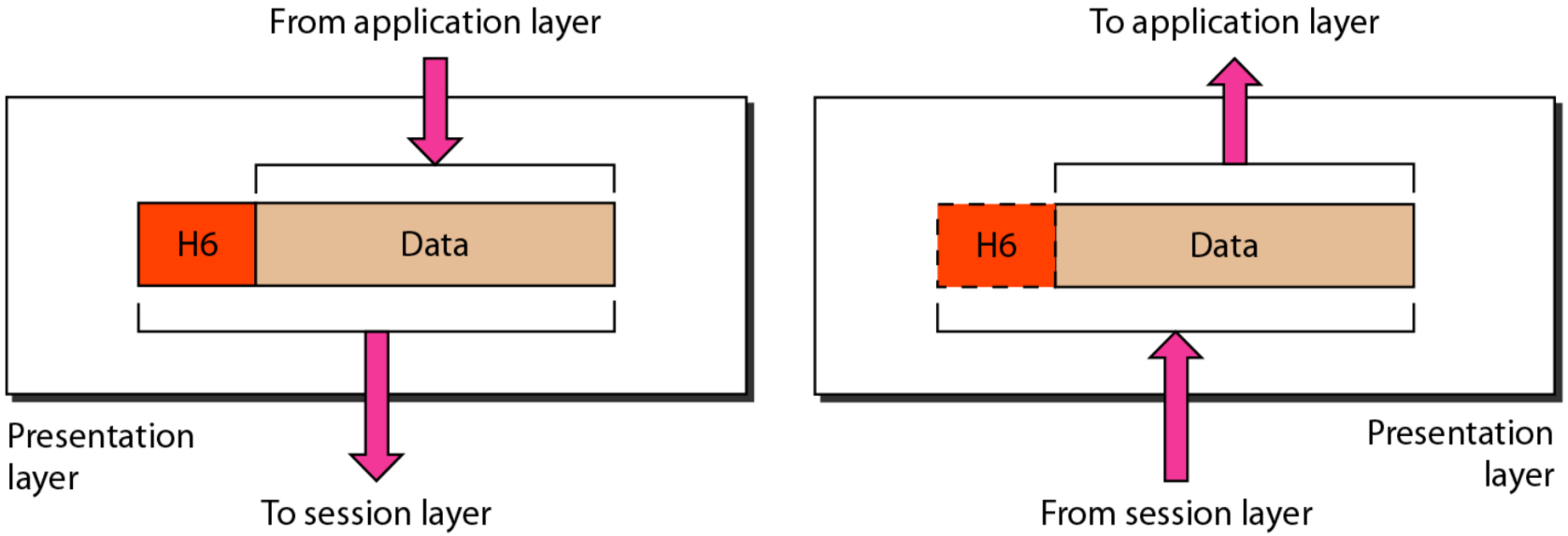
The Session Layer

- The session layer is responsible for dialog control and synchronization.



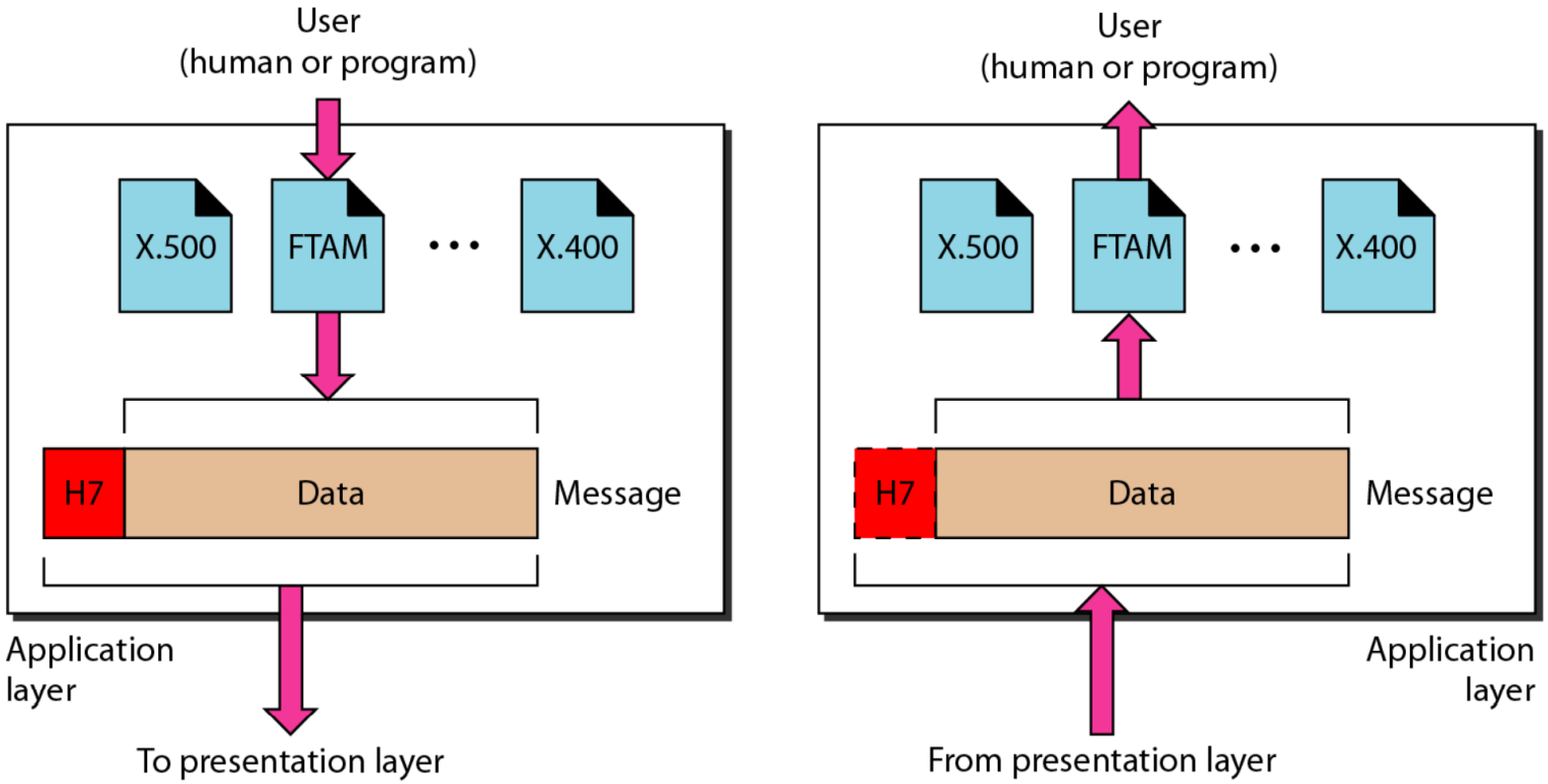
The Presentation Layer

- The presentation layer is responsible for translation, compression, and encryption of data.

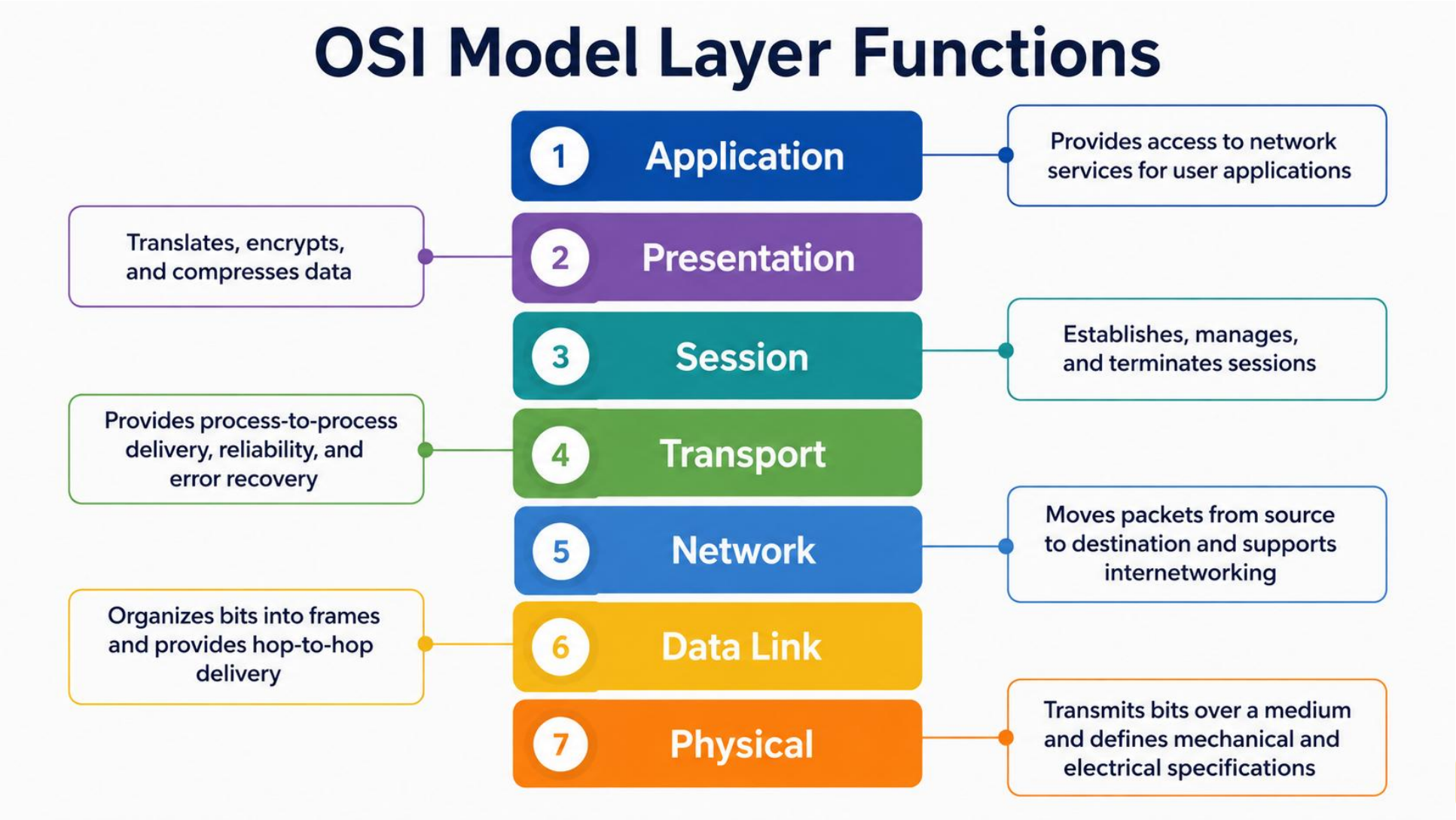


The Application Layer

- The application layer is responsible for providing services to the user.

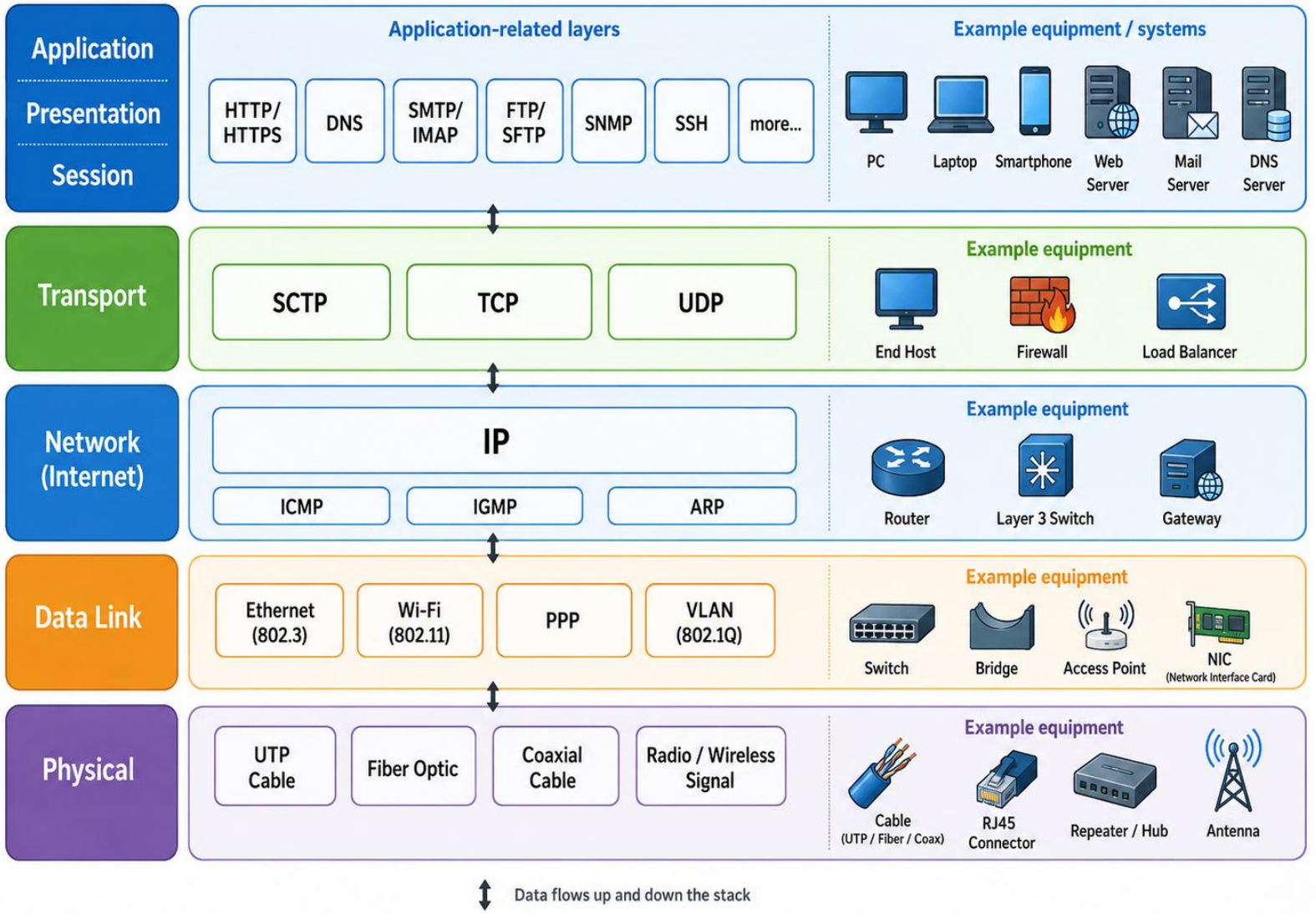


Summary of Layer Responsibilities



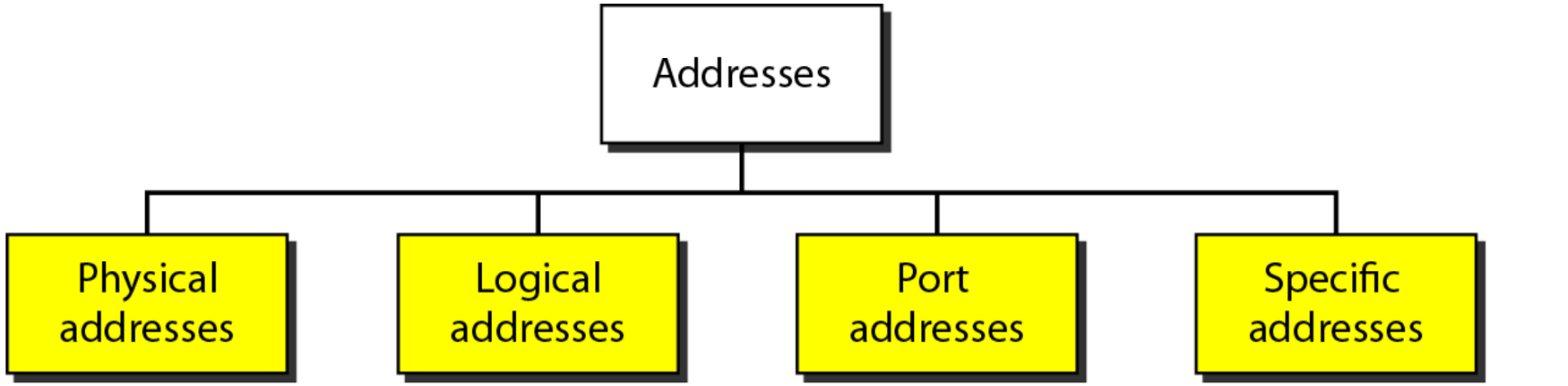
TCP/IP Protocol Suite

- A package of protocols used in TCP/IP model

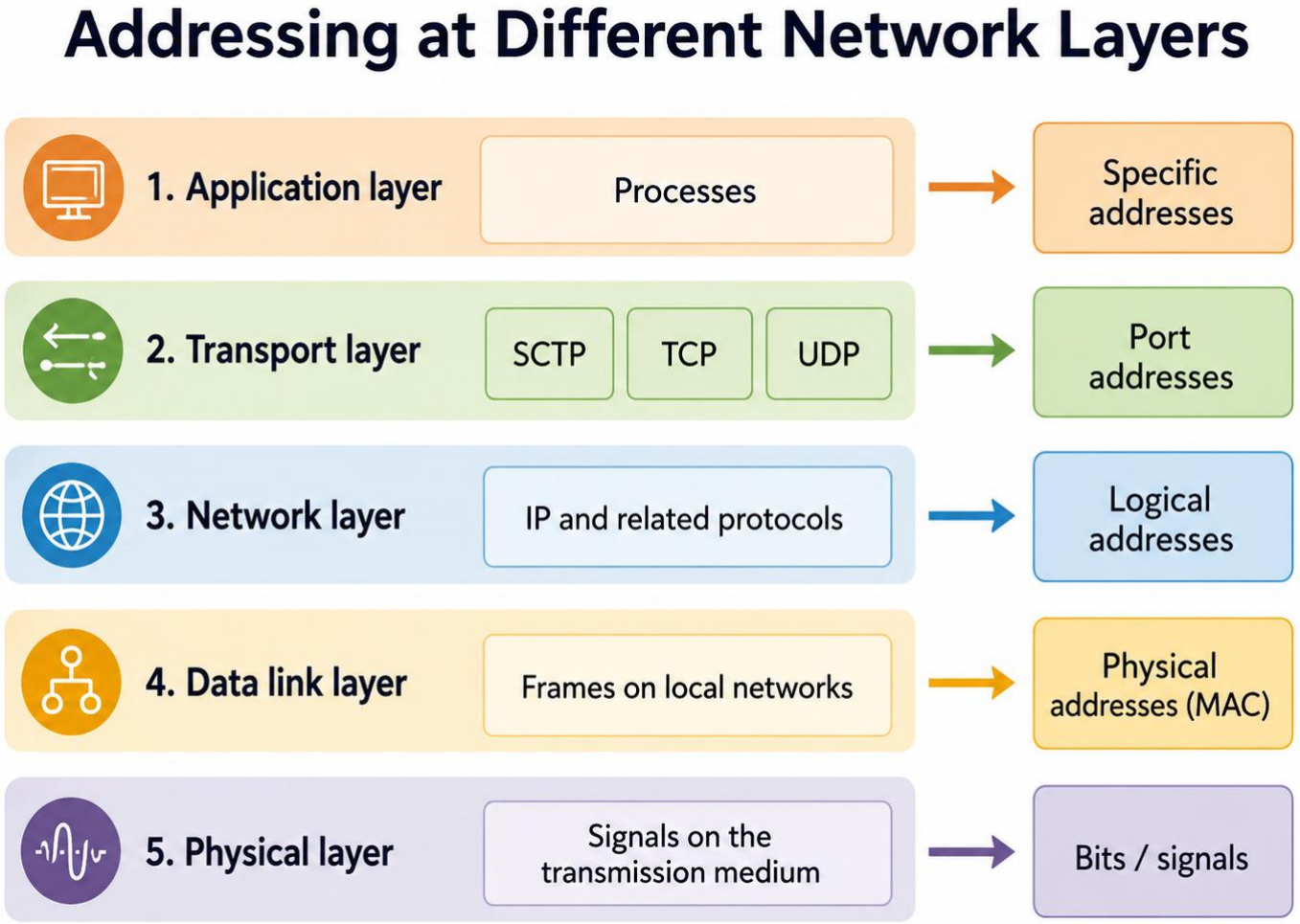


Addressing

- Multiple hosts can communicate with each other by using unique addresses.
- Since different layer is independent from other layer, some network in a layer is hidden from the network in another layer.
- Addressing for each layer is required.

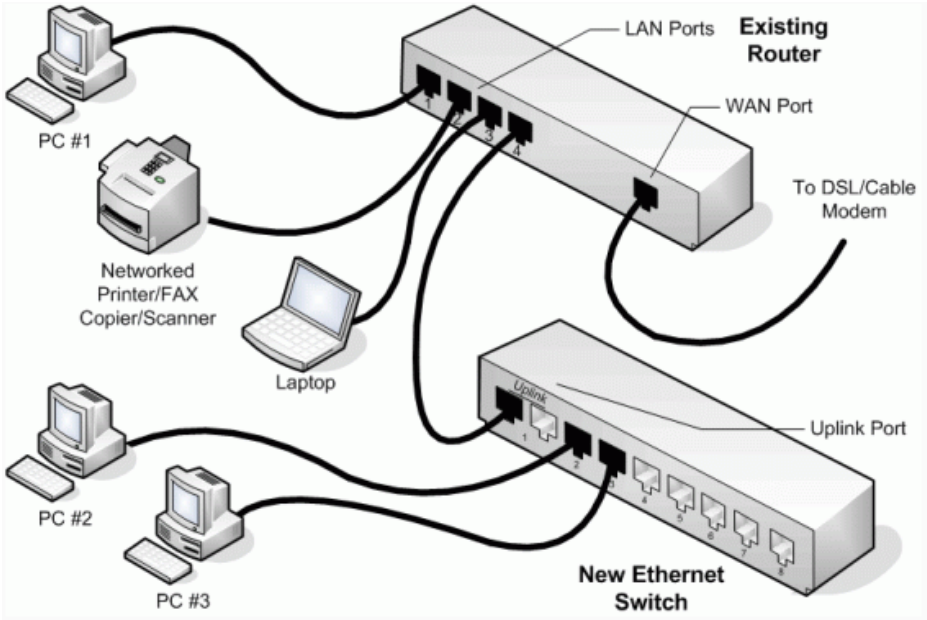


Layered Addressing



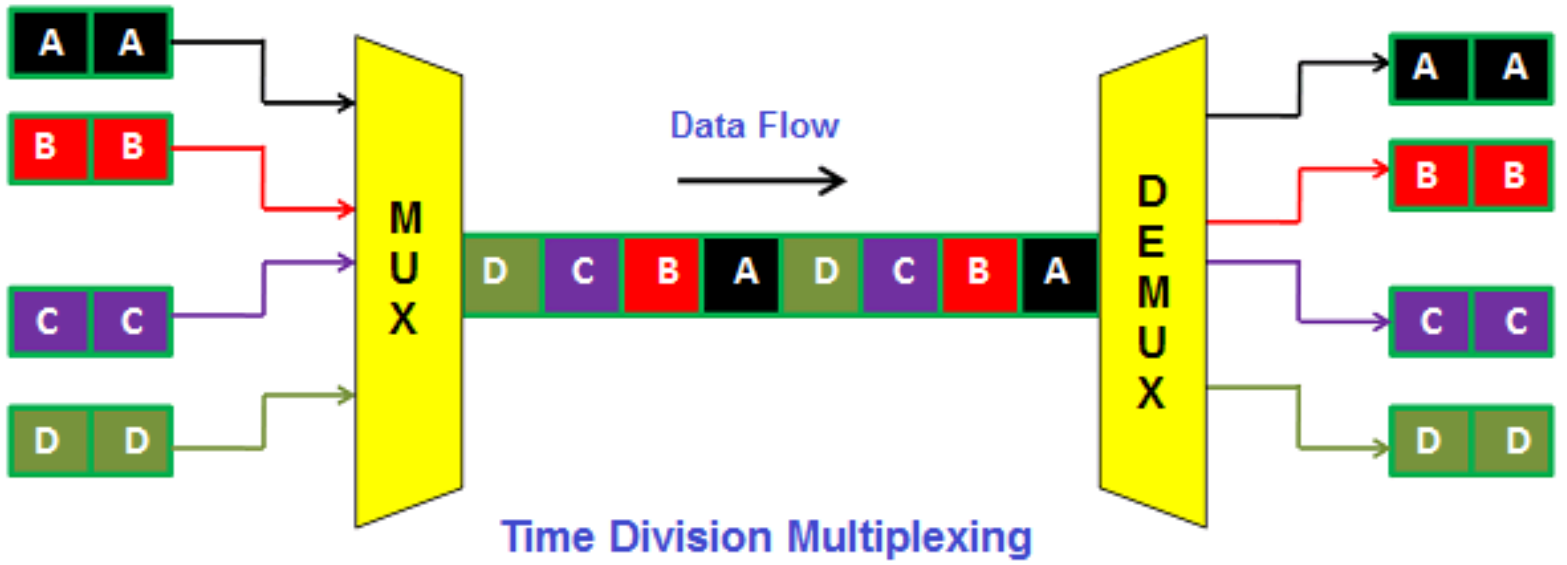
Addressing in Physical Layer

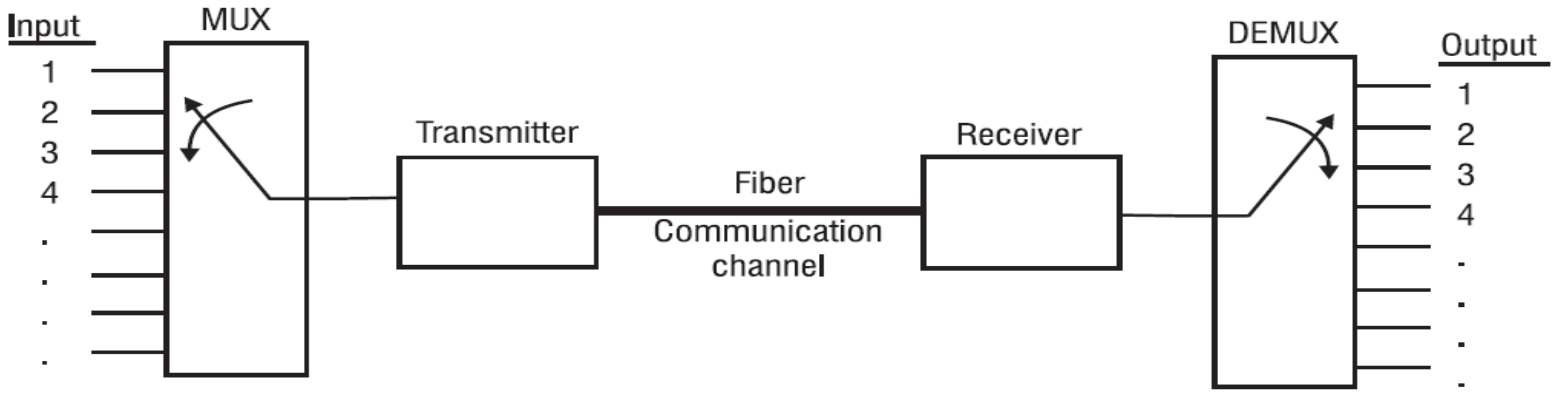
- Usually in the modern day, connection in the physical layer is one to one.
  No addressing is needed.
- However, when multiplexing is used,
  - **Time division multiplexing (TDM):** the addressing is the station label in front of the data frame. (divide by time) → **Slot number**
  - **Frequency division multiplexing (FDM):** The address is the frequency specified or used when sending the signal. (divide by frequency) → **Frequency range (channel number)**



Multiplex = combine multiple signals together

Time Division Multiplexing (TDM)



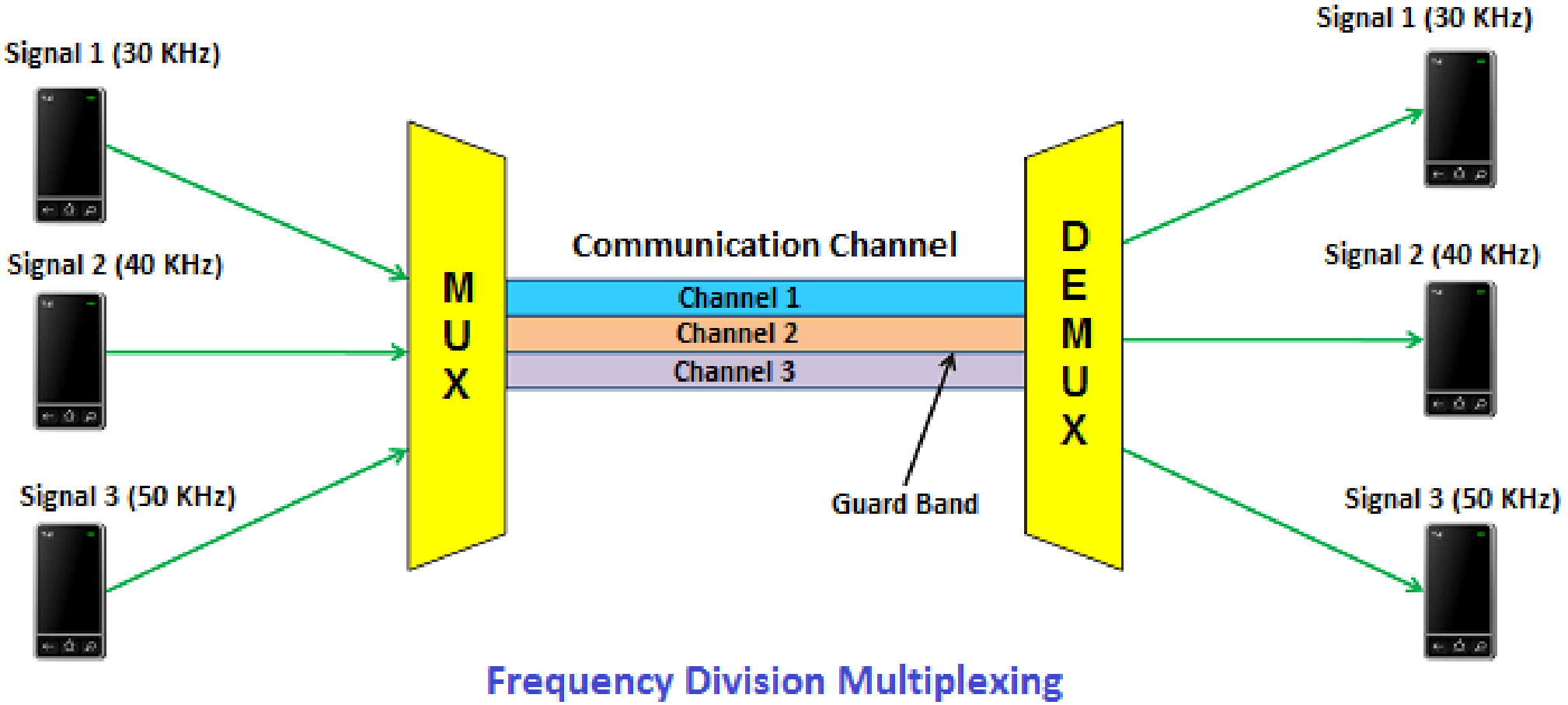


Start frame	Input 1	Input 2	Input 3	Input 4	Input 5	Input 6	Input 7	Input 8	Stop frame

Transmitted frame

Frequency Division Multiplexing (FDM)

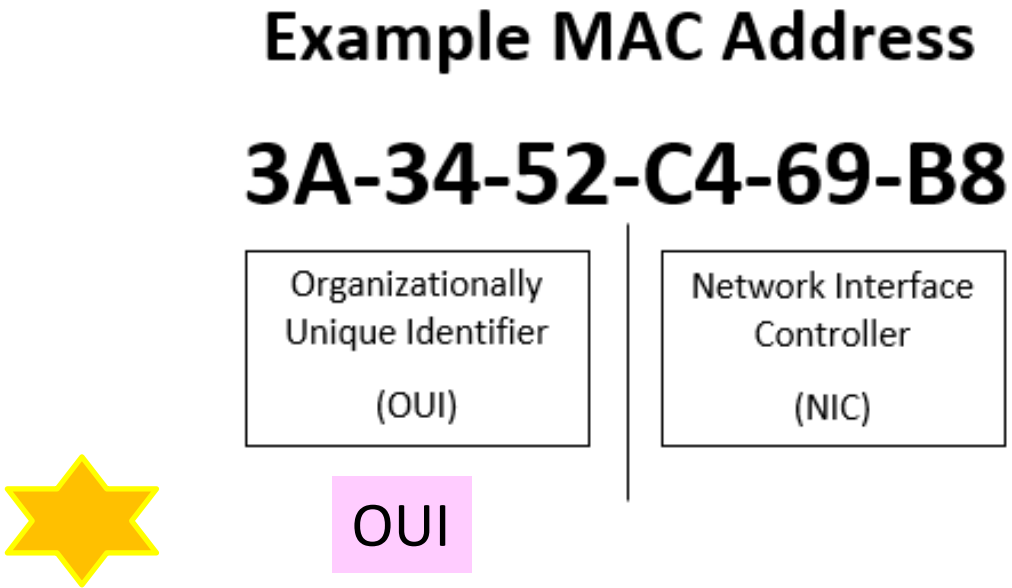
1G



Frequency Division Multiplexing

Addressing in Data Link Layer

- Data Link Layer usually concerns with **local area networks** or physical point-to-point data communication
- In order to identify a communication partner, an address is needed. We call this address as "MAC Address" (Media Access Control Address)
- The MAC address is a **48-bit number** (12 numbers of **hex** or base-16)



MAC Address

- The MAC address was originally designed to identify a specific hardware interface.
- Today, we can spoof or change the MAC address of our interface in order to prevent other people to learn about our hardware manufacturer and identify ourselves (privacy issue).
  - Apple iPhone and some Android phones change the device's MAC address every time it makes a new connection.
- The MAC address is normally **used internally within a local area network**.
- The switch handles the forwarding of messages inside a LAN using MAC addresses.

Using MAC Address

- For Example, you want to change settings (IP address association(DHCP Reservation), network speed limit/ QoS, external access, port forwarding/virtual server, blacklist/access control, firewall, etc.) of devices in the network.



- References:
  - ✓ https://www.downloads.netgear.com/files/GDC/WNR2000V4/WNR2000v4_UM_14Mar2014.pdf
  - ✓ https://www.columbia.edu/acis/networks/aptnet/netgear/index.html

How to Find my MAC Address?

- We can find our MAC address by looking up network settings of hardware devices such as
  - ✓ https://carleton.ca/its/help-centre/how-to-find-your-mac-address/

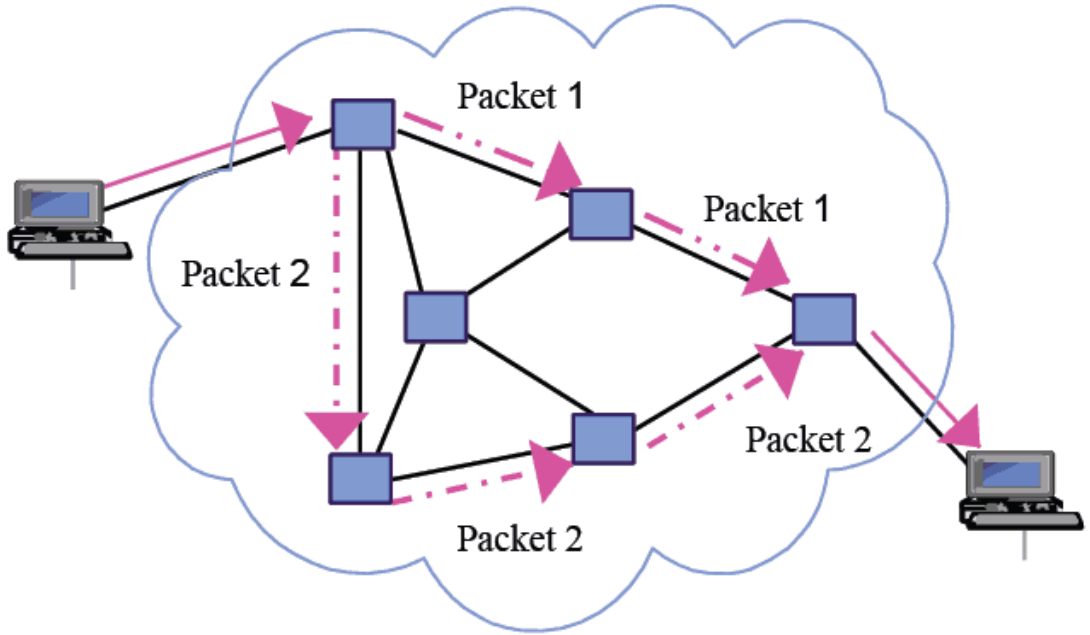
- Windows' Cmd → type **`ipconfig /all`**
  - ✓ Then, find "**Physical Address**" of your Ethernet adapter
    - Use the first 3 groups (six alphabets)

MAC Address Lookup (Block Large)
https://regauth.standards.ieee.org/standards-ra-web/pub/view.html#registries
https://standards-oui.ieee.org/
Wireshark's OUI lookup
https://www.wireshark.org/tools/oui-lookup
Registration Authority
https://standards.ieee.org/products-programs/regauth/

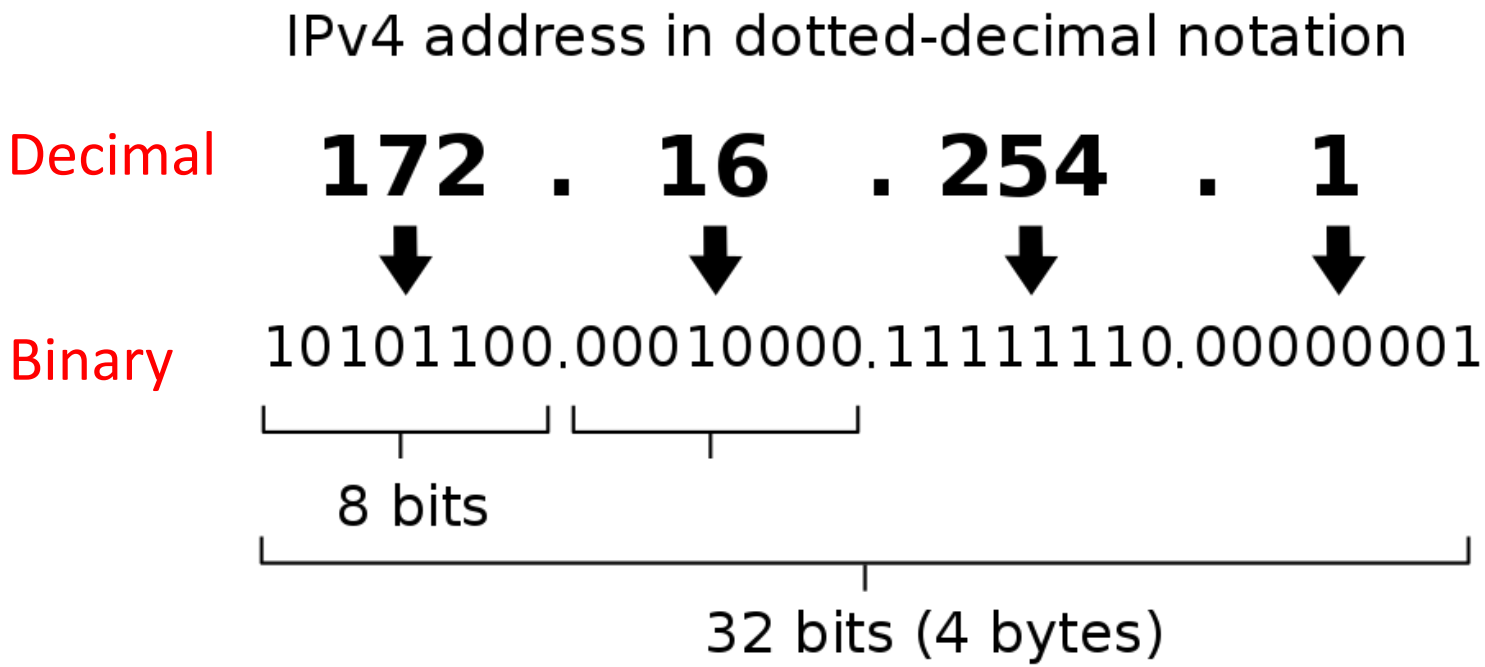
IP Address

- IP address (Internet Protocol address) is a **32-bit** number (IPv4) or **128-bit** number (IPv6) identifying a machine in the Internet.
  - How many bytes?

- There are sophisticated network routing algorithms that can be applied to the IP addresses in order to allow routers to efficiently route packets in the Internet by reading some information in the IP address (just like how the postal service read a zip-code).

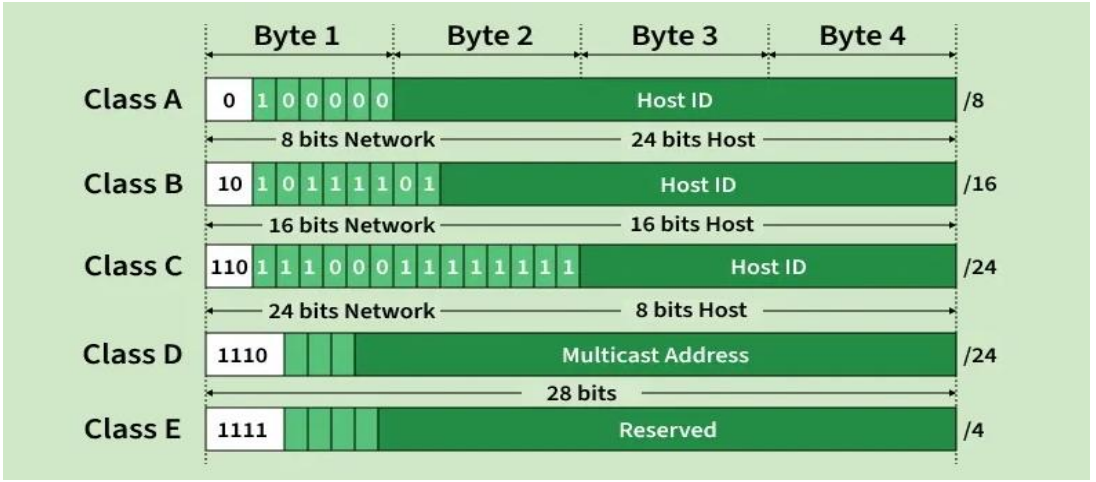


IP Address


IPv4 address in dotted-decimal notation
Decimal
172 . 16 . 254 . 1
Binary
10101100.00010000.11111110.00000001
8 bits
32 bits (4 bytes)

IP Address Classes

CLASS	LEADING BITS	START ADDRESS	END ADDRESS
CLASS A	0	0.0.0.0	127.255.255.255
CLASS B	10	128.0.0.0	191.255.255.255
CLASS C	110	192.0.0.0	223.255.255.255
CLASS D	1110	224.0.0.0	239.255.255.255
CLASS E	1111	240.0.0.0	255.255.255.255



A private IP address is used just within an organization

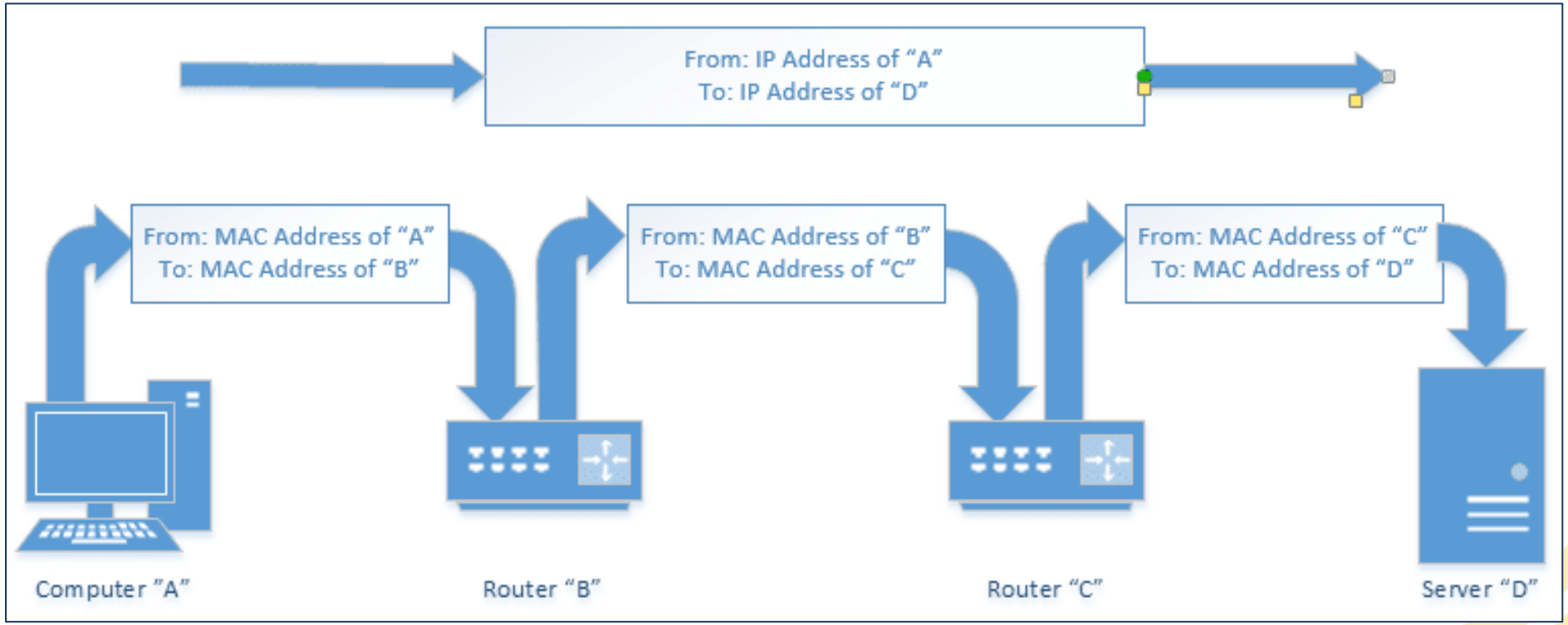
Private IP address space	
From	**To**
10.0.0.0	10.255.255.255
172.16.0.0	172.31.255.255
192.168.0.0	192.168.255.255

References:

- ✓ https://rangespeenshi.blogspot.com/2017/06/what-are-private-ip-address-ranges.html
- ✓ https://www.geeksforgeeks.org/introduction-of-classful-ip-addressing/

MAC Address and IP Address

- Use **MAC address** to route and forward **frames**
- Use **IP address** to route and forward **packets**
- Source and Destination IP addresses on the headers are not change, but the MAC addresses will be changed!



ipconfig command

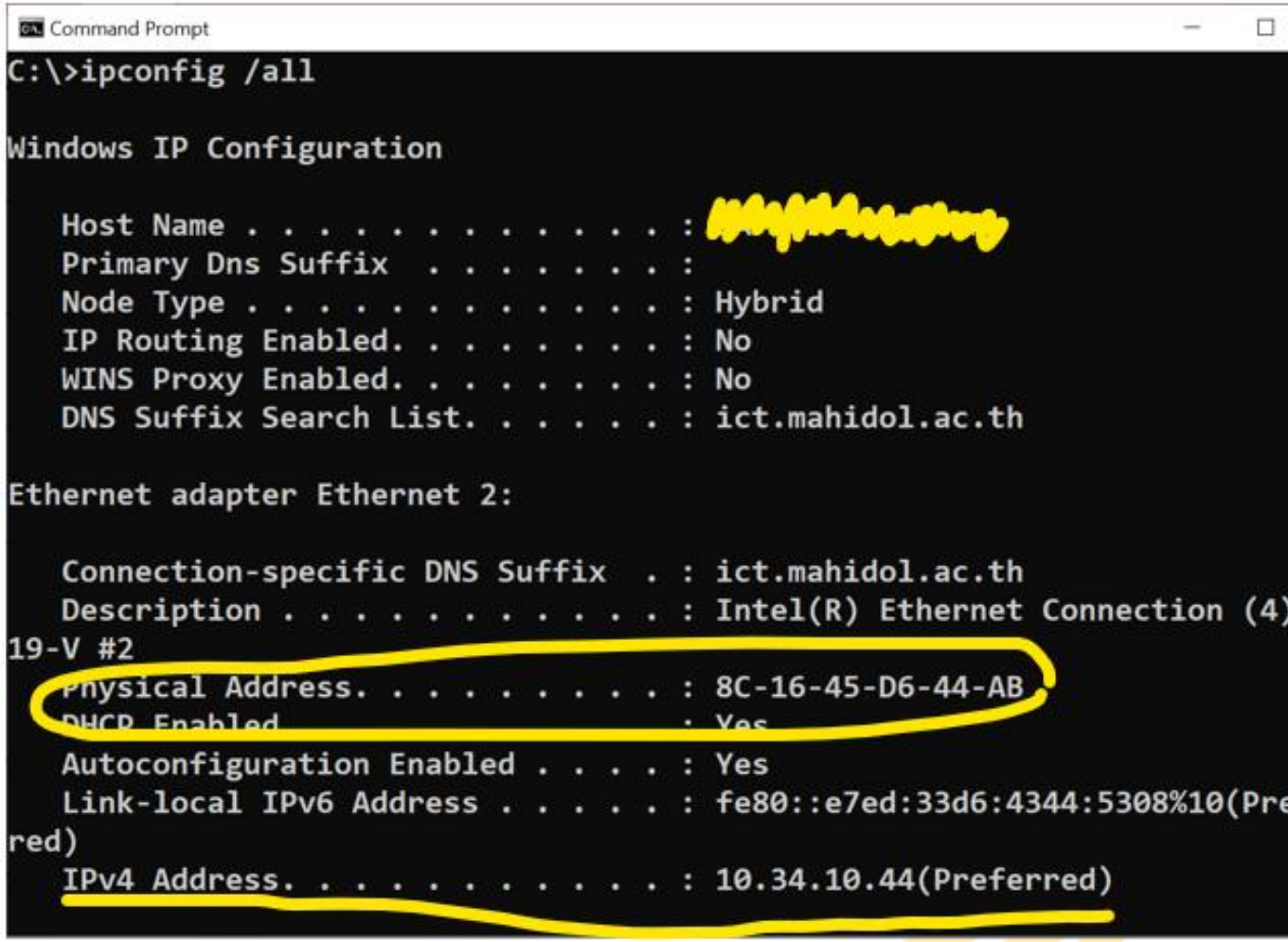
- To display the usage,

C:\> ipconfig /?

- Two common commands: display full configuration information

C:\> ipconfig

C:\> ipconfig /all



ifconfig command

- This command is similar to the ipconfig command, but running on Unix (Linux, Mac)
- Open a Terminal on Unix
- Type, ifconfig or ifconfig -a (-a means all)



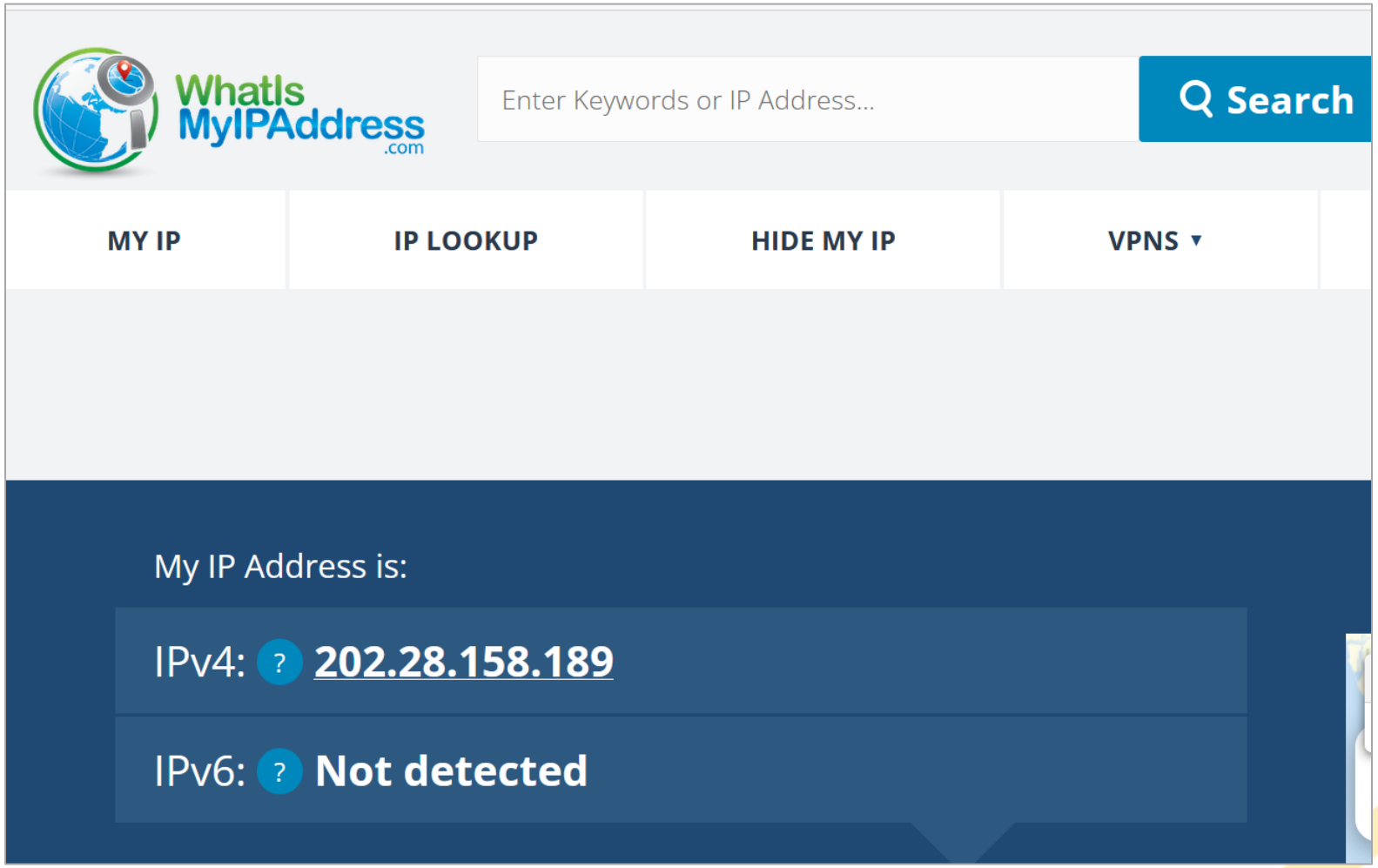
- To show the manual, use the command 'man ifconfig'
  - Enter to see the next line
  - Push the space bar to see the next page
  - Ctrl + C to stop the man command

ifconfig command

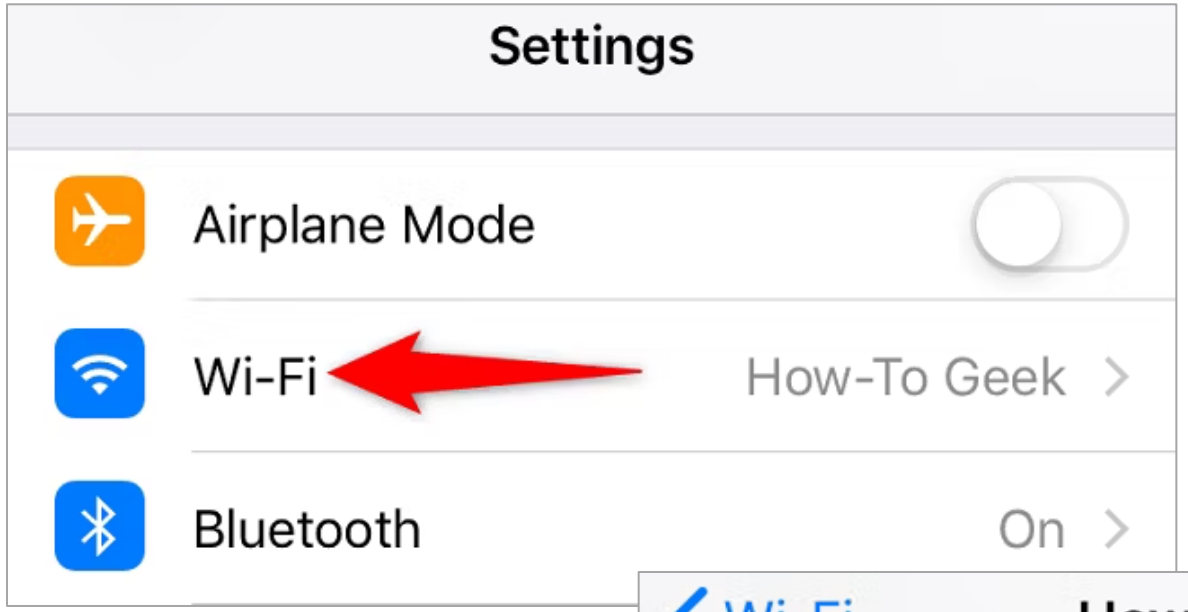


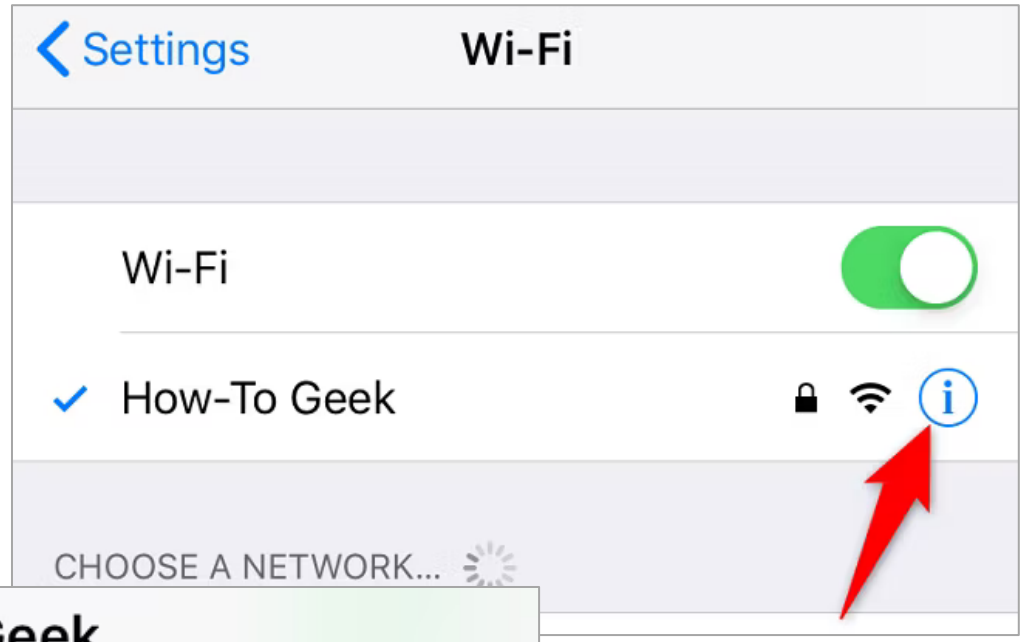
```
File  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ ifconfig -a
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.12  netmask 255.255.255.0  broadcast 10.0
        inet6 fe80::f8b7:502e:d1bb:4588  prefixlen 64  scopei
        ether 08:00:27:2c:3c:ea  txqueuelen 1000  (Ethernet)
        RX packets 1  bytes 590 (590.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 29  bytes 4654 (4.5 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisi
```

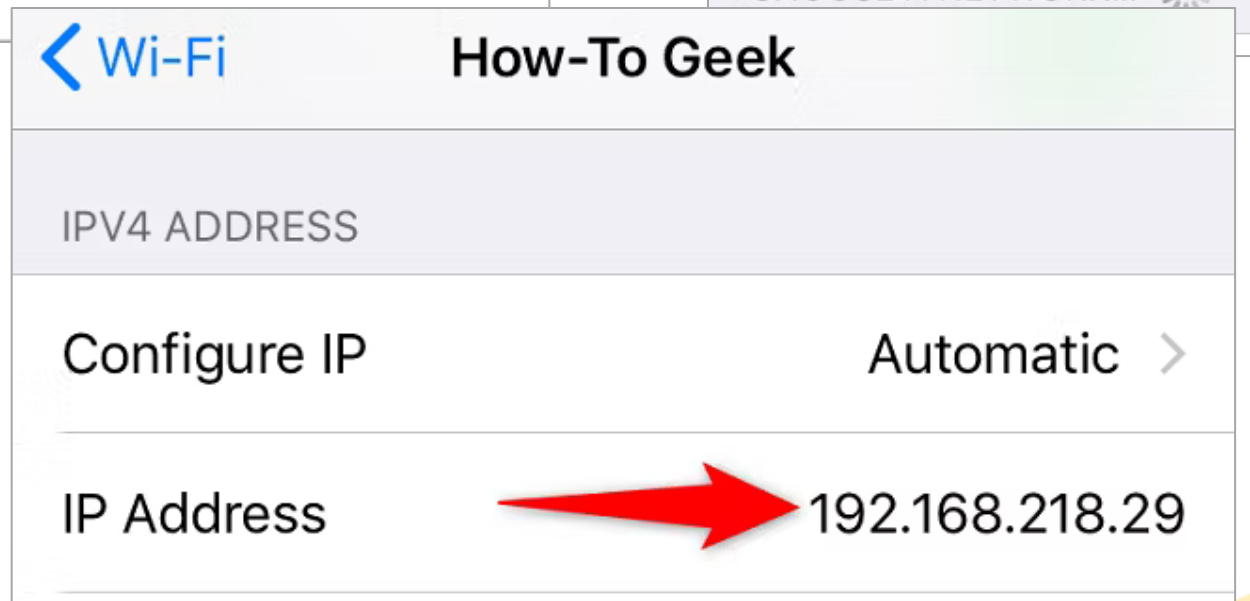
https://whatismyipaddress.com/



iphone's IP Address

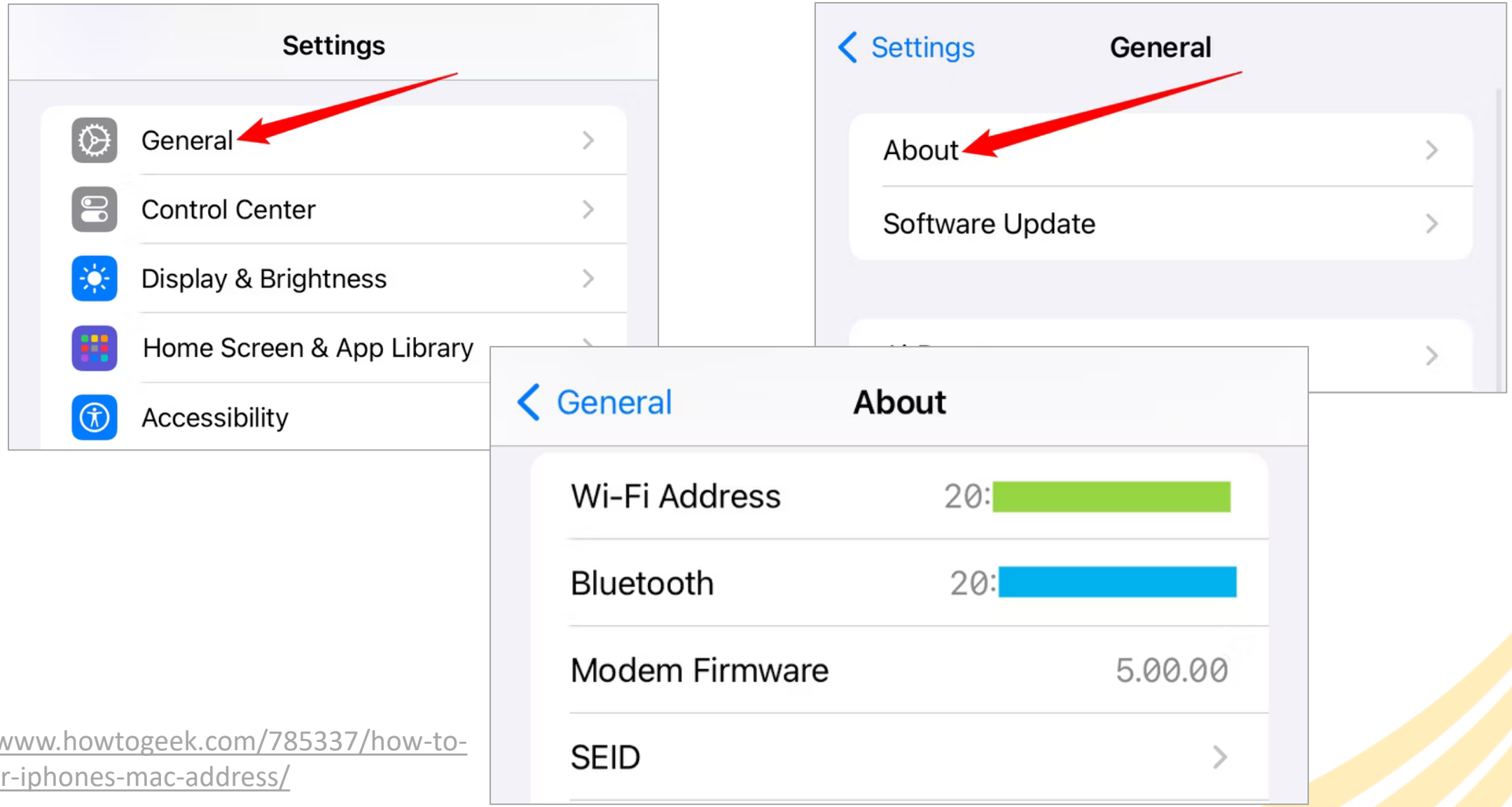






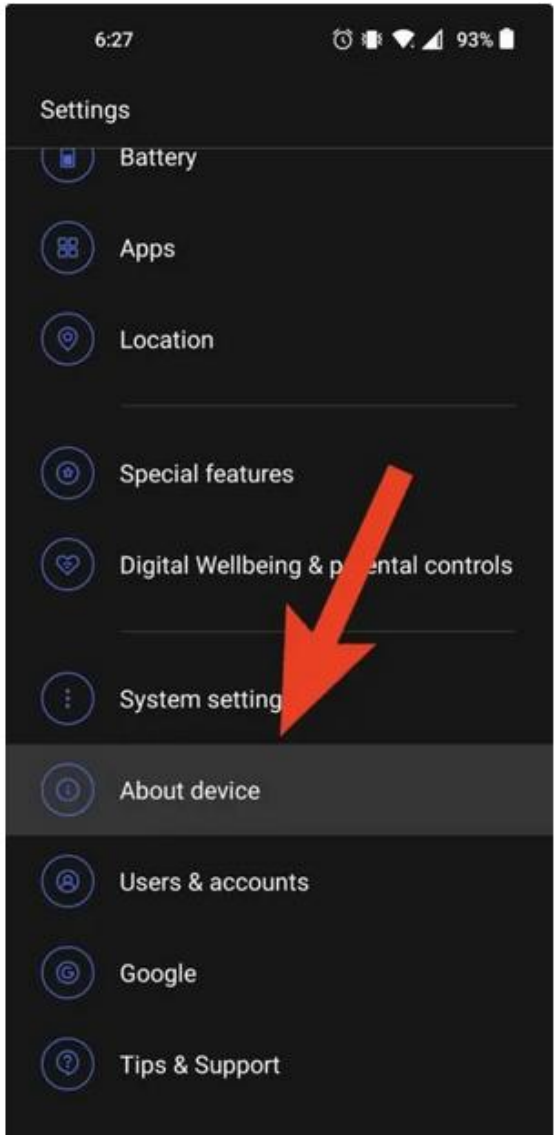
https://www.howtogeek.com/796854/iphone-ip-address/

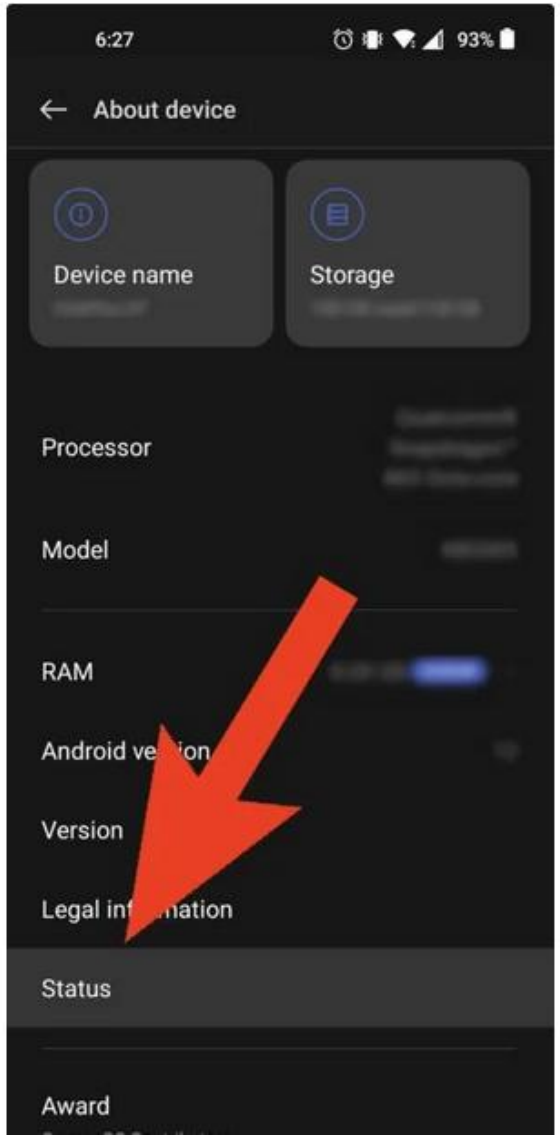
iphone's MAC Address



https://www.howtogeek.com/785337/how-to-find-your-iphones-mac-address/

Android's IP and MAC Address







https://www.androidauthority.com/find-mac-address-android-3207022/

Summary of Today Keywords (1st Half)

- Layered Network
  - Modularity
  - Divide-and-conquer approach
- OSI model: 7 layers
- TCP/IP model or Internet model: 4 or 5 layers
- Function of each layer
- Multiplexing
- MAC address
- IP address
- ipconfig /all or ifconfig -a

Let's do the exercise!



Let’s do the exercise!



Warriors of the Net by Ericsson's Media Lab



Cisco Packet Encapsulation
https://youtu.be/ntVii_EJc6s

https://youtu.be/PBWhzz_Gn10

Today Agenda

- Advantages of Layered Data Communication
- Functions of each layer
- Addressing
  - Physical layer address
  - MAC address and IP address

- - Break -

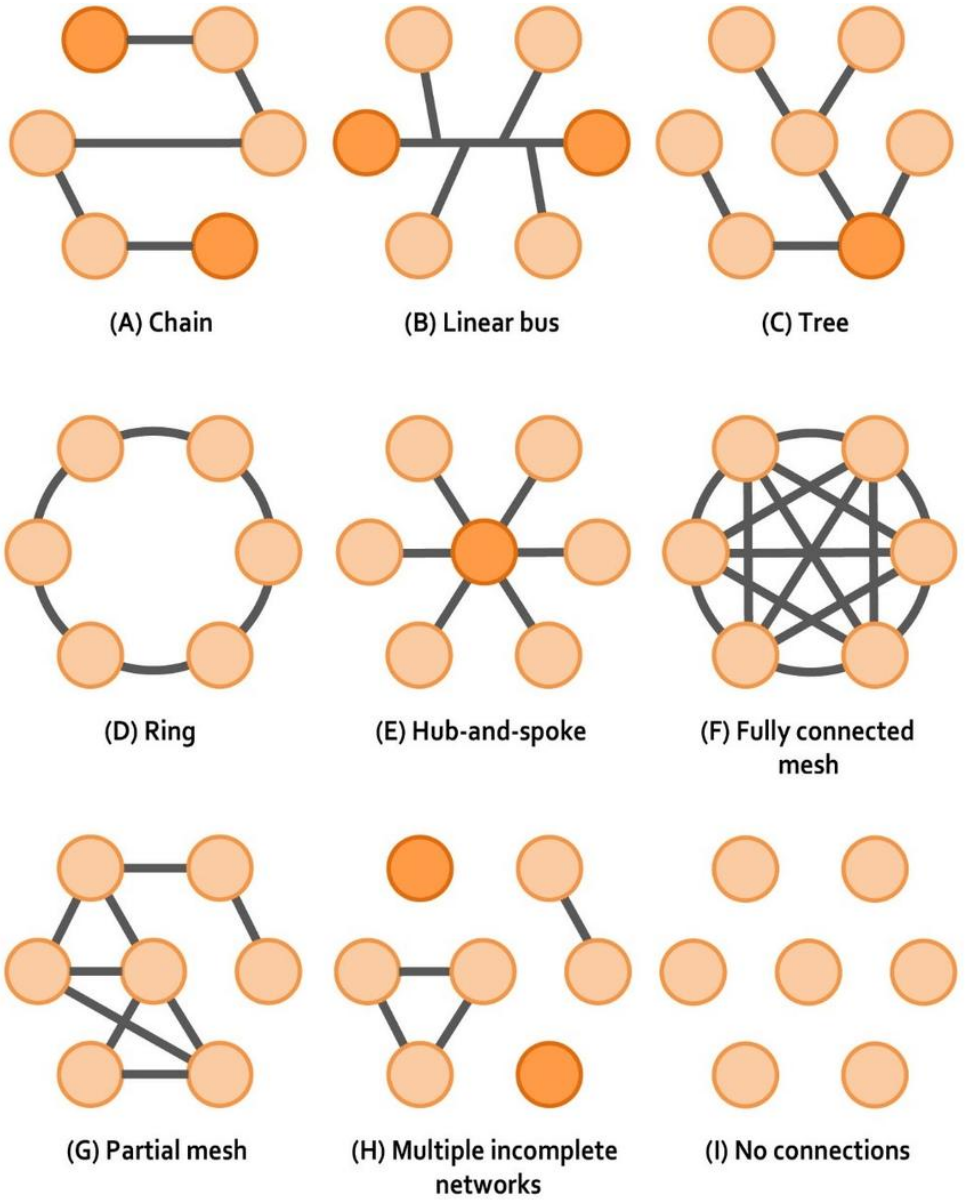
- Network Topology
- Data and Signal
- Equation of Sine wave signal
  - Phase, Frequency, Amplitude

Network Topology

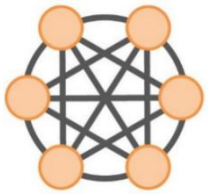
Topology:

- Network topology is the arrangement of the various elements (like computers, routers, and switches) of a communication network.

- Different way of connectivity

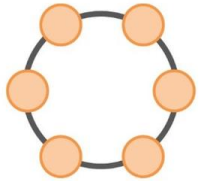


Mesh Topology



- In mesh topology each device is connected to every other device on the network through a dedicated point-to-point link. Lets say we have n devices in the network then each device must be connected with (n-1) devices of the network. Number of links in a mesh topology of n devices would be n(n-1)/2.
- Advantages
  - No data traffic issues as there is a **dedicated link** between two devices which means the link is only available for those two devices.
  - Mesh topology is **reliable** and **robust** as failure of one link doesn't affect other links and the communication between other devices on the network.
  - Mesh topology is **secure** because there is a point to point link thus unauthorized access is not possible.
  - **Fault detection** is easy.
- Disadvantages
  - **Amount of wires** required to connected each system is tedious and **headache**.
  - Since each device needs to be connected with other devices, number of I/O ports required must be huge.
  - **Scalability issues** because a device cannot be connected with large number of devices with a dedicated point to point link.

Ring Topology

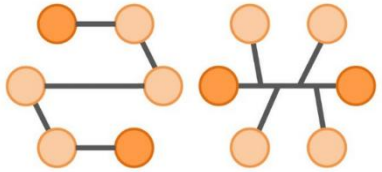


- The transmission of data takes place by **token** and data travels in one direction.
- Every node is connected to two nodes on each side. Nodes are connected in a way that make a **close loop**, hence the name **ring** topology.

- Advantages
  - Network server doesn't need to control the connectivity among workstations.
  - Information **flows in one direction**. Data flow and connectivity efficiency is smooth resulting in **decreased collision**.
  - Every system has **equal access** to resources every time.

- Disadvantages
  - Information and data is passed through many intermediate nodes (**slow**).
  - If any one node goes down, then **whole network will collapse**.
  - The whole network is connected through wire connection so network is completely dependent on these connections.

Bus Topology



- All computers or servers are connected to **single cable wi**re. Every single workstation sends and receives signals through this single cable; the bus. It is a linear topology.

- Advantages
  - Bus network is **easy to extend** and many other servers and systems can add in.
  - It is very **cheap** in terms of costing.
  - It is **best for small network** setup and effectively works for LAN.

- Disadvantages
  - Whole **network collapses** if a single cable encounters some kind of problem.
  - Since all networks are connected to central cable bus then it becomes **difficult to detect a problem** in specific server or system.
  - Bus topology's efficiency decreases when number or nodes increases in number.
  - **Security is not very good** as all computers receive same signal from the common bus.

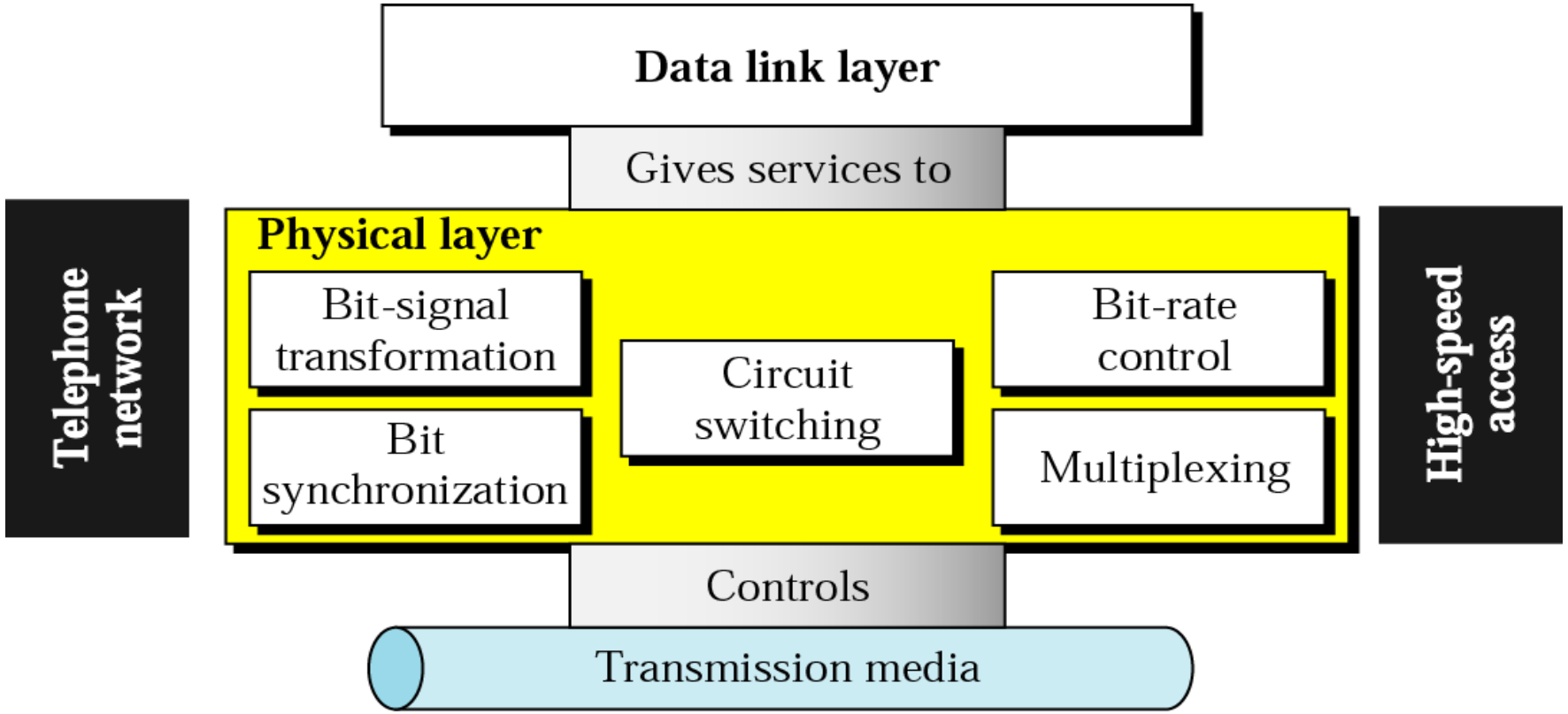
Star Topology



- All computers are connected to **central hub**. In star topology, there is point to point connection as every system and server is connected individually and directly to the device. Every signal computer is controlled and regulated by **the central hub or switch** that enables the hub to act as a connection junction. It can work as repeater or signal booster as well.
- Advantages
  - New devices can be **connected easily** and any device can be removed from the topology easily.
  - In star, the **central junction controls and monitors all activities**.
  - Since every node is individually and directly connected so problem in **one device cannot affect the rest of the network**.
- Disadvantages
  - Network cost increases with the increased number of devices.
  - **Capacity of central device is crucial** that affects the performance of all the connected devices.
  - In this network, efficiency of all other devices is compromised if the hub junction is not capable of supporting that number of devices

Introduction to Physical Layer

- The physical layer consists of the following components



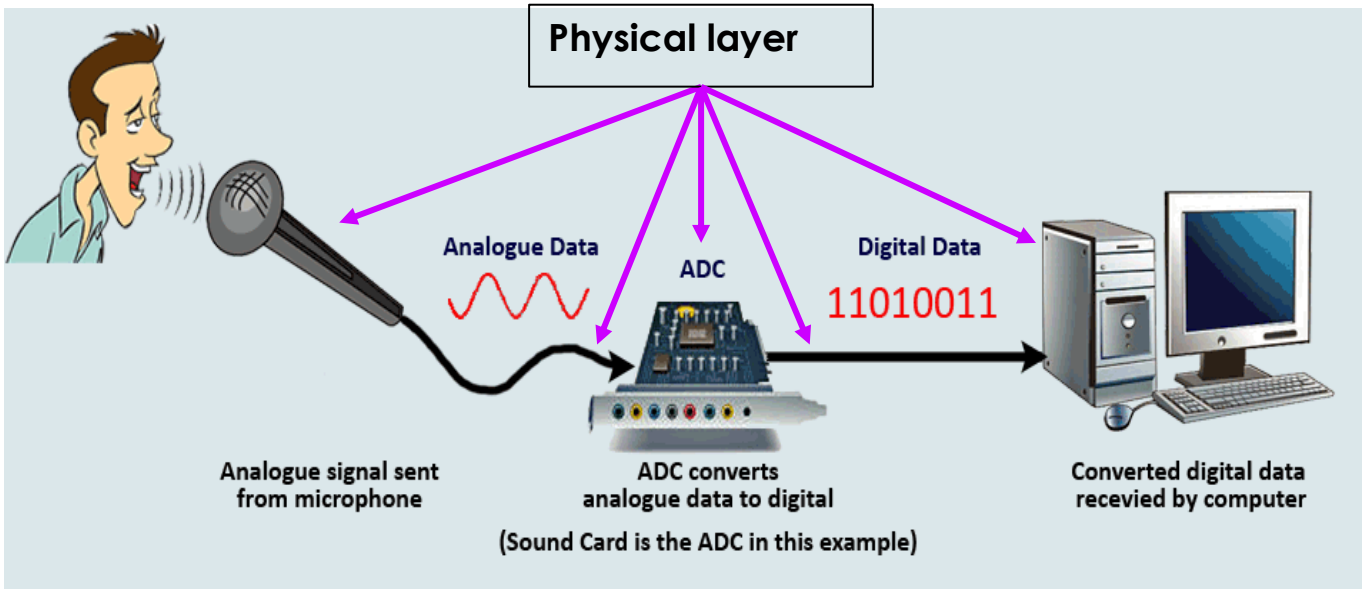
The Physical Layer

Layer #	Layer Name	Protocol	Protocol Data Unit	Addressing
5	Application	HTTP, SMTP, etc…	Messages	n/a
4	Transport	TCP/UDP	Segments/ Datagrams	Port #s
3	Network or Internet	IP	Packets	IP Address
2	Data Link	Ethernet, Wi-Fi	Frames	MAC Address
1	Physical	10 Base T, 802.11	Bits	n/a



Physical Layer

- The physical layer is responsible for carrying individual bits in a frame across the link in the form of signals.
  - For transmission, **data needs to be converted to signals** (e.g., electrical or light signals) which are suitable for transmission in a specific network medium type. The transmission media conducts signal energy along a physical path.
- The physical layer is also concerned with how the medium such as cable and wireless channel is implemented to provide support for efficient data transmission.

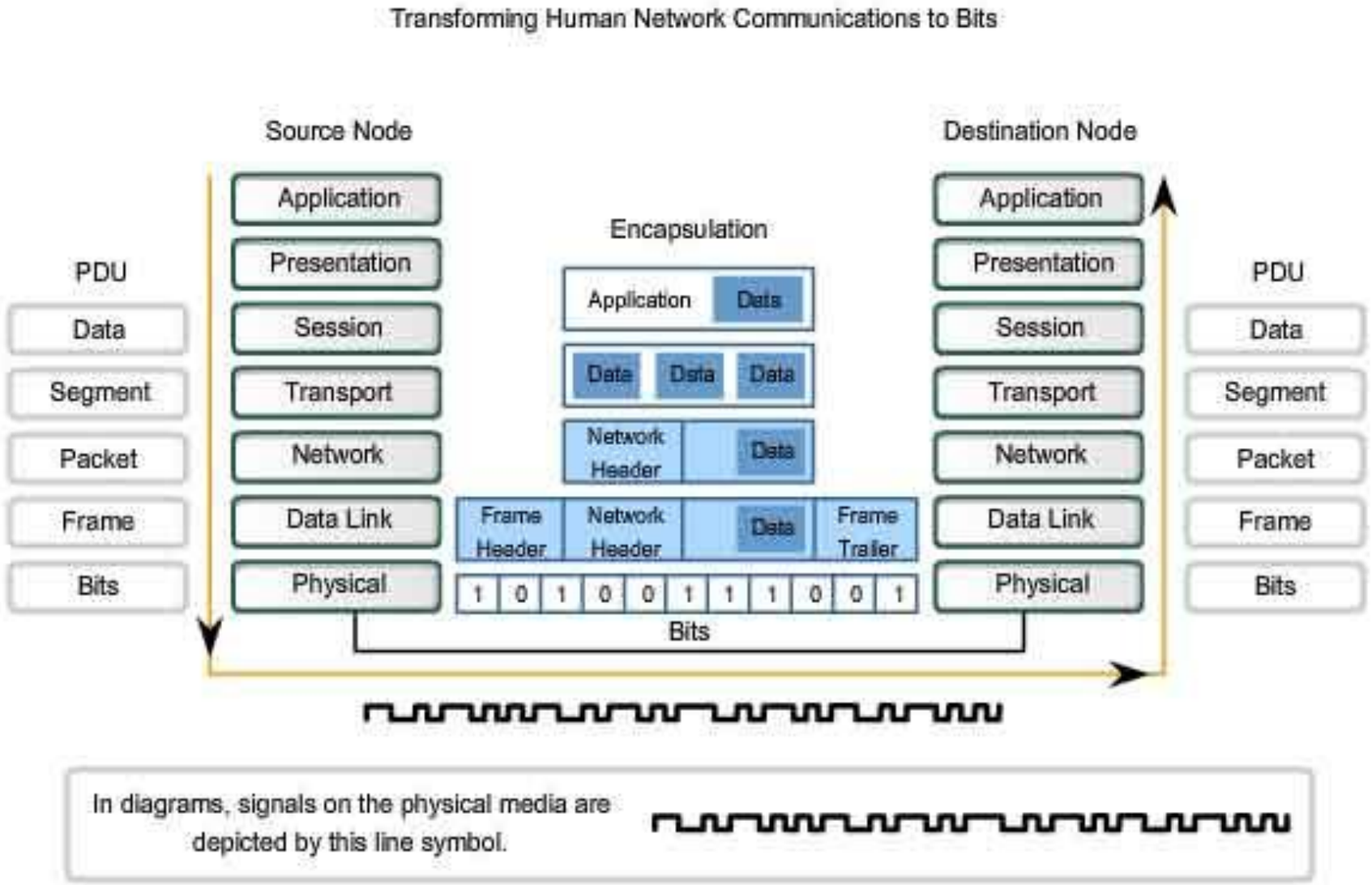


The Physical Layer

- The physical layer consists of the electronic circuit transmission technologies of a network.
- The physical layer defines the means of transmitting raw **bits** over a physical data link connecting network nodes.
- The bitstream may be grouped into code words or symbols and converted to a physical signal that is transmitted over a transmission medium.
- The physical layer provides a standardized electrical, mechanical, and procedural interface to the transmission medium.
- The shapes and properties of the electrical connectors, the frequencies to broadcast on, the line code to use and similar low-level parameters, are specified by the physical layer.

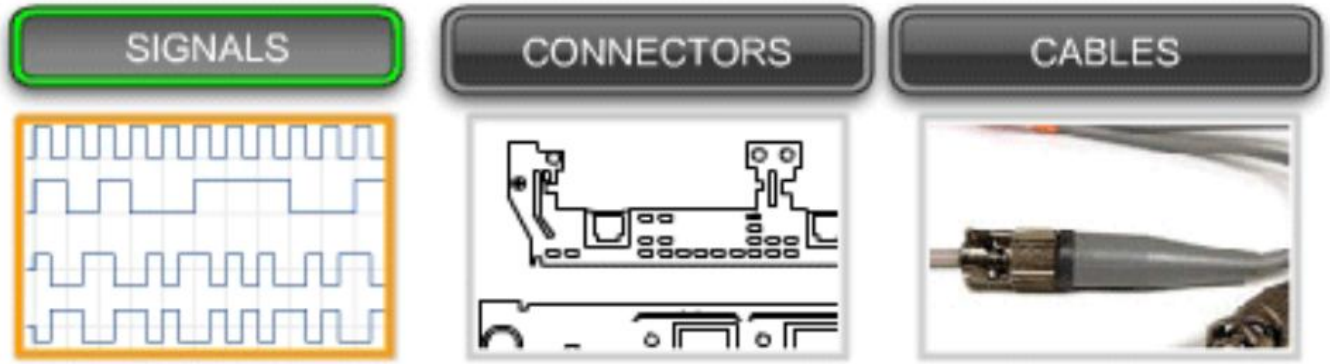


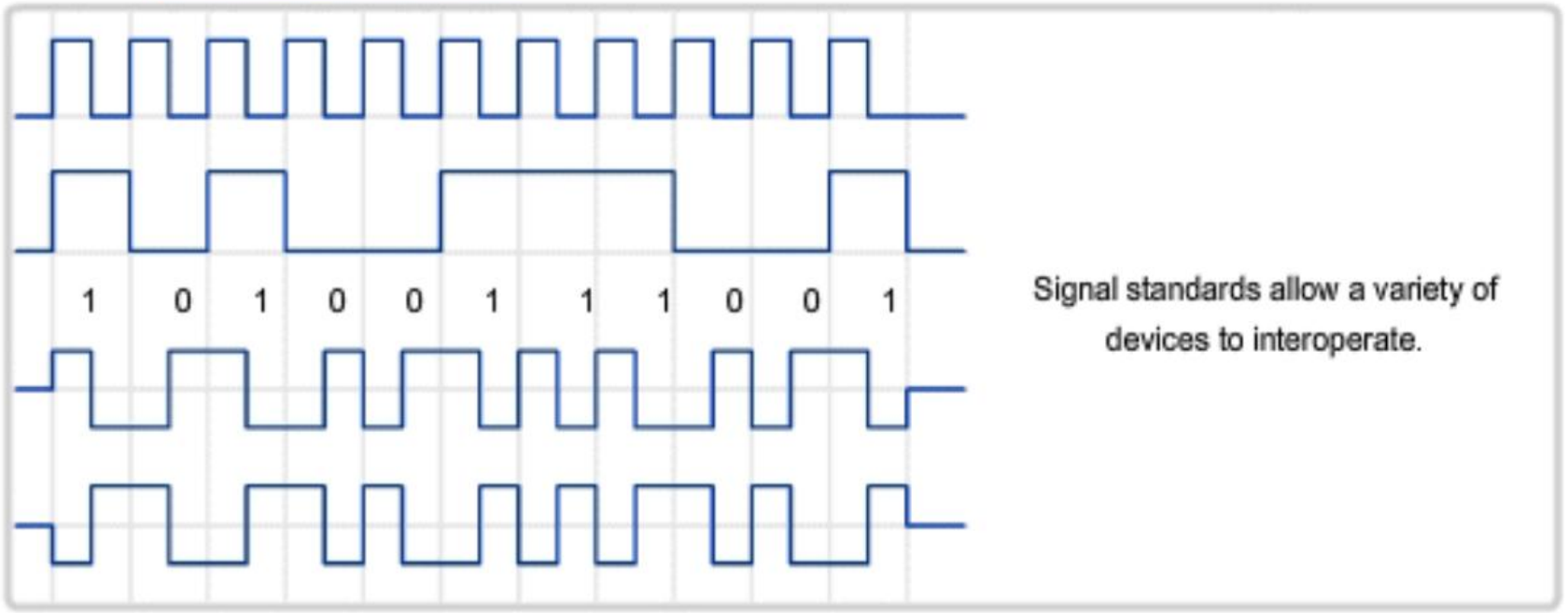
Physical Layer Signaling



Physical Layer Standards

Standards for the Physical layer specify signal, connector, and cabling requirements.

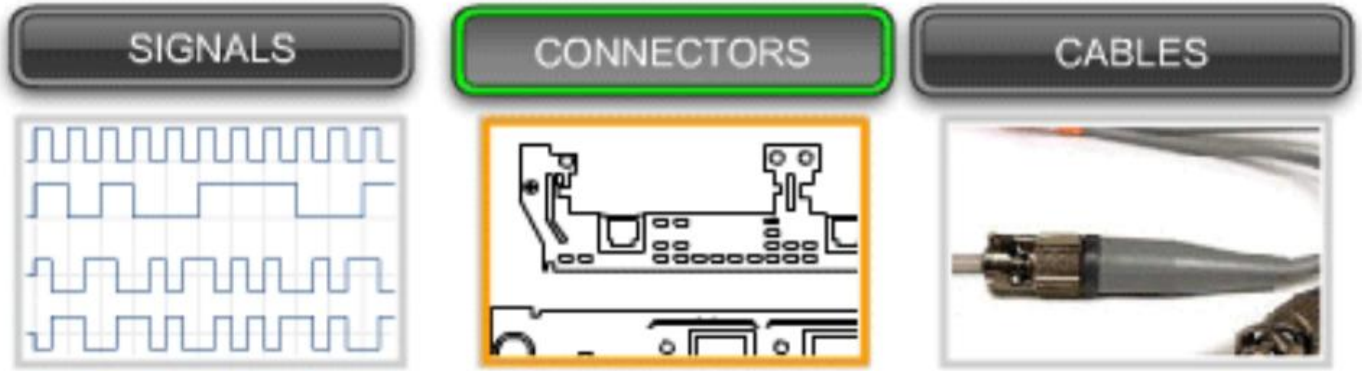


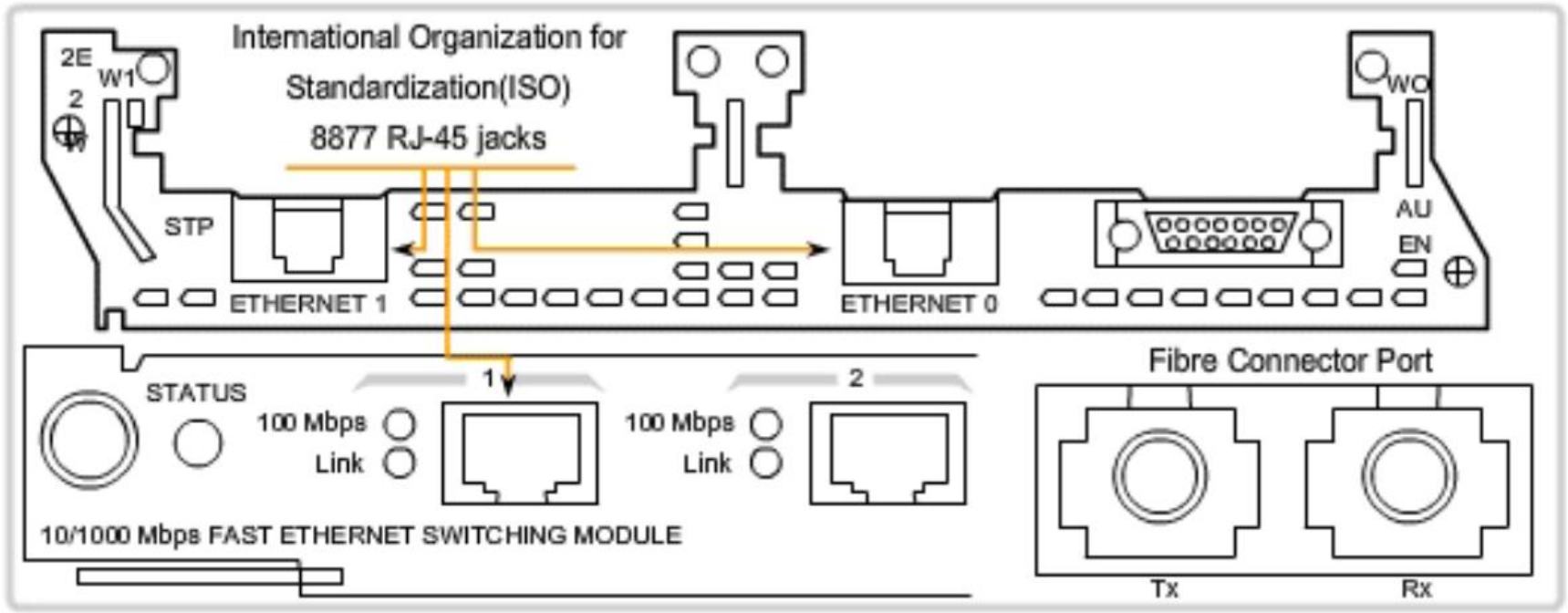


Signal standards allow a variety of devices to interoperate.

Physical Layer Standards

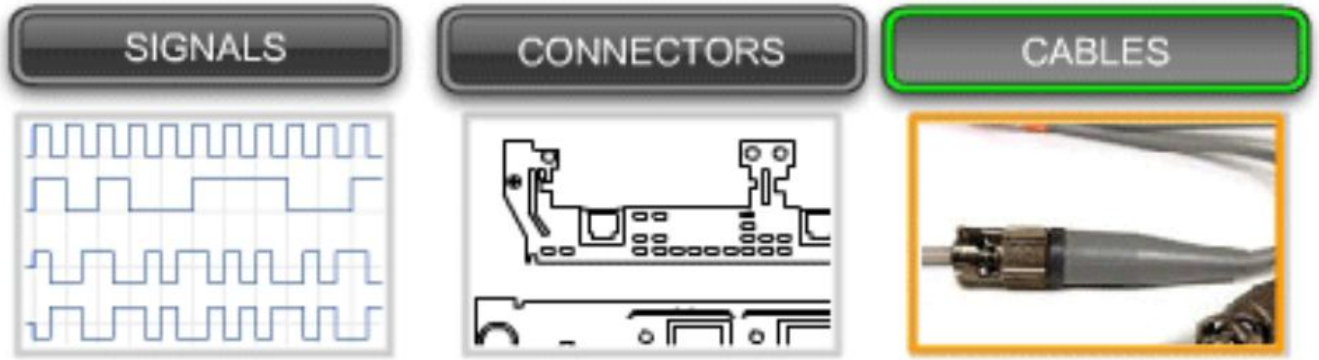
Standards for the Physical layer specify signal, connector, and cabling requirements.



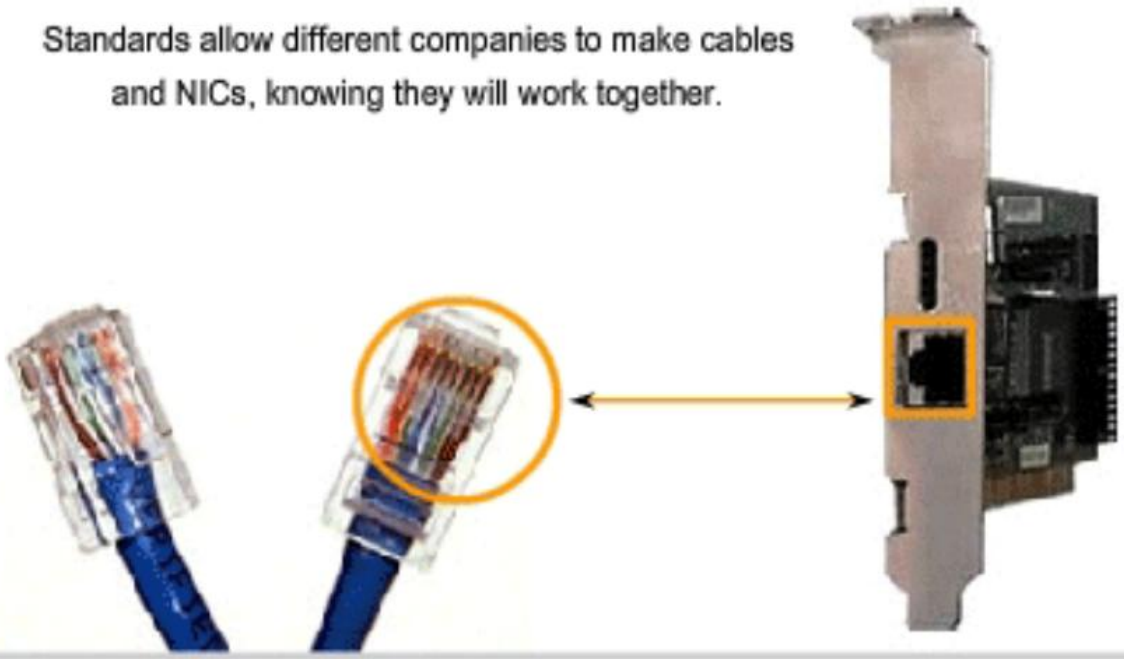


Physical Layer Standards

Standards for the Physical layer specify signal, connector, and cabling requirements.



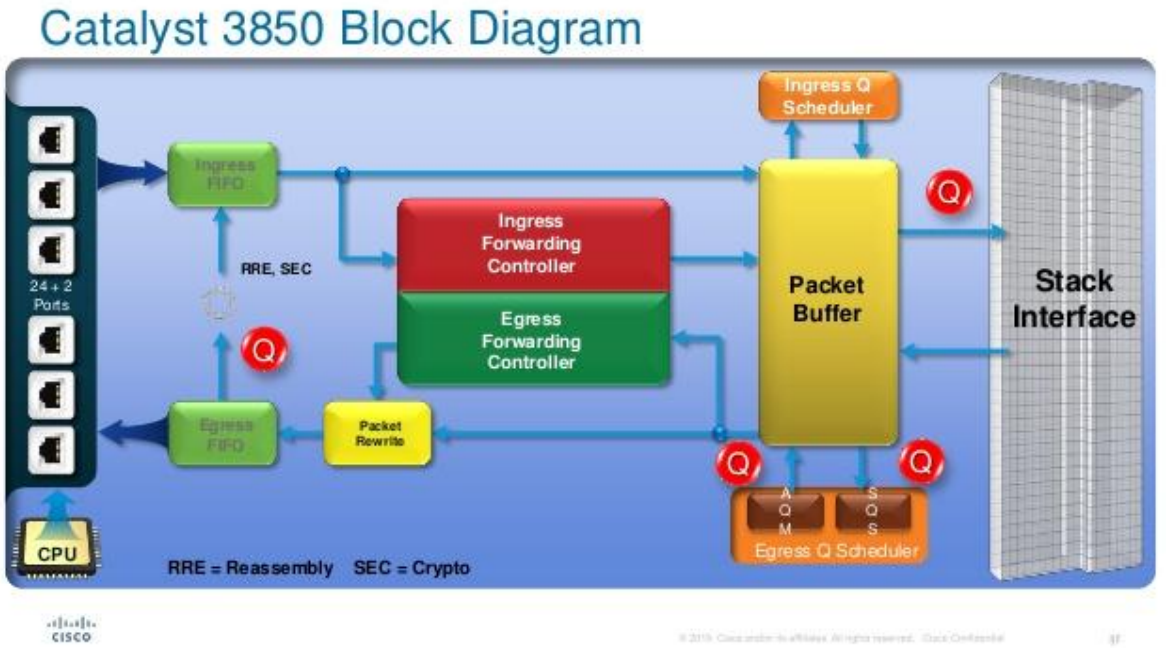
Standards allow different companies to make cables and NICs, knowing they will work together.



Physical Layer Standard

- The standard covers only when the device has to be used with other systems.
- The internal working of the network device such as electronic circuit is not standardized. Manufacturers are free to create their own devices so that they can compete on the performance of their devices.





Summary of Physical Layer Services

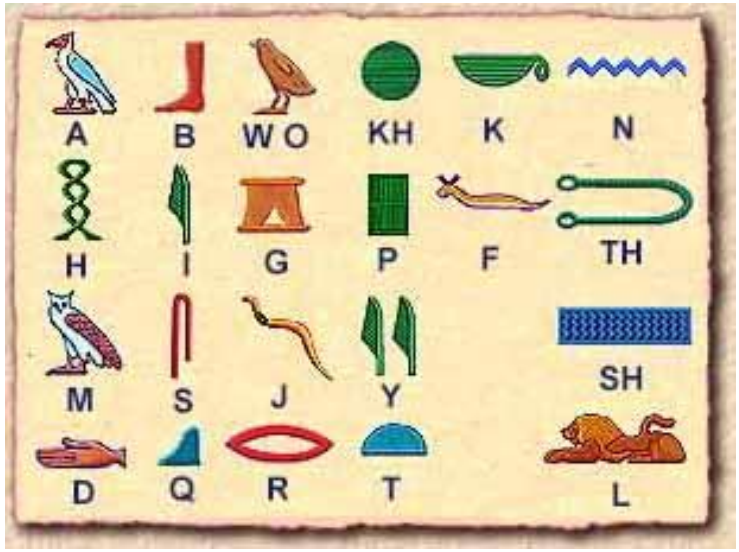
- Bit-by-bit or symbol-to-symbol delivery
- Standardized transmission medium including
  - **maximum cable length**, an electrical specification of transmission line signal level and impedance
  - **medium characteristics**: **electromagnetic** compatibility including electromagnetic spectrum frequency allocation and specification of signal strength, analog bandwidth
  - **the mechanical specification** of electrical **connectors** and **cables**

- Standardized interfacing to the transmission medium
  - **Signaling protocol** (start-stop signaling, serial communication)
  - **Line coding** (digital data to digital signal conversion)
  - **Pulse Code Modulation (PCM)** (analog data to digital signal conversion)
  - **Digital Modulation** (digital data to analog signal conversion)
  - **Analog Modulation** (analog data to analog signal conversion)



Data and Symbols

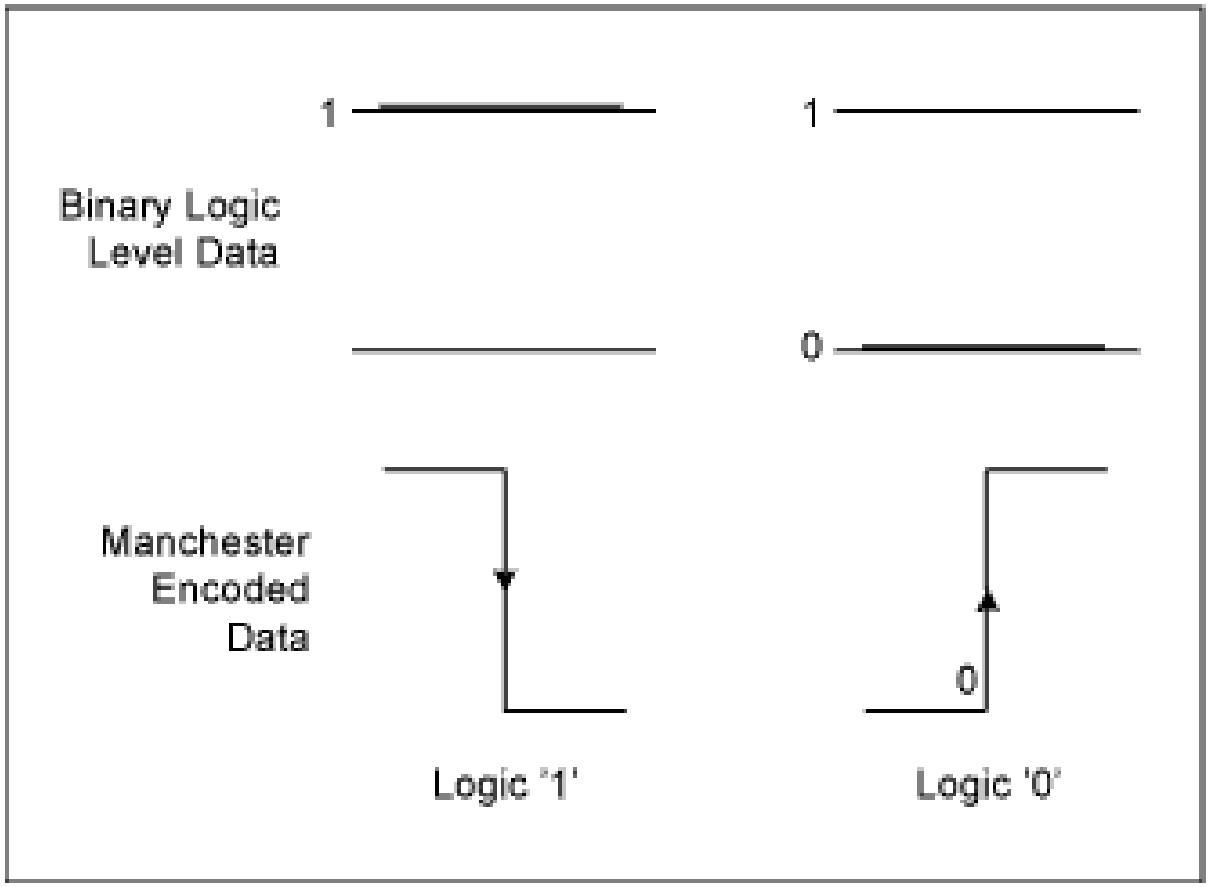
- **Data** is an abstract term which carry **meaning** (according to our understanding).
- A **symbol** can refer to anything we would like to represent.
- A symbol is a representation of information.





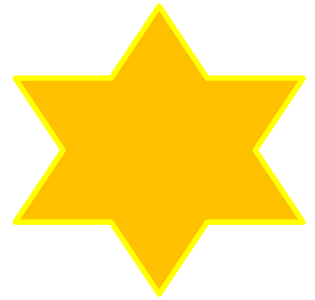
Electrical Symbols

- For electrical signal, the symbol is the pattern of voltage since it is what we can measure.
- Example:



Digital Signals

Data and Signals



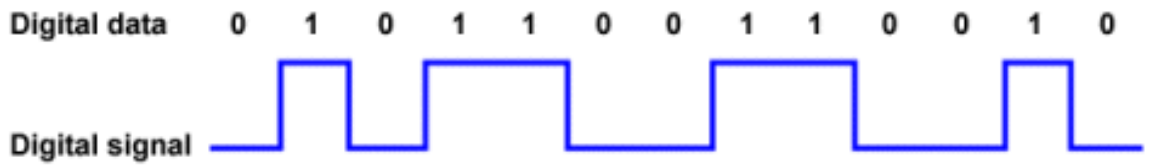
- There are two types of data and signals depending on our interpretation
  - Digital
  - Analog

- Data: data are a representation of information or works.
  - Digital data are data that are discrete, discontinuous. Whole numbers and language letters are commonly used as digital data representations.
  - Analog data are data that are **continuous**. Information in the physical world are often represented as analog data since the measured values are continuous.

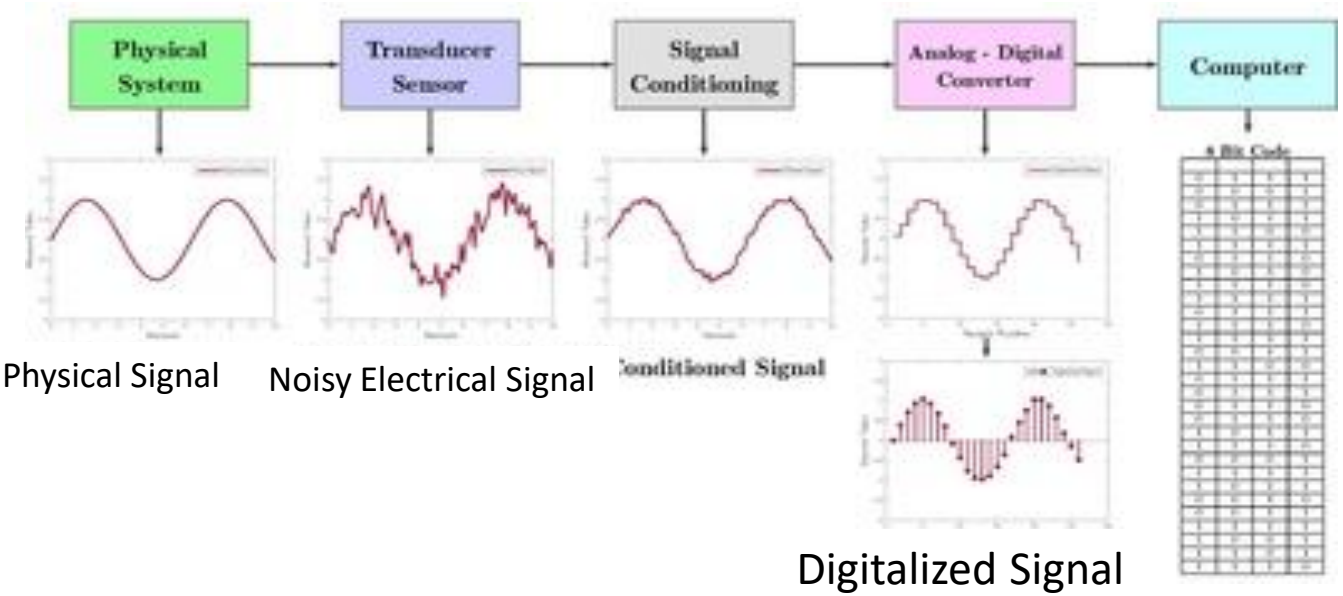
- Signal: signal is the data or information converted into an electrical or optical form suitable for transmission.
  - Digital signals are discrete or step-wise variation of electrical or optical voltage or current over time.
  - Analog signals are continuous variation of electrical or optical voltage or current over time.

Data vs. Signal

- Data is what we understand. Data conveys information. Information conveys knowledge. Knowledge leads to wisdom.
- Signal is what we can measure. *It is often continuous*.
- We usually interpret signal into data.



Digital data to digital signal

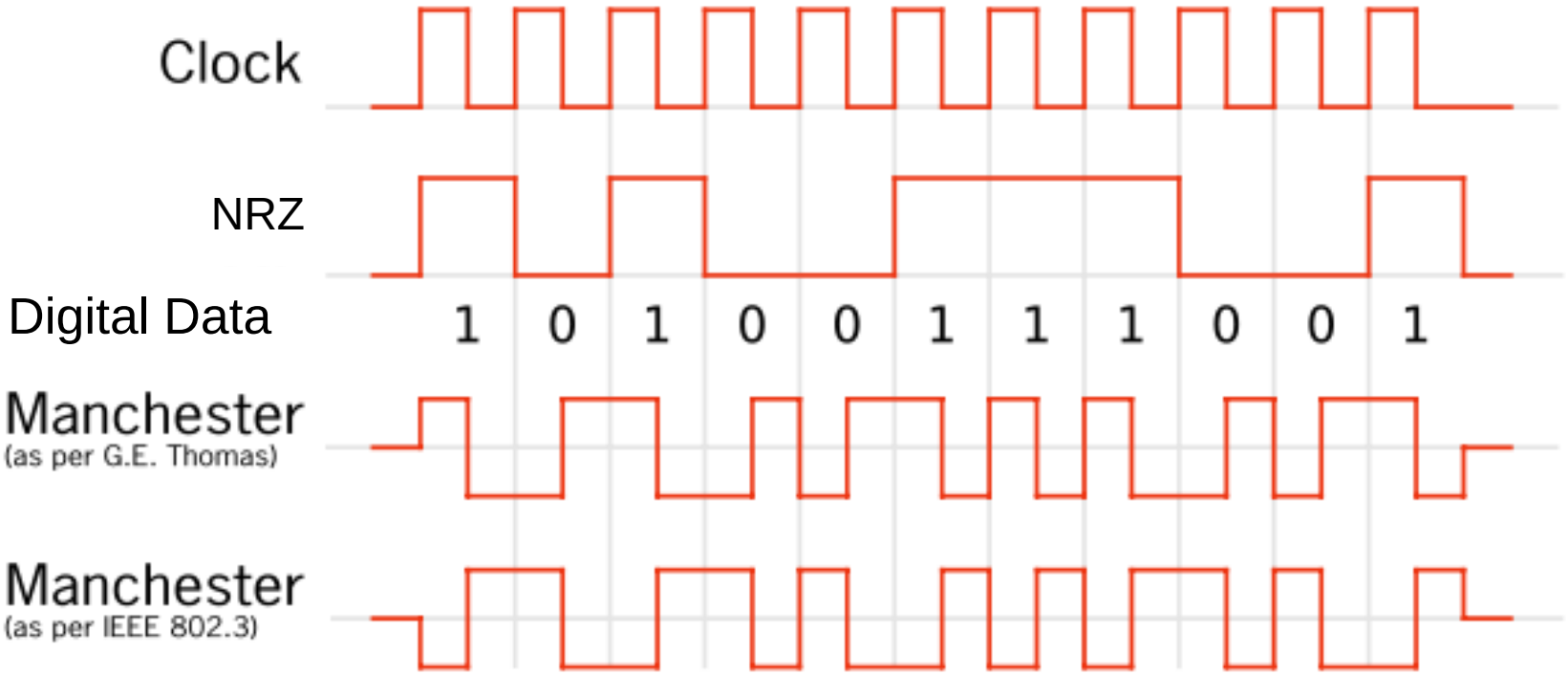




Analog data to digital data

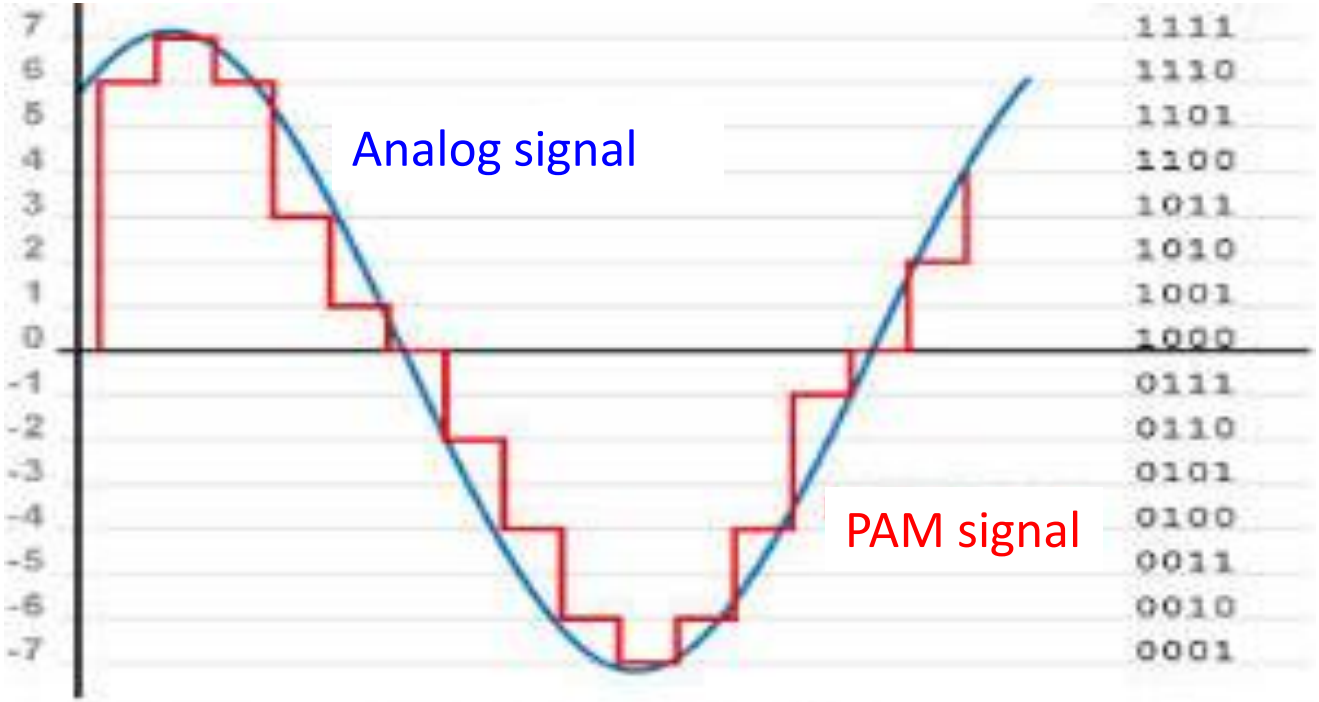
Data to Signal Conversion

- Most symbols are designed to have equal width. So, they have to be read against clock signals (human standard criteria to measure signals and retrieve information)
- Data can be represented as signal in many patterns (symbols).
- Connected symbols form semantic or vocabulary.
- Example of digital data to digital signal



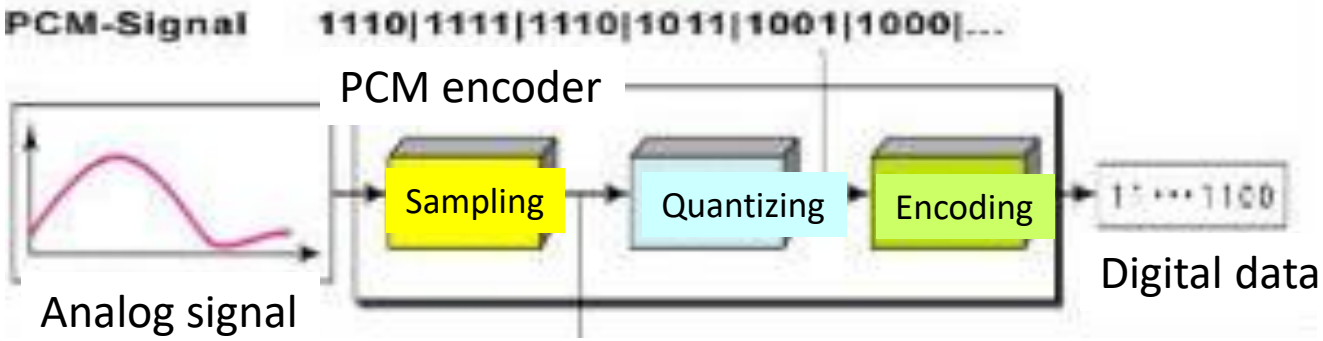
Data to Signal Conversion

- We can also convert analog signal to digital data. We will study this system later.



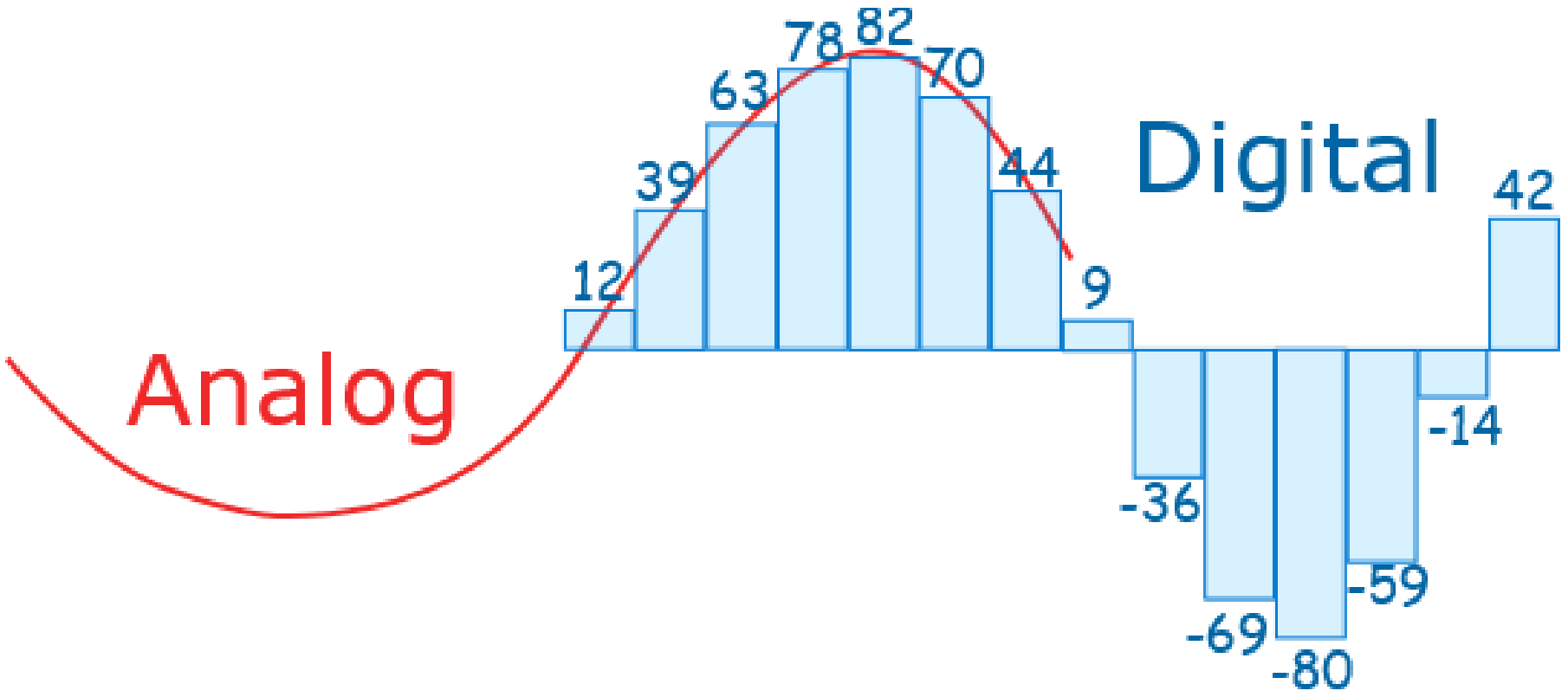


MP3 bit rate



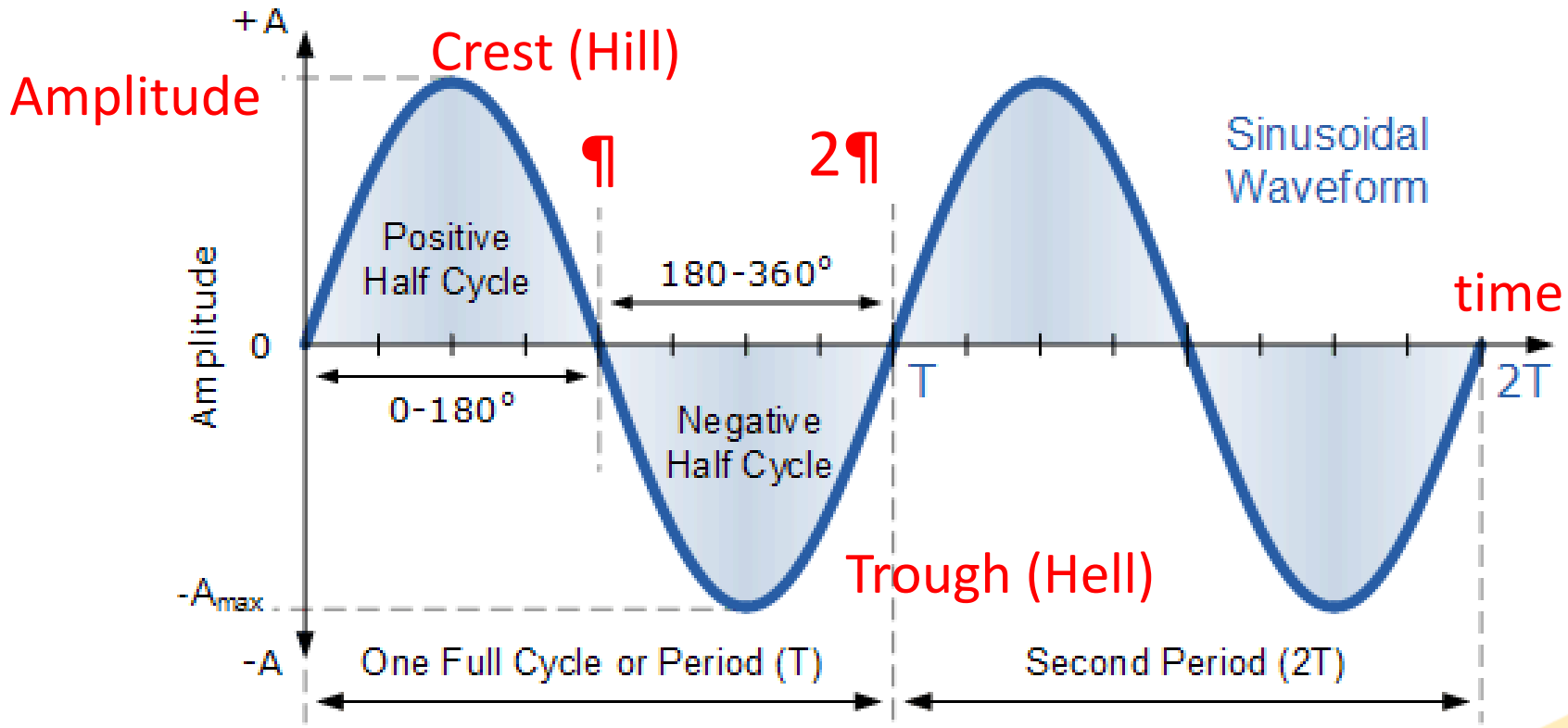
Data to Signal Conversion

- Analog data to digital signal to digital data



Signal

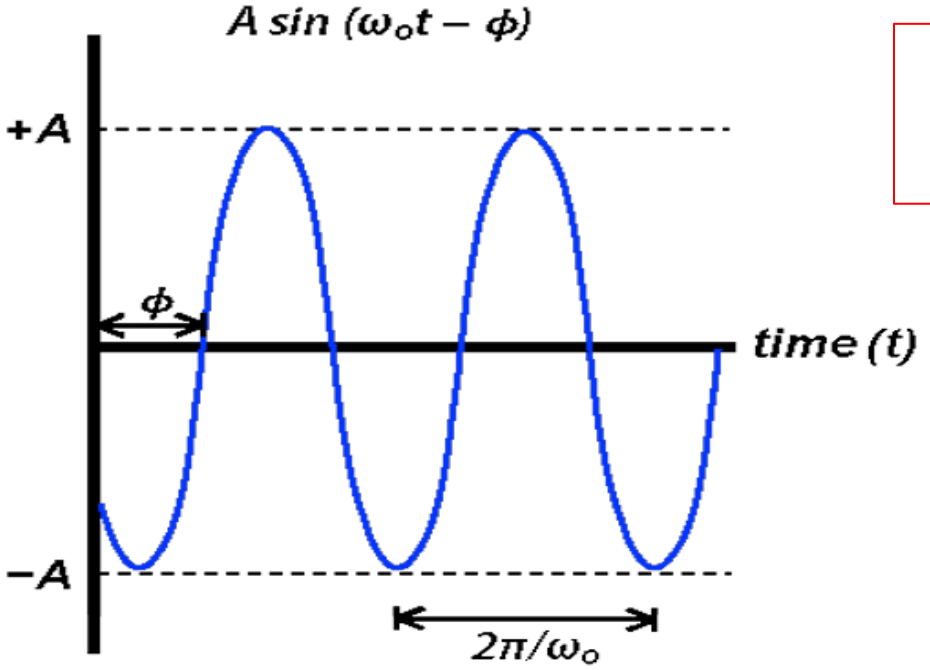
- A signal is a movement of energy in a medium
- **Sine wave** is considered to be the most natural and fundamental wave that represent an analog signal.
- **All types of signals can be constructed based on sine waves.**



Sine Wave

$y(t) = 10 \sin(4t + ¶)$

- Simple **properties** of sine waves are as follows.
  - The amplitude (A) of a sine wave represents the amount of energy that the wave has.
  - The frequency (f) of the wave represents the speed at which the wave changes its form over a period of time (the number of times the signal repeats itself in one second or cycles per second)
  - The period (T) is amount of time it takes the signal to complete one cycle $T = 1/f$
  - The phase (p) describes the difference in timing between two otherwise similar signals (usually compare against signal that starts at the origin going upward)



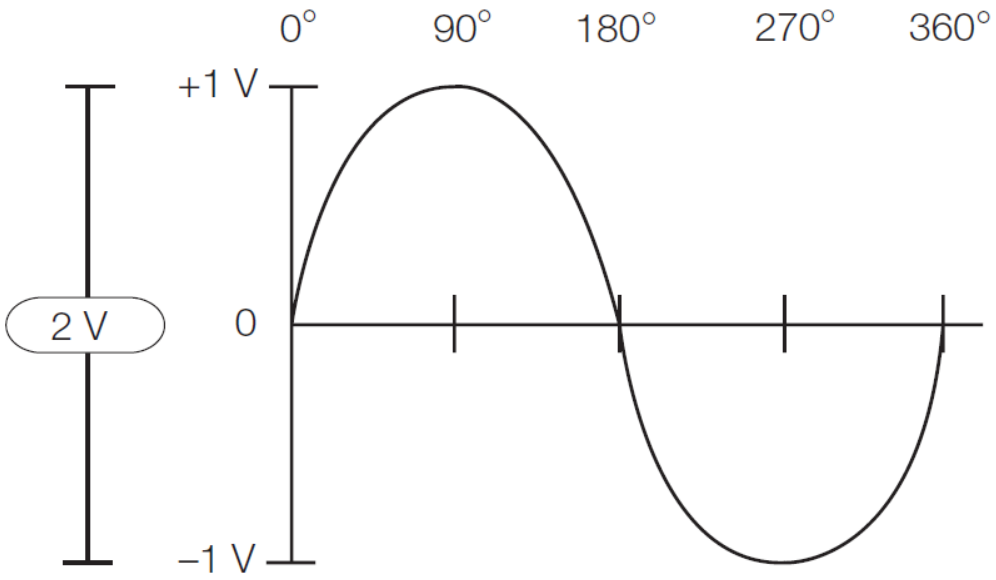
$$y(t) = A\sin(2\pi ft + \varphi) = A\sin(\omega t + \varphi)$$

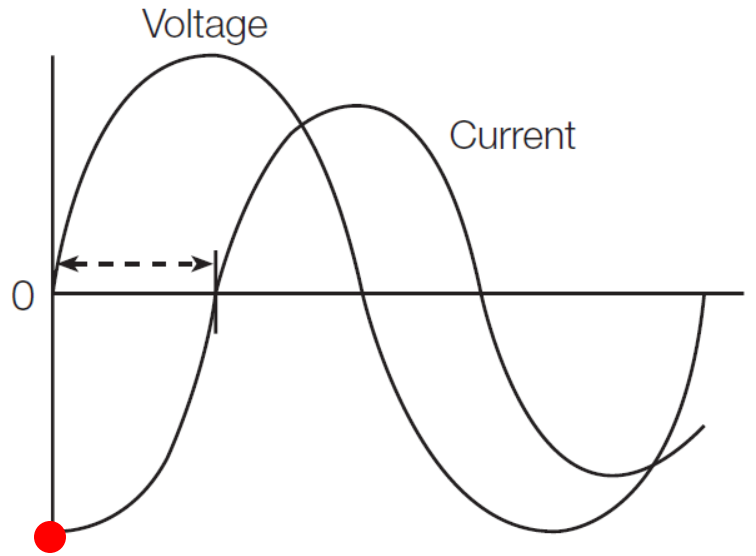
$2¶f = \omega \rightarrow f = \omega/2¶$

https://en.wikipedia.org/wiki/Sine_wave

Sine Wave

90 degree = ¶/2
180 degree = ¶
270 degree = 3¶/2
360 degree = 2¶





Phase = 270 degree or 3¶/2

Sine Wave

```
import numpy as np
import matplotlib.pyplot as plt
import math

# Assign frequency and phase
freq = 1
phase = 0

# Visualize y(t)
fig = plt.figure(figsize=(10, 6))

t = np.arange(-2, 6, 0.05)

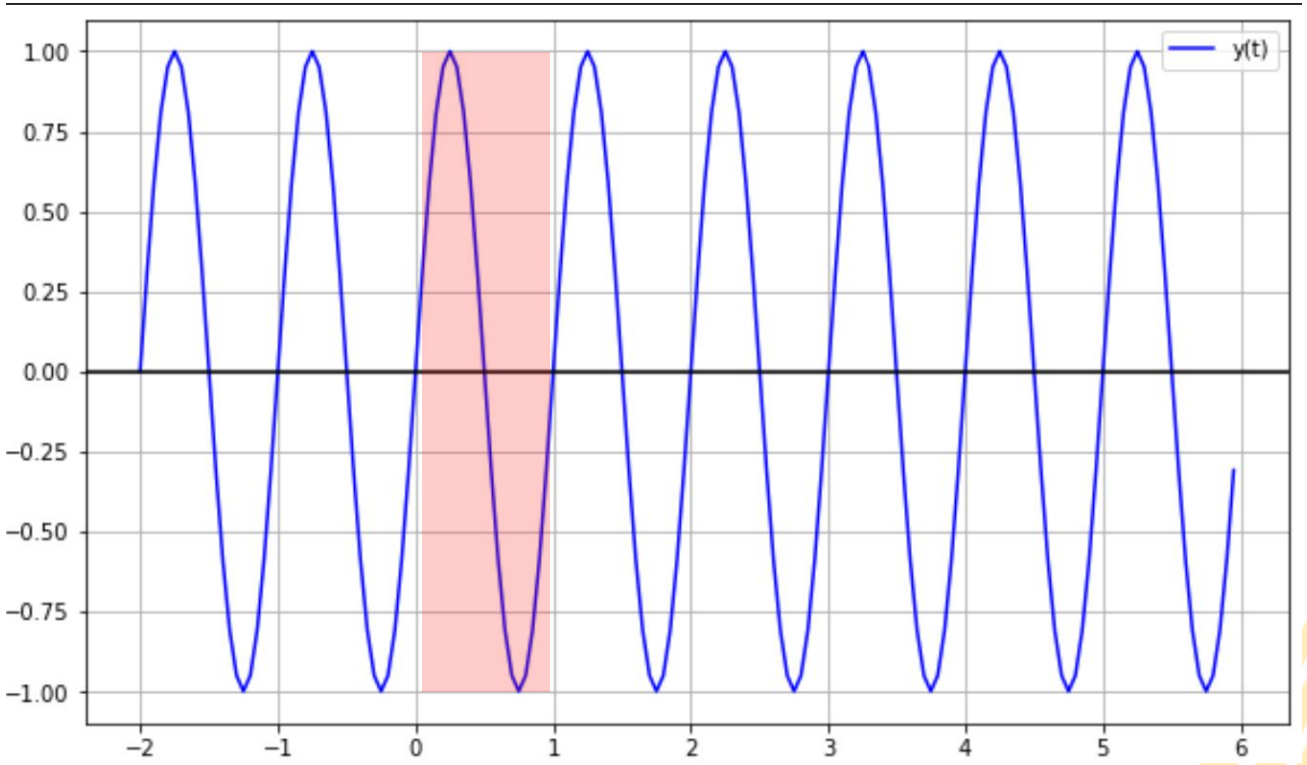
y = np.sin(2 * math.pi * freq * t + phase)

plt.plot(t, y, label='y(t)')
plt.grid(True, which='both')
plt.axhline(y=0)
plt.xlabel('Time (t)')
plt.ylabel('Amplitude')
plt.title('Sine Wave')
plt.legend()

plt.show()
```

https://matplotlib.codeutility.io/

$y = \sin(2\pi f t + \Phi)$ $f = 1\ (Hz), \Phi = 0$



Sine Wave

```
import numpy as np
import matplotlib.pyplot as plt
import math

# Assign frequency and phase
freq1 = 1
freq2 = 0.5
freq3 = 0.25
phase = 0

# Visualize signals
fig = plt.figure(figsize=(10, 6))

t = np.arange(-2, 6, 0.05)

y1 = np.sin(2 * math.pi * freq1 * t + phase)
y2 = np.sin(2 * math.pi * freq2 * t + phase)
y3 = np.sin(2 * math.pi * freq3 * t + phase)

plt.plot(t, y1, label='y1(t)')
plt.plot(t, y2, label='y2(t)')

# Uncomment this line to display y3
# plt.plot(t, y3, label='y3(t)')

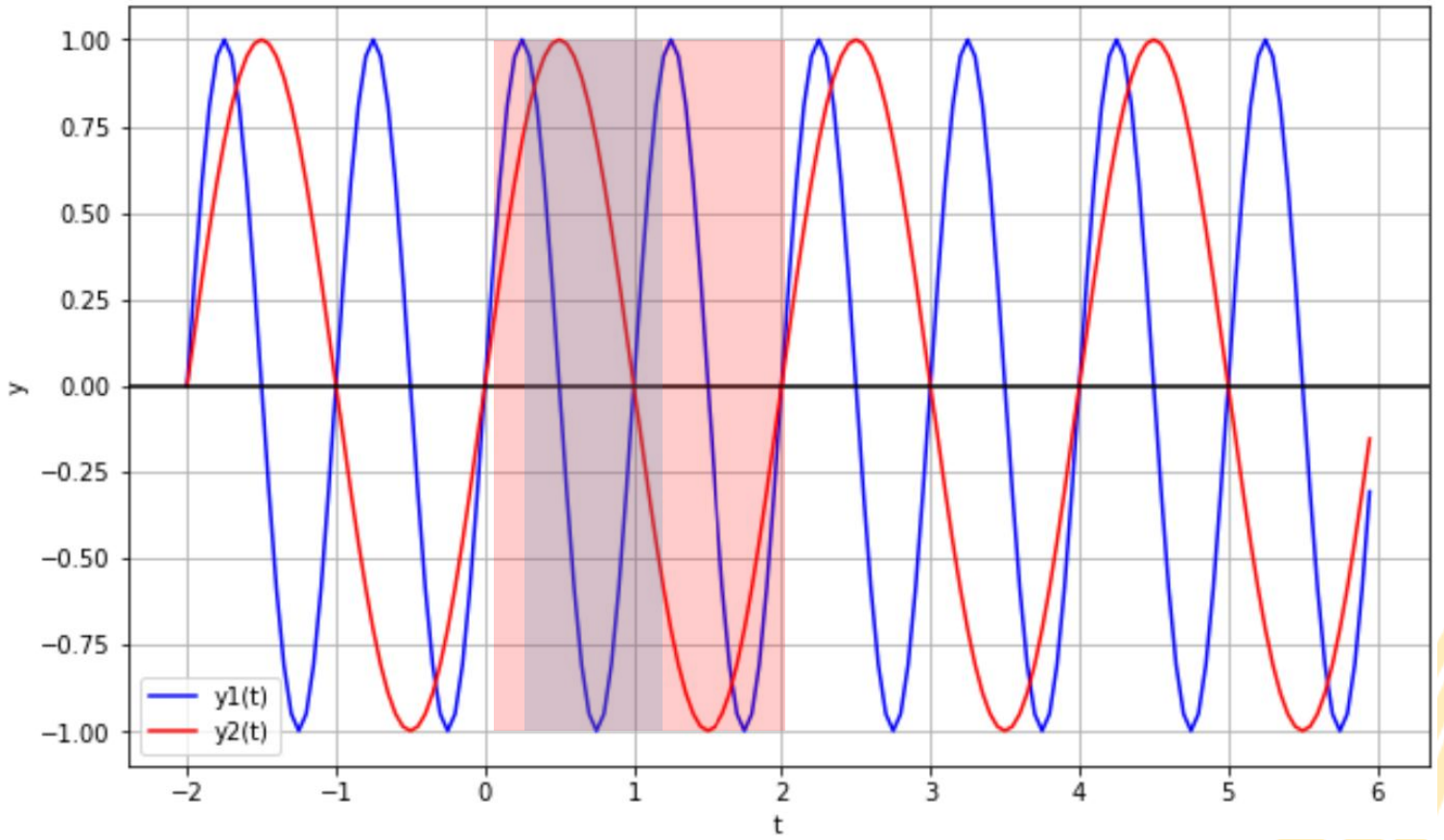
plt.grid(True, which='both')
plt.axhline(y=0)

plt.xlabel('t')
plt.ylabel('y')
plt.legend()

plt.show()
```

https://matplotlib.codeutility.io/

$y = \sin(2\pi f t + \Phi)$ $f = 0.5$



Sine Wave

```
import numpy as np
import matplotlib.pyplot as plt
import math

#Assign frequency and phase
freq1 = 1
freq2 = 0.5
freq3 = 0.25
phase = 0

#Visualize signals
fig = plt.figure(figsize=(10, 6))

t = np.arange(-2, 6, 0.05)

y1 = np.sin(2*math.pi * freq1 * t + phase)
y2 = np.sin(2*math.pi * freq2 * t + phase)
y3 = np.sin(2*math.pi * freq3 * t + phase)

plt.plot(t, y1, label='y1(t)')

#Uncomment this line to display y2
#plt.plot(t, y2, label='y2(t)')

plt.plot(t, y3, label='y3(t)')

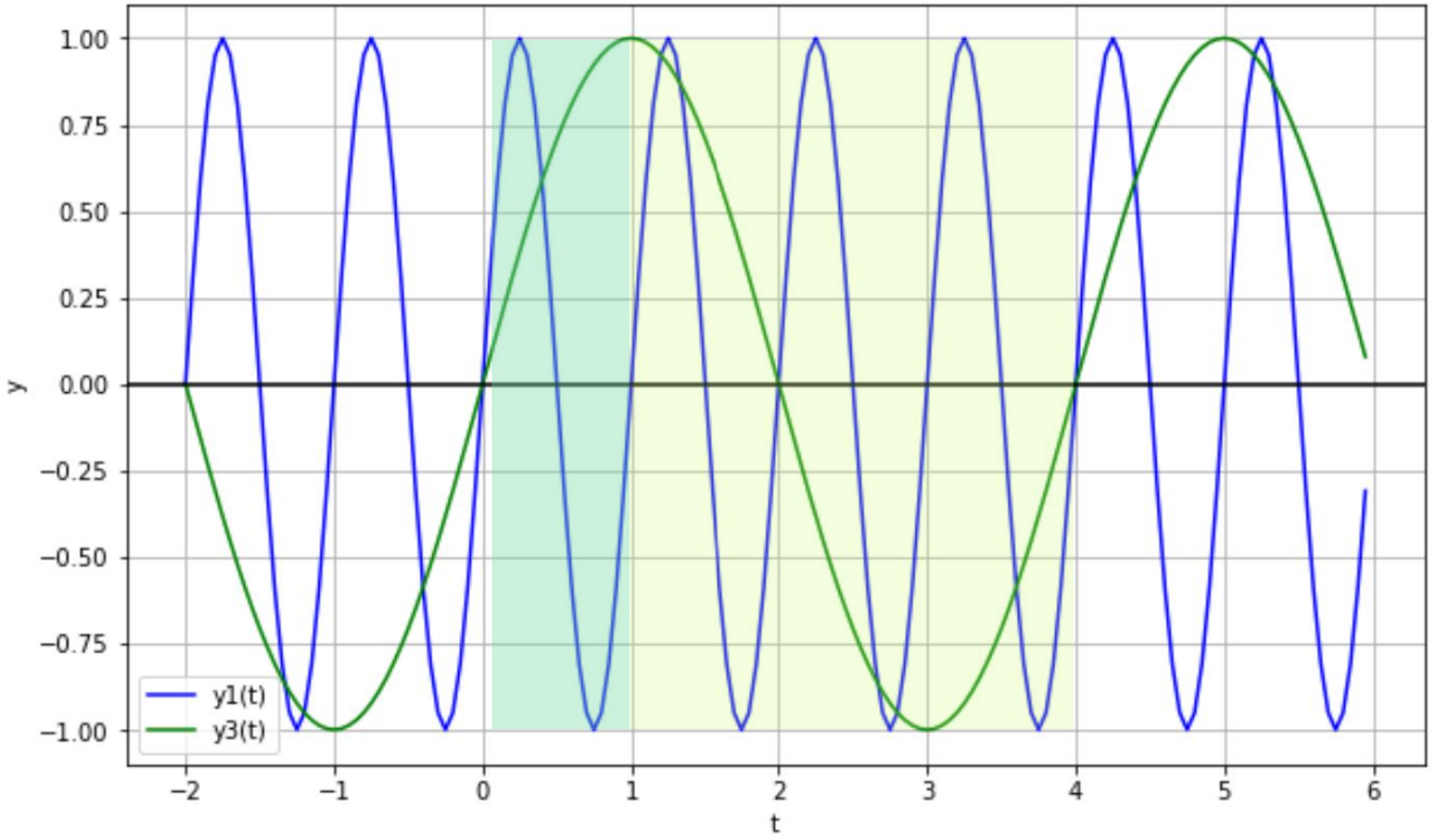
plt.grid(True, which='both')
plt.axhline(y=0)

plt.xlabel('t')
plt.ylabel('y')
plt.legend()

plt.show()
```

https://matplotlib.codeutility.io/

$y = \sin(2\pi f t + \Phi)$ $f = 0.25$



Sine Wave

https://matplotlib.codeutility.io/

$$y = \sin(2\pi f t + \Phi) \qquad \Phi 1 = 0, \Phi 2 = 1, \Phi 3 = -1$$

```
import numpy as np
import matplotlib.pyplot as plt
import math

# Assign frequency and phase
freq1 = 1
freq2 = 0.5
freq3 = 0.25

phase1 = 0
phase2 = 0.5
phase3 = 1
phase4 = -0.5
phase5 = -1

# Visualize signals
fig = plt.figure(figsize=(10, 6))

t = np.arange(-2, 6, 0.05)

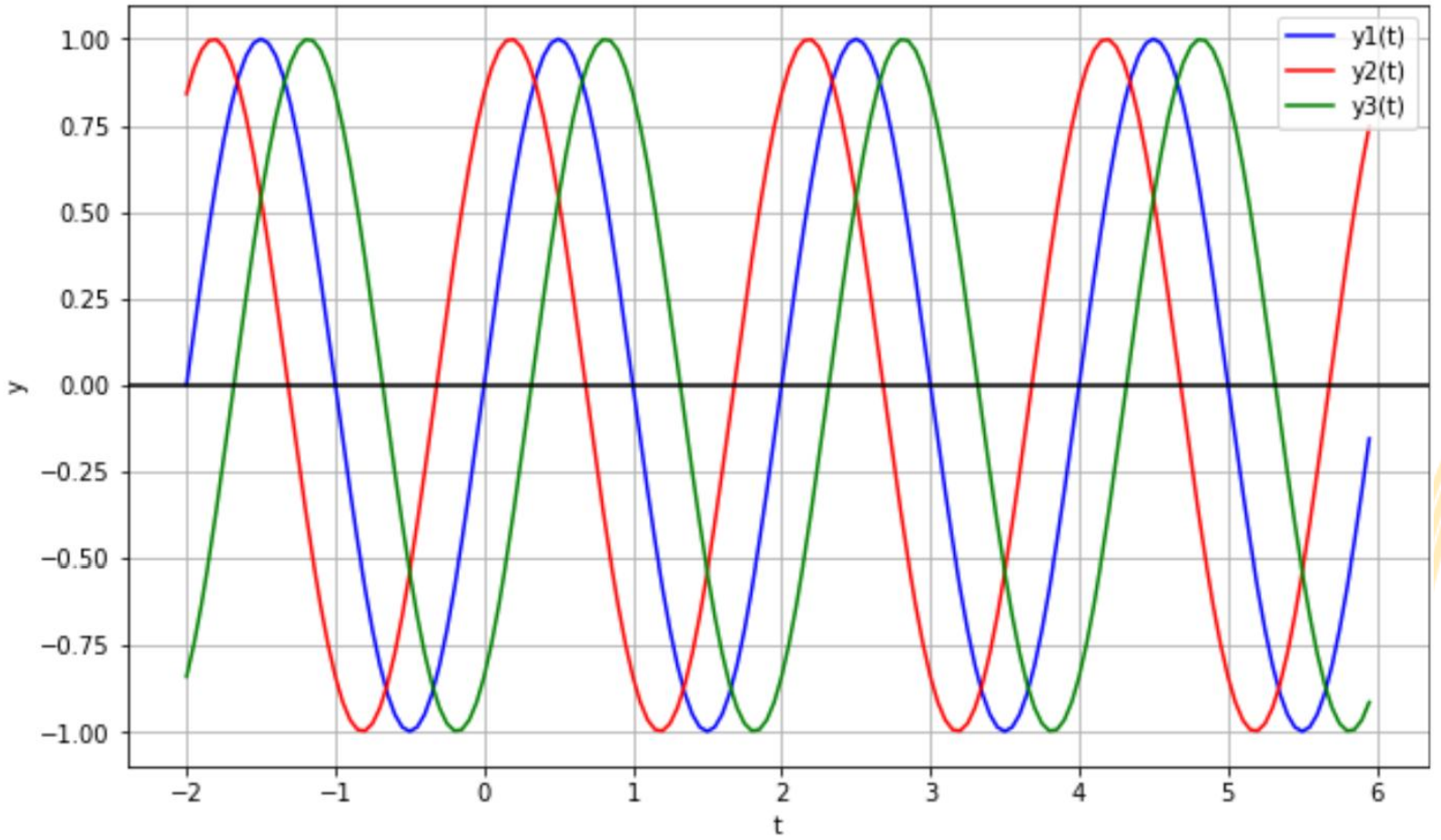
y1 = np.sin(2 * math.pi * freq2 * t + phase1)
y2 = np.sin(2 * math.pi * freq2 * t + phase3)
y3 = np.sin(2 * math.pi * freq2 * t + phase5)

plt.plot(t, y1, label='y1(t)')
plt.plot(t, y2, label='y2(t)')
plt.plot(t, y3, label='y3(t)')

plt.grid(True, which='both')
plt.axhline(y=0)

plt.xlabel('t')
plt.ylabel('y')
plt.legend()

plt.show()
```



Sine Wave : Converting Phase to Degree

$$y = \sin(2\pi f t + \Phi) \qquad \Phi 1 = 0, \Phi 2 = 1, \Phi 3 = -1$$

$$\omega = 2\pi f$$

For this sine wave $f = 1$

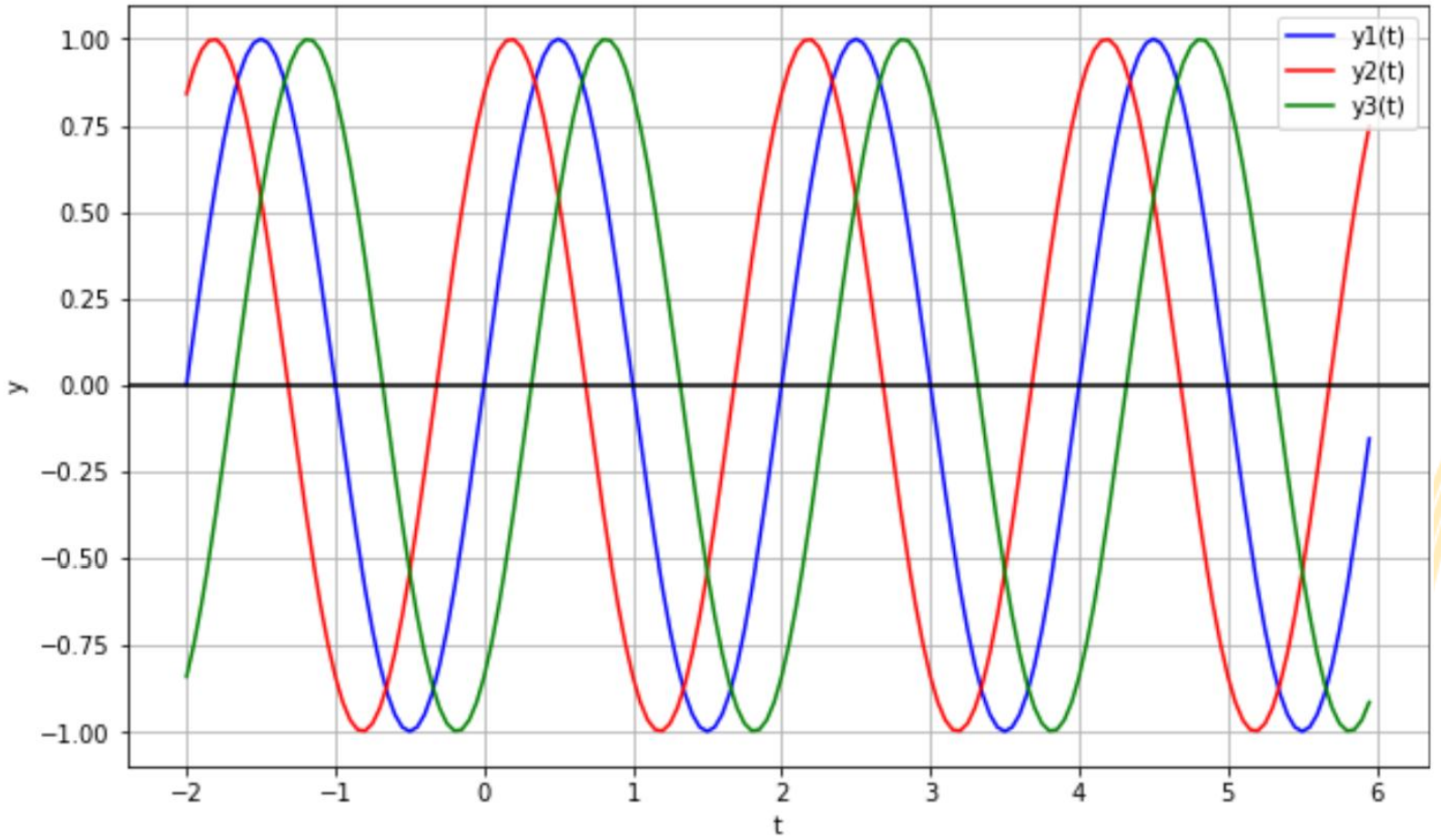
So, we have $\omega = 2\pi(1) = 2\pi$

Since, we designate that $\omega = 360°$

So, the phase = -1 in degree will be

$$2\pi = 360°$$

$$\Omega = 1 = \frac{1}{2\pi} \times 360° = 57.29°$$



Sine Wave : Playing with Phases

```
import numpy as np
import matplotlib.pyplot as plt
import math

#Assign frequency and phase
freq1 = 1
freq2 = 0.5
freq3 = 0.25

phase1 = 0
phase2 = 0.5
phase3 = 1
phase4 = -0.5
phase5 = -1
phase6 = np.pi/2
phase7 = -np.pi/2

#Visualize signals
fig=plt.figure(figsize=(10,6))

t = np.arange(-2,6,0.05)

y1 = np.sin(2*math.pi * freq2 * t + phase1)
y2 = np.sin(2*math.pi * freq2 * t + phase6)
y3 = np.sin(2*math.pi * freq2 * t + phase7)

plt.plot(t, y1, label='y1(t)')
plt.plot(t, y2, label='y2(t)')
plt.plot(t, y3, label='y3(t)')

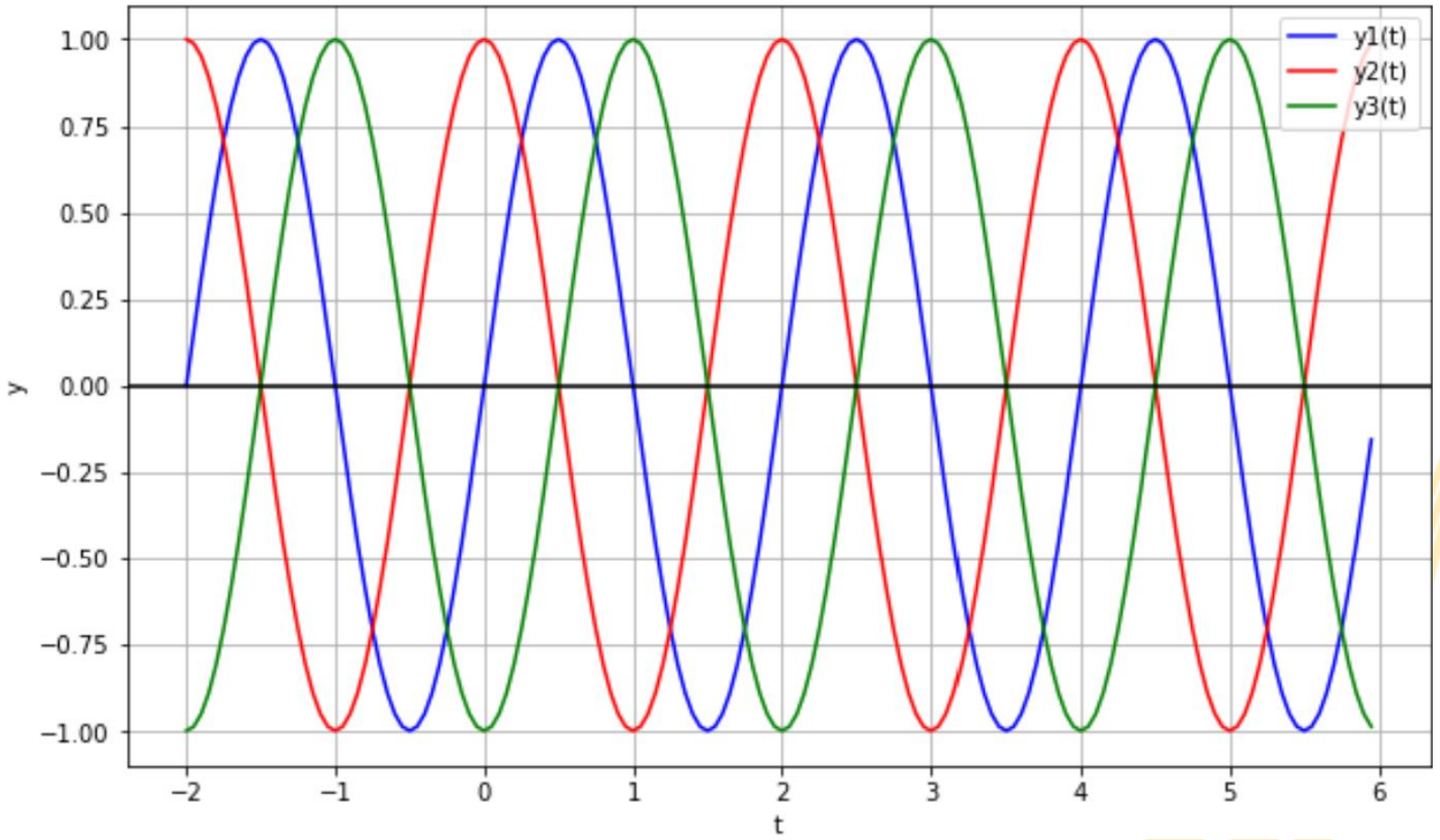
plt.grid(True,which='both')
plt.axhline(y=0)

plt.xlabel('t')
plt.ylabel('y')
plt.legend()

plt.show()
```

https://matplotlib.codeutility.io/

$y = \sin(2\pi f t + \Phi)$ $\Phi 1 = 0, \Phi 2 = \pi/2, \Phi 3 = -\pi/2$



Sine Wave : Playing with Phases

```
import numpy as np
import matplotlib.pyplot as plt
import math

#Assign frequency and phase
freq2 = 0.5

phase1 = 0
phase7 = -np.pi/2
phase8 = -np.pi

#Visualize signals
fig=plt.figure(figsize=(10,6))

t = np.arange(-2,6,0.05)

y1 = np.sin(2*math.pi * freq2 * t + phase1)
y2 = np.sin(2*math.pi * freq2 * t + phase7)
y3 = np.sin(2*math.pi * freq2 * t + phase8)

plt.plot(t, y1, label='y1(t)')
plt.plot(t, y2, label='y2(t)')
plt.plot(t, y3, label='y3(t)')

plt.grid(True,which='both')
plt.axhline(y=0)

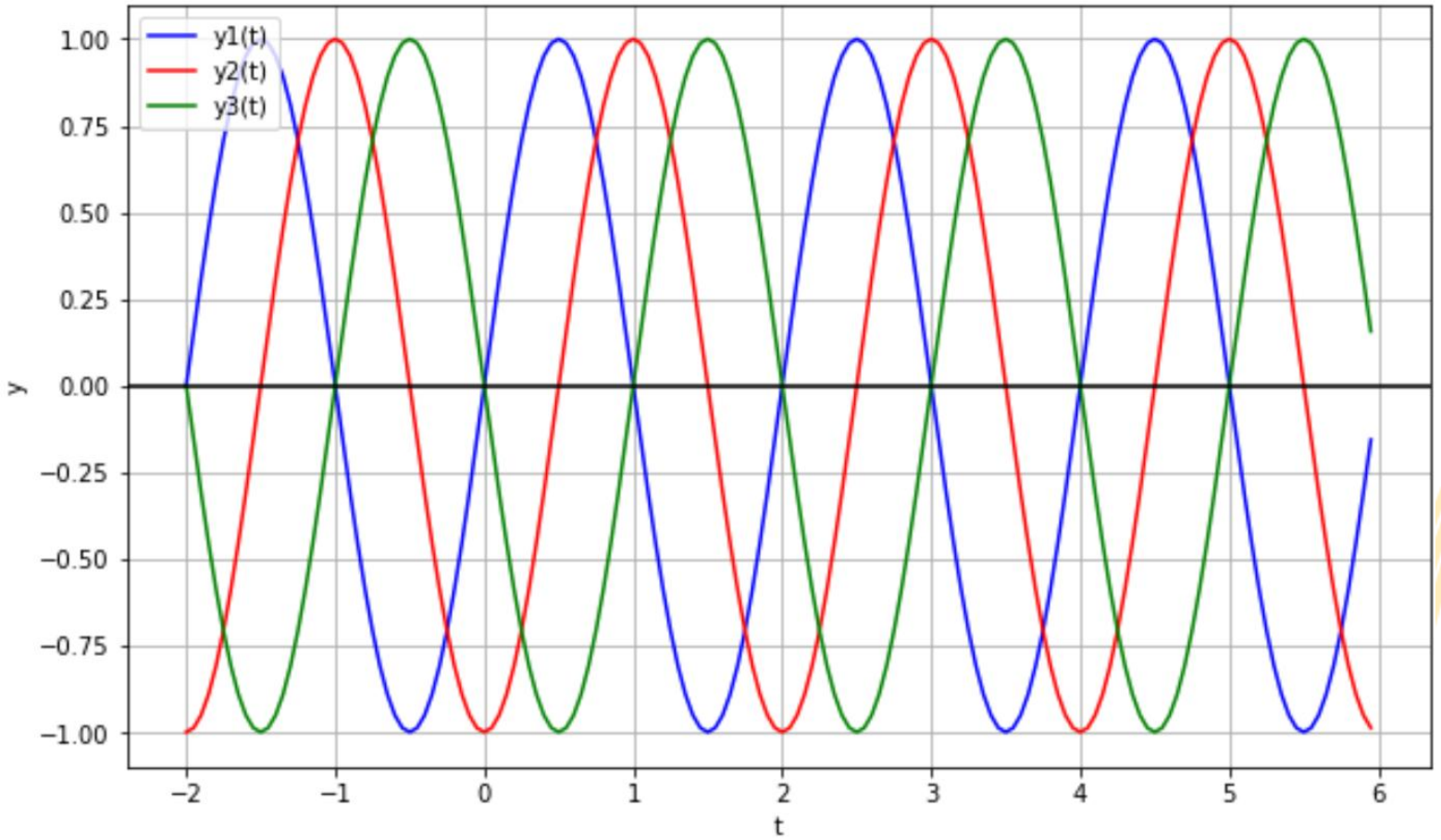
plt.xlabel('t')
plt.ylabel('y')
plt.legend()

plt.show()
```

https://matplotlib.codeutility.io/

$y = \sin(2\pi ft + \Phi)$

$\Phi 1 = 0, \Phi 2 = -\pi/2, \Phi 3 = -\pi$



Square Wave

- A square wave is a composite of infinite sine waves

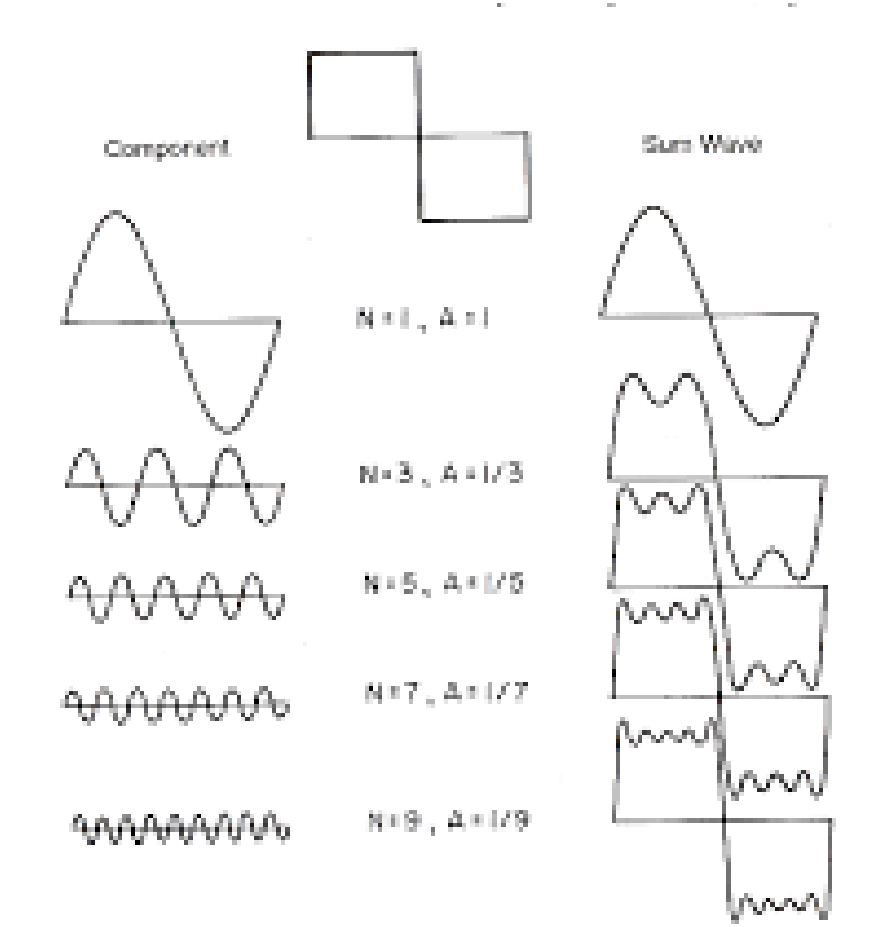
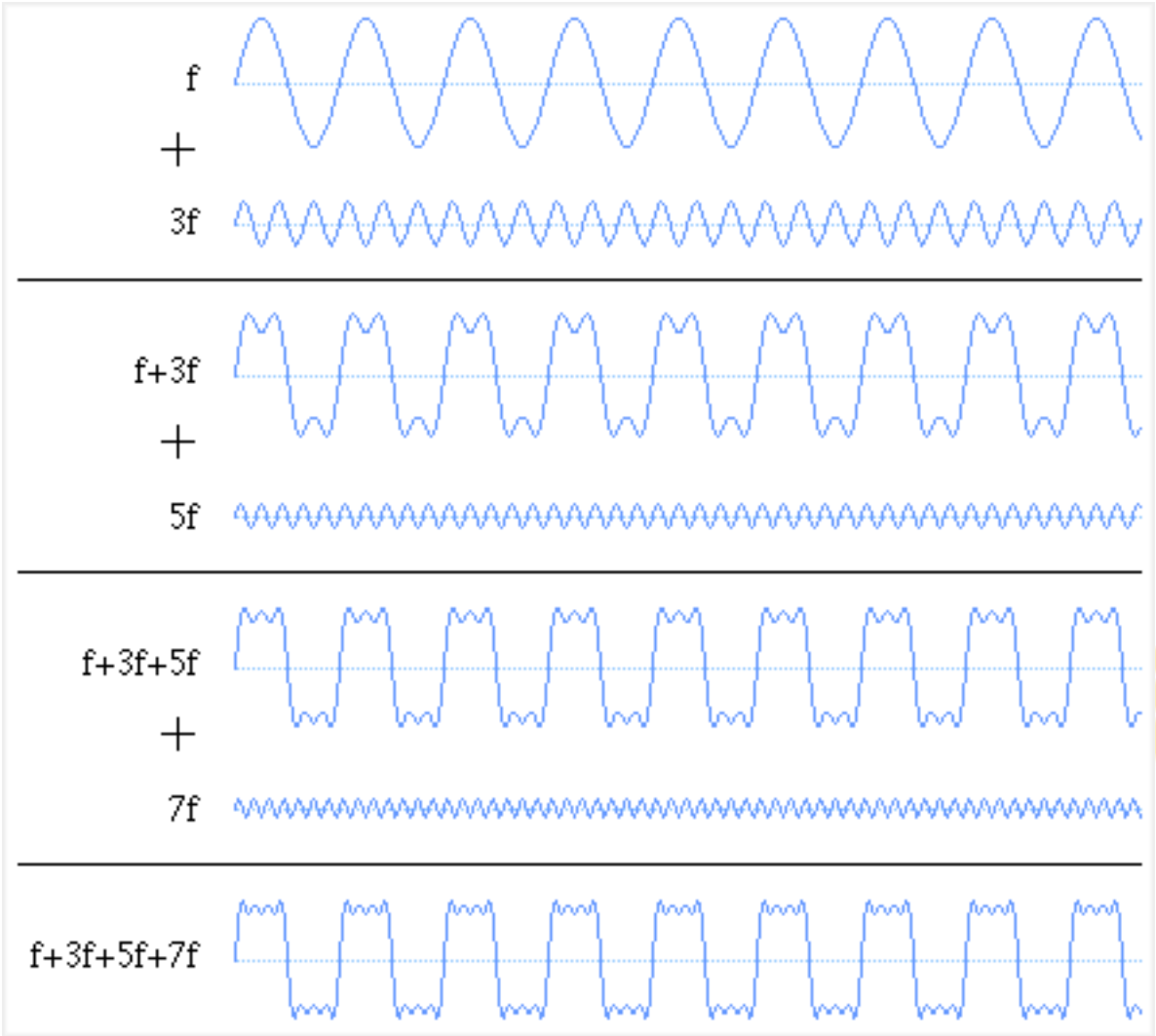
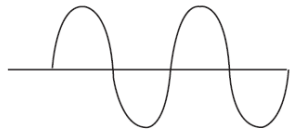


Figure 4-5 Fourier synthesis of a square wave. At the left are the successive harmonics; at the right are the sum waves including each successive harmonic. The graph at the top is the wave being synthesized.

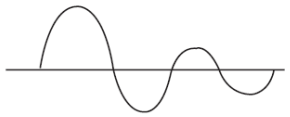


http://www.slack.net/~ant/bl-synth/4.harmonics.html

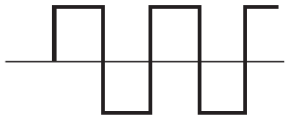
Types of Signal Waveforms



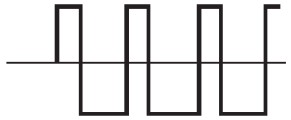
Sine Wave



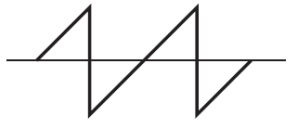
Damped Sine Wave



Square Wave



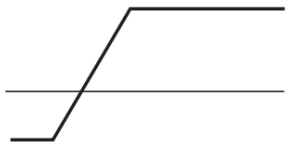
Rectangular Wave



Sawtooth Wave



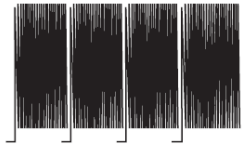
Triangle Wave



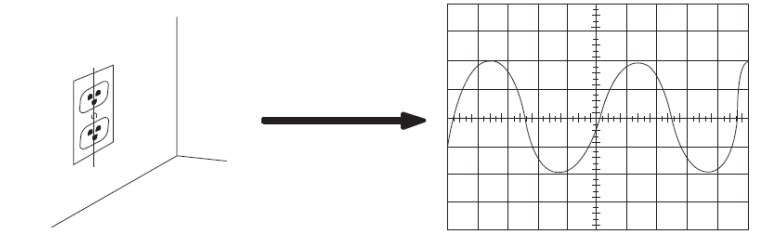
Step

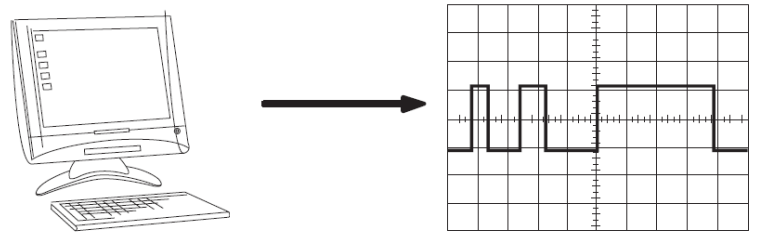


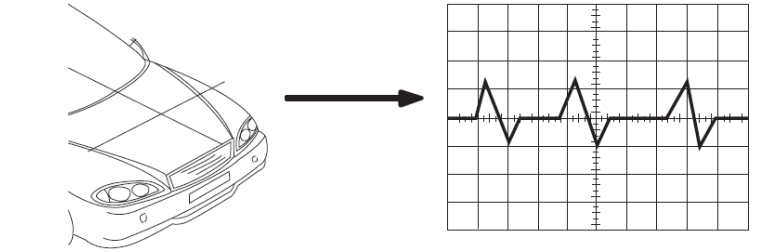
Pulse

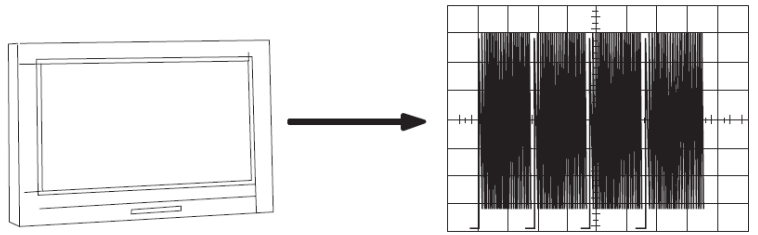


Complex

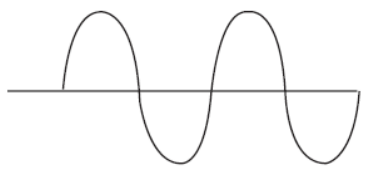




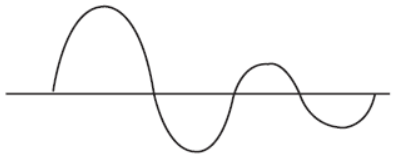




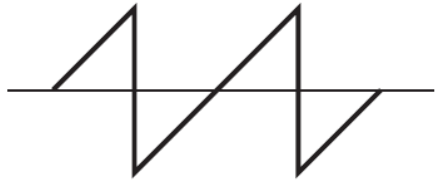
Types of Signal Waveforms



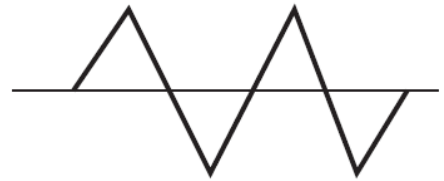
Sine Wave



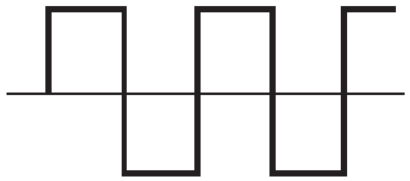
Damped Sine Wave



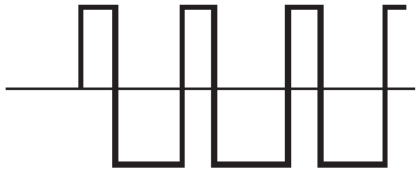
Sawtooth Wave



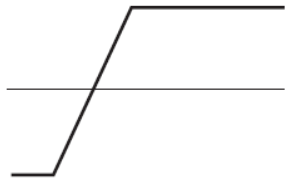
Triangle Wave



Square Wave



Rectangular Wave



Step



Pulse



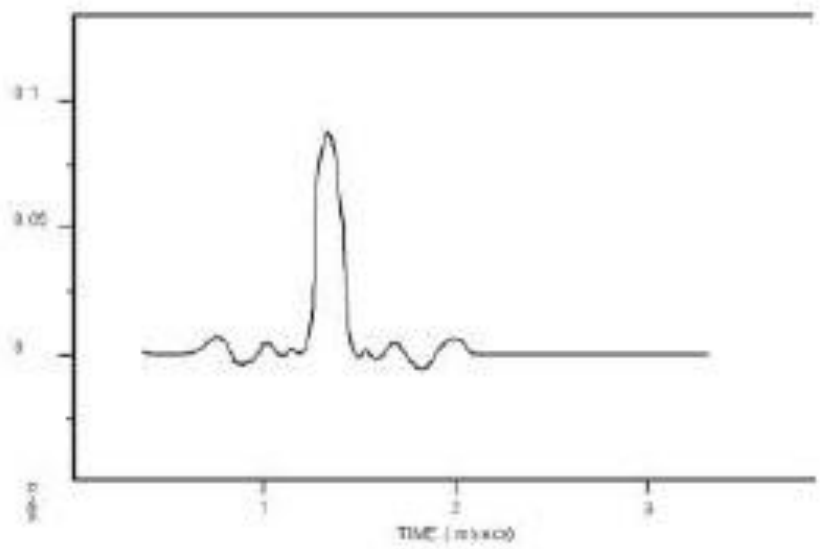
Pulse Train

Periodic vs. Aperiodic Signals

- Periodic signals are signals that have repeated patterns.



Periodic signal



Non-periodic signal

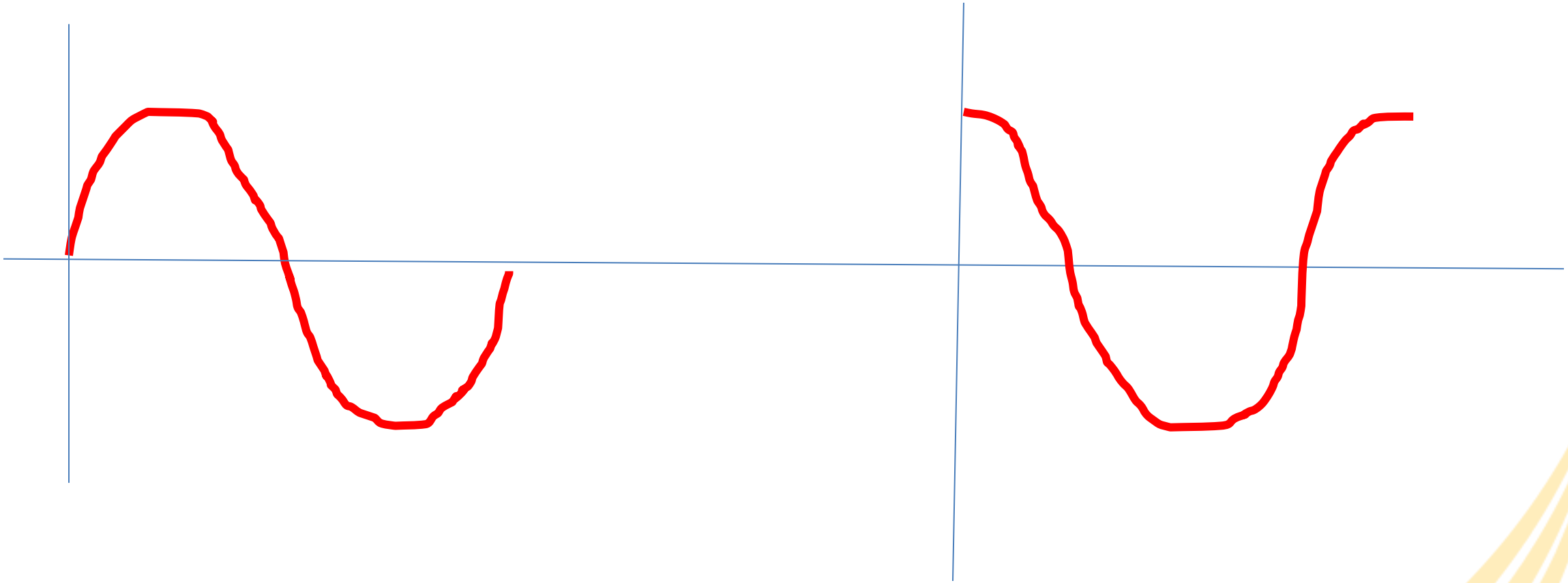
Today Agenda

- Advantages of Layered Data Communication
- Functions of each layer
- Addressing
  - ✓ Physical layer address
  - ✓ MAC address and IP address

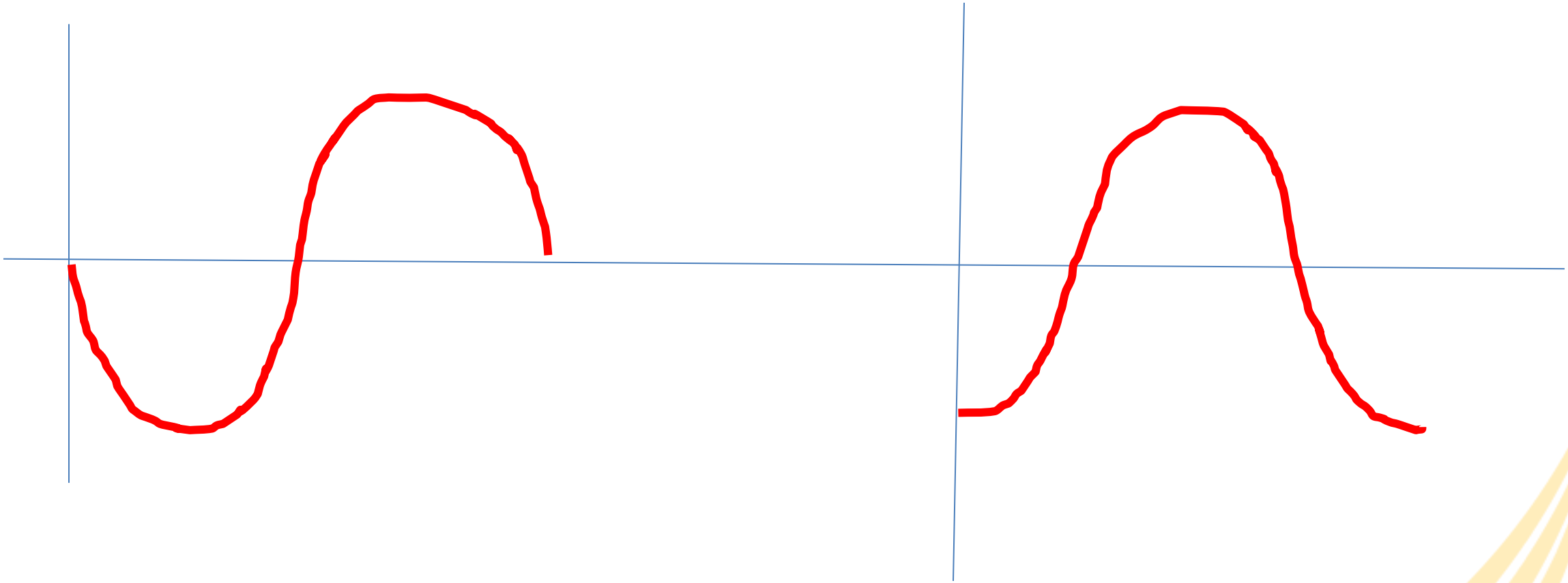
- - Break -

- Network Topology
- Data and Signal
- Equation of Sine wave signal
  - ✓ Phase, Frequency, Amplitude

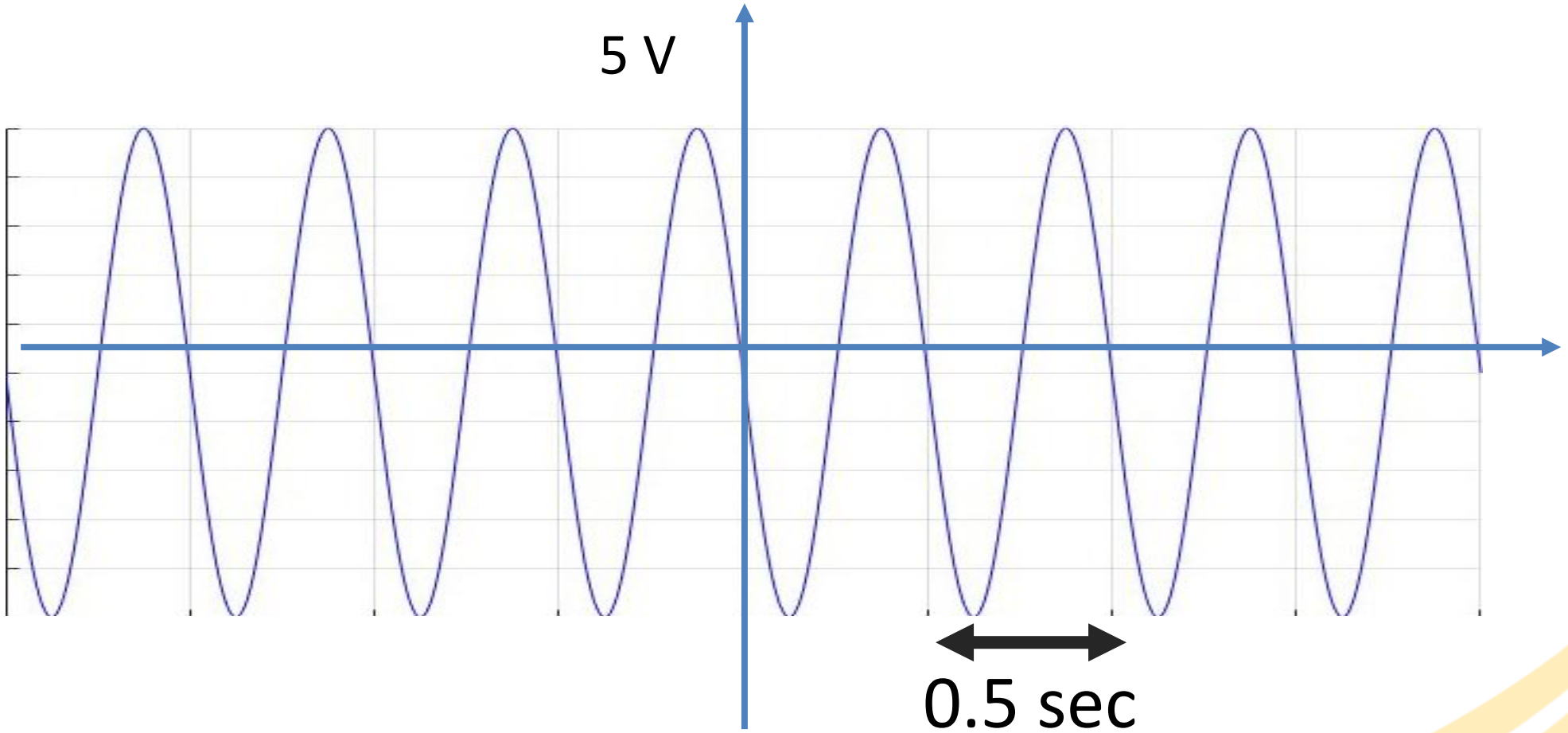
Exercise 1: Phase – How many degree? (1)



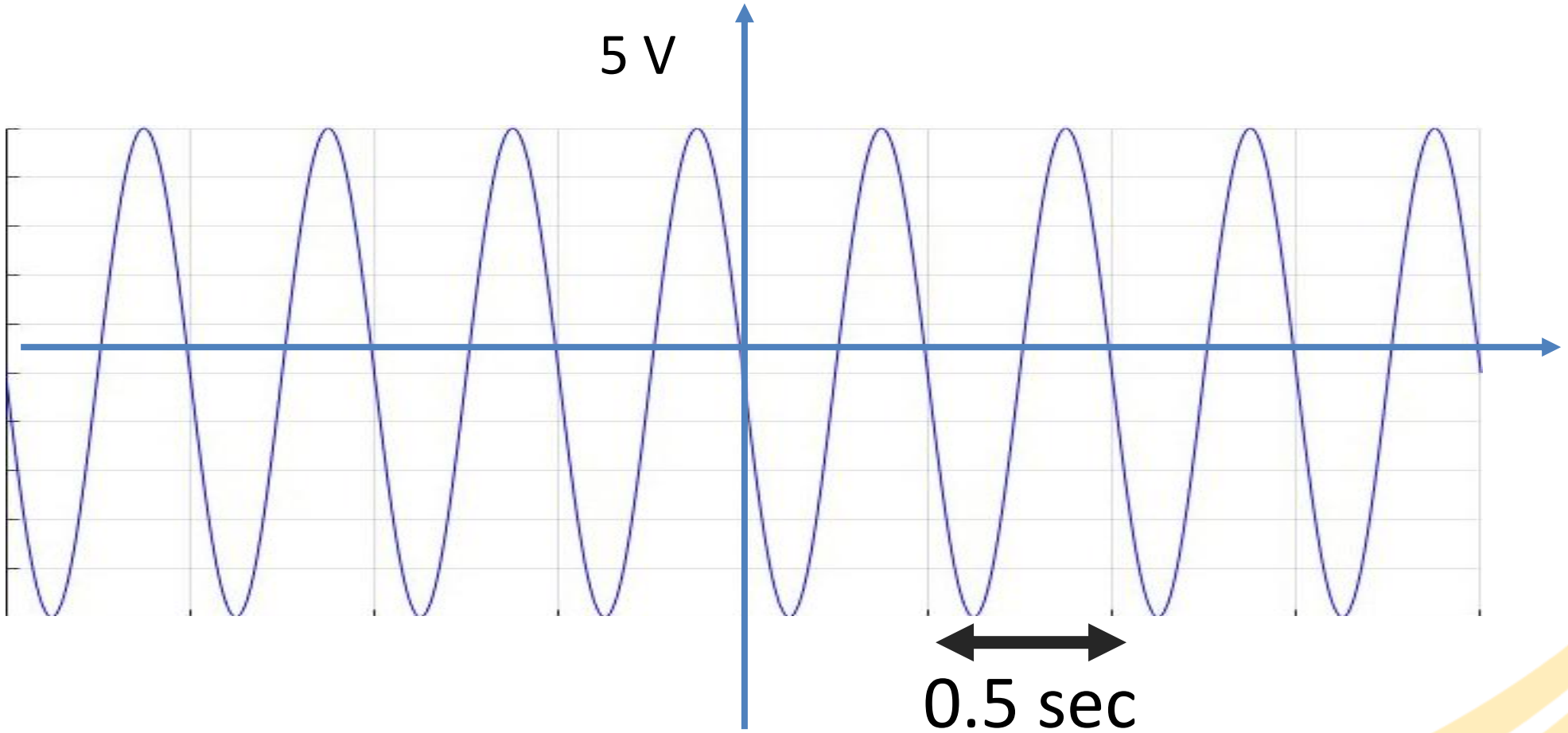
Exercise 2: Phase – How many degree?



Exercise 3: What is the Frequency of this signal? (Hz)



Exercise 4: What is the Phase of this signal? (Degree)



That's All for Today



- Lets do the exercise and end the class

Quiz #1 (Lectures 1 & 2 & 3)

Thu 27 August 2026

Be honest and Do it by Yourself!

